AMERICAN INSTITUTES FOR RESEARCH®

The Impact of eMINTS Professional Development on Middle School Teacher Instruction and Student Achievement

Year 3 Report

Coby Meyers
Ayrin Molefe
Sonica Dhillon
Bo Zhu

April 2015

The Impact of eMINTS Professional Development on Middle School Teacher Instruction and Student Achievement

Year 3 Report

April 2015

Coby Meyers
Ayrin Molefe
Sonica Dhillon
Bo Zhu

AMERICAN INSTITUTES FOR RESEARCH®

1000 Thomas Jefferson Street NW
Washington, DC 20007-3835
202.403.5000 | TTY 877.334.3499

www.air.org

1527_02/15

Contents

Executive Summary

In 2010, the eMINTS (enhancing Missouri's Instructional Networked Teaching Strategies) National Center received an Investing in Innovation (i3) validation grant to implement the eMINTS Comprehensive Program in rural middle schools and test the efficacy of the program in a randomized controlled trial. The program is based on four underlying research-based components: inquiry-based learning, high-quality lesson design, a community of learners, and technology integration. The program provides teachers with approximately 240 hours of professional development spanning two years and support that includes monthly classroom visits for coaching. The eMINTS National Center also developed a third year of professional development with the Intel® Teach Program to build on what teachers learned during the first two years of the eMINTS Comprehensive Program. The third year combines additional professional development and Intel's suite of Web-based teaching tools to expand teachers' use of inquiry-based learning. The key evaluation objectives are as follows:

- Employ experimental methods to rigorously examine the program's impact on middle school teacher practices and student achievement, particularly for middle schools in rural settings.
- Examine the impacts of a third year of professional development, using Intel Teach Program courses and tools, on middle school means of teacher and student outcomes.

In spring 2011, the study randomized 60 high-poverty rural schools in Missouri to one of three groups: (1) the traditional, two-year eMINTS Comprehensive Program, (2) the eMINTS Comprehensive Program plus a third year of professional development using Intel Teach Program courses, or (3) business as usual. This report examines implementation and impacts on middle schools after three years. Group 1 schools were one year removed from the completion of the eMINTS Comprehensive Program. Group 2 schools were completing their year of Intel Teach, or their third straight year receiving professional development (eMINTS+Intel).

The study investigated the following confirmatory research questions (RQs):[1]

1. What is the impact of the eMINTS Comprehensive Program and the eMINTS Comprehensive Program plus Intel Teach on school mean 7th and 8th grade performance in mathematics and communication arts?
2. What is the impact of the eMINTS Comprehensive Program and the eMINTS Comprehensive Program plus Intel Teach on school mean 7th and 8th grade measure of 21st century skills (including communication, technology literacy, and critical thinking)?
3. What is the impact of the eMINTS Comprehensive Program and the eMINTS Comprehensive Program plus Intel Teach on school mean 7th and 8th grade measure of student engagement?

[1] Confirmatory research questions are the main questions that this study is designed to address. This study was designed using an experimental design to provide rigorous estimates of the causal effects of the eMINTS Comprehensive Program and the eMINTS Comprehensive Program plus the Intel Teach Program.

The study also examined the following exploratory questions on impact variation and impacts on teacher practices:

4. Do the impacts of the eMINTS Comprehensive Program and the eMINTS Comprehensive Program plus Intel Teach on school mean performance in mathematics and communication arts, 21st century skills, and engagement vary across subgroups of students?

5. What are the impacts of eMINTS Comprehensive Program and the eMINTS Comprehensive Program plus Intel Teach on seventh- and eighth-grade teachers' instructional practices?

This Year 3 report provides results on the above questions on the basis of different analytic samples. The mathematics and communication arts analytic samples (RQ1) each consisted of about 3,000 students, and the analytic samples for 21st century skills and student engagement (RQ2 and RQ3) each had about 2,300 students. These analytic samples also were used to assess variations in impact across student subgroups. Between 98 and 117 teachers were used for the analysis of teacher outcomes.

By design, the eMINTS Comprehensive Program is a two-year professional development program, and eMINTS+Intel adds a third year to the original program length. In this third year of implementation, teachers assigned to the eMINTS only did not receive additional professional development. The eMINTS+Intel group received the same comprehensive two-year eMINTS program plus a third year of training using the Intel Teach Program course training. As a result, this Year 3 study compares the each treatment group to the schools assigned to the business as usual group, as well as to each other.

eMINTS Program Implementation

After two years of the eMINTS Comprehensive Program, the implementation results are encouraging. Across all eMINTS and eMINTS+Intel schools, eMINTS technical and professional development staff provided essential resources, professional development, and guidance to support implementation at the district, school, and classroom levels. Twenty-five of the 38 treatment schools[2] (65.8 percent)[3] demonstrated high levels of overall implementation fidelity, and 13 of the 38 (34.2 percent) treatment schools demonstrated moderate levels of overall implementation fidelity. Seventeen of the 38 treatment schools (44.7 percent) reported 85 percent or greater attendance on the teacher professional development component (which includes formal professional development and in-classroom coaching and mentoring), which is considered the most important component of the eMINTS Comprehensive Program. All of the schools (100 percent) reported 80 percent or greater attendance on the administrative support components (which includes formal professional development and school walk-throughs), which are considered the second most important component of the eMINTS Comprehensive Program.

[2] Two treatment schools dropped out before the first year of implementation.
[3] Not including ongoing technology support, all schools had high implementation in at least three of the four remaining components.

It is important to note that the technology resources provided to classrooms in the eMINTS Comprehensive Program are purposely staggered in both installation and implementation. The full suite of technology resources (hardware and software) was fully installed and functional in the treatment schools by mid-February 2012. Thus, students had full access to the technology for approximately three months of the first year of the eMINTS Comprehensive Program. By contrast, students had access to the entire suite of technology for the full school year beginning in the second year. During Years 1 and 2, 35 of the 38 treatment schools (92.1 percent) reported high levels of technology infrastructure implementation, and 22 of the 38 treatment schools (57.9 percent) reported high levels of technology use. Ongoing technology support was reportedly high in 18 of the 38 treatment schools (47.4 percent).

For the 20 eMINTS+Intel schools who received the Intel Teach Program, implementation results are mixed for Year 3. The eMINTS National Center provided teacher professional development, coaching sessions, and unit plan reviews. However, many schools did not reach high levels of implementation fidelity. Of the 20 eMINTS+Intel schools, nine (45 percent) demonstrated high levels of overall implementation fidelity, two (10 percent) demonstrated moderate levels of overall implementation fidelity, and nine (45 percent) demonstrated low levels of overall Intel program implementation.

Impacts on Achievement, 21st Century Skills, and Engagement

In Year 3, eMINTS and eMINTS+Intel treatment schools both significantly outscored control schools in mathematics. The differences in average achievement were notable, albeit relatively small, with effect sizes of 0.13 (eMINTS) and 0.18 (eMINTS+) standard deviations. eMINTS+ had a higher estimated impact than eMINTS did, but the difference was not significant. No significant differences were found among eMINTS, eMINTS+, and control schools in terms of students' communication arts, technology, and student engagement scores. There were statistically significant variations in estimated impacts across student subgroups in mathematics and student engagement.

Impacts on Instructional Practices

Teacher self-report surveys and classroom observations were used to measure changes in teachers' instructional practices. No significant differences were found among eMINTS, eMINTS+Intel, and control schools in terms of teacher survey responses to the Community of Learners domain. But classroom observation results indicate significant and medium positive effects of eMINTS (0.49 standard deviations) and eMINTS+Intel (0.52 standard deviations) versus the control schools.

eMINTS and eMINTS+Intel schools had significantly higher average scores than control schools in terms of teachers' inquiry-based learning. Teacher survey data revealed large positive effects (0.73 and 0.96 standard deviations, respectively). Classroom observation data also revealed medium positive effects (0.68 and 0.56 standard deviations, respectively).

eMINTS and eMINTS+Intel schools also had significantly higher average scores than control schools in terms of technology integration. Teacher survey data revealed very large positive effects (1.43 and 1.55 standard deviations, respectively). Classroom observation data had medium to large positive effects (0.58 and 0.78 standard deviations, respectively).

High-quality lesson design could not be observed, but teacher survey data showed no significant differences between eMINTS and control schools. Moreover, teacher survey data showed that eMINTS+Intel schools had significantly higher average scores in this domain than control schools (an effect size of 0.53 standard deviations) but no different than eMINTS schools.

Chapter 1: Introduction

Overview

Missouri's rural K–12 student population is nearly 250,000 students, which is the 18th largest in the United States. Poverty is high, with 44 percent of Missouri rural students qualifying for free- or reduced-price lunch (Strange, Johnson, Showalter, & Klein, 2012). The Institute of Education Sciences (2010) reported that only 19 percent of Missouri students eligible for free- or reduced-price lunch attained proficiency or better on the 2009 National Assessment of Educational Progress mathematics test. Student mobility in Missouri rural schools ranks 14th highest in the United States, with 12.4 percent of families reporting changing residences in the 15 months prior to being surveyed (Strange et al., 2012). Per-pupil expenditure for Missouri rural schools is the 13th lowest in the United States (Strange et al., 2012). The attributes of many rural communities (isolation, a low tax base, an aging population, and higher poverty levels) contribute to the scarcity of qualified teachers for rural schools nationally and in Missouri specifically (Monk, 2007). Among the promising practices that rural schools might use to recruit and retain high-quality teachers is improving a school's culture and working conditions (McClure & Reeves, 2004).

Middle schools (typically Grades 6–8) provide additional challenges. Middle school represents a critical time for students to develop the knowledge and skills they will need to achieve college and career readiness. Olson (2006) found that the reading and mathematics skills needed for success in the workplace are comparable to those needed for success in the first year of college. Unfortunately, middle school also is the period in which students may begin to lag in academic performance. Lembke and Gonzalez (2006) reported that the performance of U.S. middle school students is lower than that of peers in other countries, particularly when tested on tasks embedded in 21st century skills. Tasks of this nature commonly require skills cited by groups such as the Partnership for 21st Century Skills (Bellanca & Brandt, 2010) as being associated with problem solving, communication, collaboration, creativity and innovation, and the use of information technology.

Schoolwide and teacher professional development programs focused on developing students' 21st century skills have recently emerged to help prepare students for college and the workforce. According to the Partnership for 21st Century Skills, successful professional development programs should emphasize a comprehensive approach that includes updates to standards and assessments and incorporates the following components:

- Ensure that educators understand the importance of 21st century skills and how to integrate them into daily instruction.
- Enable collaboration among all participants.
- Allow teachers and principals to construct their own learning communities.
- Tap the expertise within a school or a district through coaching, mentoring, and team teaching.
- Support educators in their role of facilitators of learning.
- Use 21st century tools (Partnership for 21st Century Skills, 2006).

Most programs include professional development to improve specific aspects of effective teaching, such as standards-based lesson design, data-driven decision making, differentiation, inquiry- or project-based instruction, collaborative learning structures, and technology integration; few, however, incorporate a comprehensive approach to 21st century skill development to the extent that the eMINTS Comprehensive Program does.

The eMINTS National Center at the University of Missouri, in partnership with the Missouri Department of Higher Education (DHE) and the Missouri Department of Elementary and Secondary Education (DESE), offers the eMINTS professional development program to teams of educators, especially those serving high-need students. eMINTS professional development generates buildingwide reform by helping teachers master the translation of Common Core or state standards and information from assessments into engaging classroom practices that employ technology. The program is based on four underlying research-based components: inquiry-based learning, high-quality lesson design, a community of learners, and technology integration. It addresses issues identified as barriers in the consistent use of standards-based instruction and technology. The program provides teachers with approximately 240 hours of professional development spanning two years and support that includes monthly in-classroom coaching and mentoring visits. As part of refining and improving eMINTS professional development, eMINTS staff recently integrated the Intel Teach Program into the comprehensive program, adding a third year of professional development to help teachers sustain and build on the first two years of eMINTS professional development. The third year combines online and face-to-face professional development for teachers to expand their use of inquiry-based learning. Teachers also gain access to Intel's suite of online tools, which are designed to involve students in 21st century higher-order thinking and problem solving.

Despite years of implementing standards-based accountability systems, many teachers today lack the necessary preparation to develop standards-based instructional strategies and inform decisions based on student assessments (Drake, 2007). The eMINTS Comprehensive Program is designed to help teachers use standards-based instructional strategies and assessment in their lesson planning, implement inquiry-based instruction, facilitate a vibrant learning community, and integrate technology into their instruction. As teachers begin to apply these approaches to lesson planning and instruction, it is expected that student engagement, the application of 21st century skills, and academic performance will improve. A decade of evaluations of the eMINTS original two-year program consistently has shown promise in changing teachers' practices and raising student achievement. The evaluation studies are limited, however, in their generalizability across multiple settings and methodological rigor (Martin, Strother, & Reitzes, 2009; Martin, Strother, Weatherholt, & Dechaume, 2008; Office of Social and Economic Data Analysis [OSEDA], 2001b, 2002, 2003b, 2003c; Tharp, 2004, 2006). Prior program evaluations often focused on intermediate elementary students (Grades 3–6) in urban settings and were conducted using either nonequivalent comparison group designs or pre-post designs with no comparison groups. Numerous middle school principals in Missouri had previously expressed interest in eMINTS, and the provision of professional development at that level was determined to be a logical next step in delivery.

In 2010, the eMINTS National Center received an i3 validation grant to implement the eMINTS Comprehensive Program in rural middle schools and test the efficacy of the program in a randomized controlled trial. The key objectives of the evaluation were as follows:

- Improve the rigor of the research base addressing the effectiveness in improving teacher practices and student achievement.
- Examine the difference that a third year of professional development makes for eMINTS teachers, using Intel Teach Program courses and tools, on teacher and student outcomes.
- Expand the applicability of the evaluation results to middle school students in high-poverty rural schools.

This report examines implementation and impacts on teachers and students after three years.

About the eMINTS National Center and the Intel Foundation

The eMINTS National Center offers a wide range of professional development options to teachers, administrators, technology specialists, and other educators. Leading experts at the University of Missouri, DESE, and DHE are collaborating to produce programs intended to engage students and enrich teaching by integrating technology and student-centered instruction. The programs range from short-term, customized awareness sessions to full school or organizationwide implementations requiring a long-term commitment. Professional development is geared to the needs and interests of PK–16 educators and is delivered either by eMINTS staff or local trainers who have completed eMINTS train-the-trainer certification.

The Intel Foundation was established in 1988 to develop programs, exercise leadership, and provide grant funding to fuel innovation in science, technology, engineering, and mathematics (STEM) education. The foundation strives to increase opportunities for people and communities worldwide—especially girls, underserved youth, and underrepresented populations. Intel's premier professional development program, the Intel Teach Program, consists of a series of K–12 professional development programs designed to introduce, expand, and support student-centered approaches in the classroom through technology integration. The Intel Teach Program has a large audience to which it delivers programming, reaching more than 350,000 U.S. teachers and 6 million teachers in 50 countries worldwide since 2002. Courses include both face-to-face and online professional development and are supported by free Web resources, such as unit plans, assessment strategies, and tools to support higher-order thinking skills. Through these courses, teachers learn how to use teacher and student online resources and online thinking tools to create new strategies for encouraging classroom collaboration and enhancing critical thinking skills.

The eMINTS Comprehensive Program

The eMINTS Comprehensive Program is considered the center's flagship program. It began in 1999 as a professional development program for Missouri teachers and has expanded into nine U.S. states and Australia. The overall goal of the eMINTS Comprehensive Program is to help teachers develop student-centered, purposeful instruction fostered by technology use. The program addresses issues identified as barriers among teachers in the consistent use of standards-based instruction, student assessment information, and technology.

The eMINTS Comprehensive Program is based on four underlying research-based components: inquiry-based learning, high-quality lesson design, a community of learners, and technology integration. The program includes a specific set of school and classroom technology equipment, intensive on-site professional development, online and face-to-face professional learning communities (PLCs), and job-embedded coaching to enhance teachers' classroom practices (Joyce & Showers, 1995). Specific details on the eMINTS Comprehensive Program, including equipment requirements, the teacher professional development program, and the recently added third-year professional development option with Intel, are provided in the following subsections.

Equipment Requirements

To be properly implemented, eMINTS classrooms must first meet minimum hardware, software, Internet, and equipment connectivity requirements.[4] An official eMINTS Comprehensive Program classroom at the middle school level includes an interactive whiteboard, an LCD projector, a teacher laptop, a digital camera, a printer and scanner, and at least a 1:1 ratio of students to computers. Specific Internet and equipment connectivity requirements must be met in an eMINTS classroom to ensure proper instructional functionality. The connectivity requirements may be met using either wired or wireless configurations or a combination of the two. In addition to Internet connectivity, the following setup requirements must be met to achieve minimum instructional functionality:

- A teacher laptop or workstation must connect to a classroom interactive whiteboard.
- A teacher laptop or workstation image must simultaneously appear on the teacher monitor and interactive whiteboard.
- Students must be able to display their work on the interactive whiteboard through a shared folder system or server.

Professional Development

The eMINTS Comprehensive Program consists of professional development for school principals, district and school technology coordinators, and classroom teachers. Before the start of the school year, a certified eMINTS instructional specialist (eIS) is assigned to a collection of schools—according to his or her geographic location—to provide formal and individualized professional development to teachers and communicate with principals and technology coordinators. eIS facilitate the ongoing development of a school-based leadership team within each school to support implementation and ensure that the required technology infrastructure and equipment functionality are maintained in eMINTS classrooms.

eMINTS program coordinators train principals and technology specialists to monitor and support the instructional improvements of eMINTS teachers. Specifically, during a one-day, face-to-face professional development session early in the school year, program coordinators introduce principals to constructivist theories underlying the eMINTS model and their implications for classroom teaching and learning. They introduce principals to an observation rubric for supervising and supporting teachers' instructional performance according to eMINTS standards.

---

[4] During the spring 2011 semester before the eMINTS study began, eMINTS staff visited all the study schools to assess their readiness for eMINTS implementation. When schools assigned to implement the eMINTS Comprehensive Program (the program schools) did not meet minimum infrastructural, hardware, or software requirements, center staff paid for the necessary equipment and infrastructure upgrades using i3 grant funds.

During the winter and spring of each school year, the program coordinator or other certified eMINTS consultant returns to conduct a school walk-through with the principal to support implementation and help the school integrate eMINTS with other school initiatives. These walkthroughs increase the principals' ability to view the school and instructional practices through an eMINTS lens and identify areas for improvement. During the two years, principals receive about 12 hours of professional development. They also receive a monthly newsletter with information about important elements of the eMINTS model and how they are applied in schools and classrooms. School technology coordinators are trained to understand eMINTS pedagogy and how technology is intended to support instruction. A certified eMINTS consultant conducts two optional two-hour online sessions with coordinators (four hours total) during each school year to introduce technology coordinators to eMINTS pedagogy and train them on equipment maintenance.

Teachers receive about 240 hours of professional development across two years in which they learn to design high-quality inquiry-based lesson plans, implement inquiry-based learning strategies, build community among teachers and students, and integrate technology into classroom instruction. Each eIS works with his or her assigned schools and teachers throughout the year to secure dates and times to conduct formal professional development sessions. During the first year of professional development, teachers receive 126 hours of formal professional development in 26 sessions that are held throughout the school year. Sessions typically take place in an eMINTS classroom or a computer lab in a central location and last between four and $6^1/_2$ hours each. The first-year curriculum focuses on basic technology applications, understanding constructivist pedagogy, community-building strategies (including interaction and interdependence), inquiry-based learning strategies, technology integration, and introducing authentic learning experiences into the classroom. At the end of the first year, with the help of an eIS, teachers spend up to 12 additional hours developing a classroom website. During the second year, eIS conduct 88 hours of professional development in 20 sessions throughout the school year. The Year 2 curriculum focuses on classroom management, website enhancement, assessment, interdisciplinary teaching and learning, and the development of multimedia and online projects. One session each year is reserved for teachers to travel to a school experienced implementing eMINTS and observe a certified eMINTS teacher. During both years, eIS supplement these formal professional development sessions with nine to 10 on-site and individualized coaching sessions (about 14 hours total) and within-building communities of practice where teaching staff meet to share ideas, collaborate on online project development, and deepen their existing understanding of concepts embedded in the eMINTS instructional model. In addition, in Year 2, eMINTS measured a teacher's knowledge and ability to develop a unit that demonstrates the eMINTS instructional model. The unit plan should comprise a website, a WebQuest, and the actual unit plan that conforms to eMINTS priorities. Finally, eMINTS provides teachers with support materials and just-in-time learning opportunities through online learning communities to help teachers improve their practice over time.

Appendix A provides a detailed professional development schedule for the eMINTS Comprehensive Program. Appendix B provides detailed definitions of the four components of the eMINTS instructional model.

The Intel Teach Program

eMINTS combined the 30 hours required to complete the Intel Teach Program online course titled Project-Based Approaches with 12 hours of face-to-face professional development. The face-to-face professional development was delivered in the following sequence and included the following topics:

- **Session 1** (4 hours; late summer or early fall 2013): preparation for taking the online Intel Teach Program course, expectations for unit (project) development, and introducing the Thinking With Technology (TWT) Visual Ranking Tool
- **Session 2** (4 hours; December 2013): developing the unit (project) plan that is the product of the Intel Teach Program course and introducing the TWT Seeing Reason Tool
- **Session 3** (4 hours; late January or early February 2014): refining and completing the unit (project) plan for submission to program evaluators and introducing the TWT Showing Evidence Tool

When possible, the eMINTS eIS who provided Years 1 and 2 professional development to their assigned schools facilitated the online course at the same schools. The online professional development was delivered in fall 2014 (September, October, and November) to the eMINTS+Intel group. The course examined the features and the benefits of project-based learning. Throughout the course, teachers considered their own teaching practices as they followed a teacher new to project-based learning who discussed strategies with a mentor teacher. Teachers also considered the ways that technology supports project-based approaches:

- Principles of project-based learning
- The steps of project design
- Integrating assessment throughout a project
- Developing a project timeline and an implementation plan
- Guiding learning through questioning, collaboration, self-direction, and reflection

In addition to the face-to-face and online professional development sessions, teachers received four or five in-classroom coaching and mentoring visits, which were approximately one hour in length, from the eMINTS eIS assigned to their cohort group. The coaching and mentoring sessions occurred in fall 2013 and winter 2014. They were also asked to complete a unit plan (one of three components of the portfolio) to demonstrate their continued understanding of the eMINTS instructional model.

The eMINTS Comprehensive Program Plus the Intel Teach Model of Change

As mentioned earlier, eMINTS and the Intel Corporation recently partnered to establish a third year of professional development for eMINTS teachers to augment the current eMINTS Comprehensive Program. This combined program, referred to as eMINTS+Intel, provides a third year of professional development using the Intel Teach courses and online tools to further enhance teachers' technology integration skills and provide a specific set of online resources for teachers to support the four eMINTS components.

Figure 1.1 describes the model of change underlying the eMINTS Comprehensive Program and the eMINTS+Intel program. In the eMINTS Comprehensive Program, the technology resources, combined with intensive teacher professional development and schoolwide support, are intended to transform teachers' classroom lesson planning and instruction. Through intensive professional development and one-on-one support, teachers learn how to develop high-quality lesson plans and integrate technology in their classroom instruction to promote inquiry-based lessons and collaborative classroom communities. These instructional changes should, in turn, influence students' engagement in school and improve students' 21st century skills (i.e., critical thinking, communication, and technology literacy) and achievement on annual state-administered assessments. The third year of Intel Teach professional development was expected to accelerate outcomes for teachers and students by providing teachers with additional professional development and Web-based tools to build on what they learned during the first two years of the program.

Figure 1.1. The eMINTS Comprehensive Program Plus the Intel Teach Program Model of Change

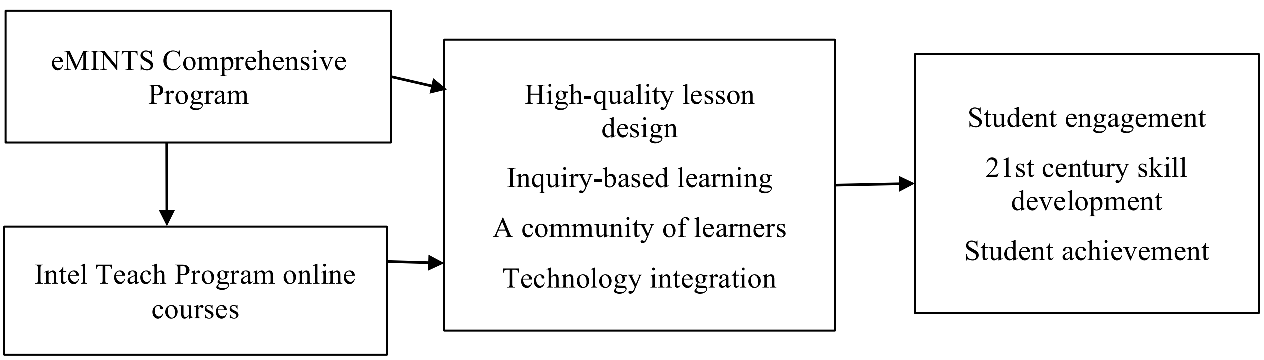


Figure 1.2 is a more detailed logic model, diagramming resources, processes, teacher outputs, and expected student outcomes that should be realized as a result of implementing the eMINTS Comprehensive Program and the eMINTS+Intel program. Building-level inputs include improvements to a school's technology equipment and infrastructure, such as computers, interactive whiteboards, digital cameras, scanners, LCD projectors, advanced software programs for teachers and students, and technology support and bandwidth.

Figure 1.2. Logic Model for eMINTS+Intel Program

RESOURCES

Technology equipment

- 1-to-1 student:laptop computer ratio
- Interactive whiteboard
- Teacher laptop
- Digital camera
- Scanner and printer

Technology infrastructure

- Data storage
- Backup and disaster recovery
- Internet connectivity
- Internet security
- Wireless connection
- Shared folder system or server

ACTIVITIES

eMINTS Comprehensive Program (Years 1–2)

Teacher professional development

- Year 1: 26 PLC sessions, 126 hours
- Year 2: 20 PLC sessions, 88 hours
- 10 classroom visits per year
- Online access to curriculum and course materials

Administrative support

- School administrator professional development
- School administrator support for eMINTS

Ongoing technology support

- Technology coordinator professional development
- Ongoing instructional technology support

Intel Teach (Year 3)
Teacher professional development

- 42 hours (face to face and online)
- Coaching (4–5 visits)
- Online resources

SHORT-TERM OUTCOMES

Increase in teacher effectiveness

- High-quality lesson design
- Inquiry-based learning strategies
- A community of learners
- Technology integration

Increase in student engagement

LONG-TERM OUTCOMES

Improvement in 21st century skills

- Critical thinking
- Communication
- Technology literacy

Increases in performance on the state achievement test

The eMINTS theory holds that school conditions should improve teachers' classroom practices when the following occur:

- Schools are provided with up-to-date technology and a high-performing technological infrastructure.
- Principals are trained to understand constructivist pedagogy and its potential for accelerating student learning.
- Principals support and monitor long-term eMINTS classroom improvements.
- Technology specialists are available in the school to support technology use and conduct regular equipment maintenance.
- School decision making occurs regularly through PLCs involving teachers, school leadership, and outside consultants (e.g., eIS).

With adequate conditions in place at the school level, classroom practices should improve when teachers can do the following:

- Understand the benefits of constructivist pedagogy for accelerating student learning.
- Learn to design standards-based lessons, implement inquiry-based learning strategies, build collaborative classroom communities, and infuse technology into their instruction.
- Receive individualized support from a specialist to reinforce what they learn in formal professional development sessions.
- Participate in a PLC to share ideas and receive feedback from their colleagues.
- Receive support and accountability from their school leadership (through walk-throughs and ongoing feedback).

With intensive and differentiated professional development grounded in constructivist pedagogy, an adequate technology infrastructure to engage teachers and students, support to understand and apply new strategies in their classrooms, and accountability from colleagues and school leadership, teachers should gradually improve the extent to which their lesson plans are aligned to standards and grounded in assessment, incorporate more inquiry-based learning strategies in their instruction, strengthen relationships with students, improve their own technology literacy, and facilitate a vibrant classroom community of learners.

As schools and teachers gradually develop and demonstrate these characteristics, students should demonstrate improvements in the following:

- Academic orientation in school
- 21st century skill development, such as communication, technology literacy, and critical thinking
- Achievement as measured by statewide achievement tests

Integration of the eMINTS Comprehensive and Intel Teach Programs

As mentioned previously, the theory supporting the eMINTS+Intel model holds that a third year of professional development should accelerate the changes observed in teacher practices and student achievement beyond the traditional two-year eMINTS model. This occurs primarily through professional development to reinforce concepts in standards-based lesson design, inquiry-based learning strategies and assessment, and technology integration.

Prior Research on the eMINTS Program

eMINTS is one of only a few professional development programs with data to support the chain of evidence from delivery of a specific program to changing teacher practices and positive impact on student achievement (Martin, Strother, Beglau, Bates, & Reitzes, 2010). Since its inception in 1999, annual external evaluations of the eMINTS program have been conducted to determine the effects of eMINTS professional development on teacher and student outcomes. Qualitative research and formative evaluations also contributed to a better understanding of the facilitating factors and challenges associated with school and classroom implementation of eMINTS. In the following subsections, we summarize 10 years of eMINTS professional development research and evaluation and assess the quality of evidence reported on the program's effectiveness to date. An eMINTS classroom refers to classes taught by certified eMINTS teachers and fulfilling eMINTS technology requirements.

Teacher Outcomes

eMINTS professional development is designed to help teachers learn how to integrate technology into their teaching, use instructional strategies that promote standards- and inquiry-based learning, and encourage collaboration and community building among students and teachers. One of the earliest reports on eMINTS from OSEDA (2001a) presented the results of surveys taken by the first cohort of eMINTS teachers and administered at three time points in two years. In these early self-reports, teachers reported improvements in their inquiry-based teaching activities, computer usage, and perception of computing skills. A second report that focused on teacher change in lesson typology through multiple observations found that after one year of eMINTS implementation, participating teachers transitioned from teacher-centered models to hybrid or student-centered models (OSEDA, 2001b). Furthermore, early evaluations (OSEDA, 2003b) demonstrated a positive relationship between eMINTS professional development on inquiry-based learning strategies and teachers' enactment of those components in their practices.

Using an observation scale to measure the classroom climate of eMINTS teachers, early evaluations also found evidence suggesting that eMINTS teachers who facilitated student-centered instruction were significantly more likely to construct a well-ordered and effective learning environment than those eMINTS teachers who were less focused on facilitating student-centered instruction (OSEDA, 2003c). Subsequent research demonstrated that eMINTS teachers' instruction became increasingly student centered, and their classrooms became increasingly linked to effective behavior management strategies (Tharp, 2004). Similar findings were observed among principals participating in the eMINTS program (Tharp, 2006), in that principals of eMINTS schools more frequently engaged students and increasingly monitored student achievement and progress.

More recent eMINTS program evaluations have focused on program fidelity and its impact on teachers' mastery (Martin et al., 2008, 2009). The Education Development Center's (EDC's) Center for Children and Technology's 2008 external evaluation (Martin et al., 2008) found high levels of fidelity in terms of program delivery, and teachers demonstrated high levels of mastery on classroom technology integration and inquiry-based learning strategies. These high levels were found in programs implemented primarily under the direction of eMINTS staff and those implemented under the direction of district-level eIS who had satisfactorily completed or were actively participating in the eMINTS train-the-trainer program (Professional Development for Educational Technology Specialists).

The evaluation also found significant positive correlations between program fidelity and teacher mastery scores on eMINTS lesson-planning procedures and the WebQuests[5] teachers submitted as part of program participation requirements (Martin et al., 2008). Specifically, the following factors of program fidelity were correlated with lesson planning at the .01 level of significance: scaffolding instruction (.263),[6] active work and learning (.296), modeling instruction (.388), technology use (.268), connection to practice (.217), and inquiry-based learning (.205). An EDC evaluation substantiated these findings, adding that "evidence [suggests] that the more closely aligned the local implementation of eMINTS is to core program goals, the greater the impact the program has on teachers' understanding of the material and on students' performance on standardized assessments" (Martin et al., 2009, p. ii). For example, in communication arts and mathematics for Grades 4 and 5, correlations between fidelity and student achievement were significant at the .05 level in both 2007 and 2008. Of the various components of fidelity, technology use and inquiry-based learning became more strongly correlated with student test scores in both communication arts and mathematics as grade levels increased (Martin et al., 2010).

Student Outcomes

eMINTS external program evaluations conducted from 2002 to 2005 used a quasi-experimental design that compared the performance of students in eMINTS classrooms with the performance of students in non-eMINTS classrooms. These evaluations consistently found that intermediate elementary (Grades 3–6) students enrolled in eMINTS classrooms significantly outperformed students enrolled in non-eMINTS classrooms on Missouri's state standardized performance assessment (the Missouri Assessment Program [MAP]) in communication arts, mathematics, science, and social studies. These results primarily pertained to students in Grade 3 communication arts and science and Grade 4 mathematics and social studies, with small sample sizes suggesting similar results may exist at Grades 5 and 6 (Hager, 2004; Hunter & Greever-Rice, 2007; OSEDA, 2002, 2003a, 2004, 2005; Tharp, Bickford, & Hager, 2005). OSEDA analyses were conducted using student achievement data from MAP to compare the percentage of students attaining proficient and advanced levels of achievement in eMINTS classrooms with the percentage of students reaching those levels in non-eMINTS classrooms. A larger percentage

[5] A WebQuest is an inquiry-oriented lesson format in which most of the information used is derived from the Internet.
[6] The numbers in parentheses represent Pearson correlation coefficients.

of eMINTS students attained proficiency or advanced levels of achievement than did non-eMINTS students in communication arts from 2002 to 2005, the difference being statistically significant at the .05 level from 2003 to 2005. Mathematics results are similar, with the only exception being 2004, when non-eMINTS students had a slightly (0.4 percent) higher rate of proficiency. The other three years of mathematics assessment data indicate statistically significant differences in favor of eMINTS students.

More recent evaluations conducted by EDC from 2006 to 2009 substantiated OSEDA's earlier findings. EDC's evaluations focused on schools that received competitive Title II.D Enhancing Education Through Technology grant awards in Missouri. The first study consisted of a sample of about 7,000 students, of whom approximately one third were in treatment classes, spread across 340 classes in 31 districts. Later reports more evenly distributed the number of students in eMINTS classrooms and non-eMINTS classrooms (approximately 6,000 students total per year) across 35 to 40 schools and fewer districts (about 10). These reports of a more established eMINTS program extended to Grades 5 and 6, where students in eMINTS classrooms consistently attained higher rates of proficiency or advanced levels in all grades (3–6) in communication arts and mathematics, with significant results at the .01 level in most comparisons, including Grades 5 and 6 (Strother, Martin, & Dechaume, 2006).

Turning to mean achievement differences on the MAP state tests, early reports (results from 2002 and 2003) indicated that students in eMINTS classrooms consistently outscored their peers in non-eMINTS classrooms as well as all other Missouri students. MAP communication arts findings revealed that eMINTS students had higher mean scores across years, with significant differences growing larger each year (from less than 1 point to more than 10 points) and producing greater effect sizes (.013 to .173). On the MAP mathematics test, the mean score differential (seven to 10 points) and effect sizes (approximately .25) remained stable and significant throughout the reports. In the first two reports (OSEDA, 2002, 2003a), eMINTS students scored higher in science but not significantly so. The results in social studies are significant, however, with effect sizes between .16 and .18.

For all subjects, the magnitude of the gap between eMINTS and non-eMINTS students by group—those with an individualized education program (IEP), in a Title I school, who qualified for the free or reduced-price lunch program, or minority—was statistically significant after their teachers had received one year of professional development. Effect sizes were consistently larger for some subgroups, especially students qualifying for free- or reduced-price lunch. For example, the OSEDA (2004) analysis of 2003 MAP data reported the following effect sizes: communication arts (.21), mathematics (.19), science (.11), and social studies (.20). Even larger effect sizes were found when student achievement in schoolwide Title I schools was analyzed: .29, .32, .16, and .25, respectively. These findings were consistent across OSEDA reports. In addition, students with IEPs and limited English proficiency (LEP) in eMINTS schools outscored their non-eMINTS peers by approximately one standard deviation in each subject, and the differences in means were statistically significant at the .001 level (Martin et al., 2008; Strother et al., 2006). "The fact that the effects were most dramatic among the highest need students suggests that the kind of environments eMINTS teachers create in their classrooms may be particularly effective for these students" (Strother et al., 2006, p. 7).

The original eMINTS intervention was a two-year program. Analyses indicated that students of second-year eMINTS teachers significantly outscored non-eMINTS students and students with first-year eMINTS teachers (Martin et al., 2009; OSEDA, 2003c). The results of perhaps the strongest evaluation of eMINTS yet conducted appear to confirm this. Martin et al.'s (2009) longitudinal analysis of student performance for two years (fall 2007 to spring 2009), using a matched-schools design, found that students assigned to eMINTS classrooms during both years significantly outperformed students assigned to non-eMINTS classrooms for both years at Grade 5 in communication arts ($p < .05$) and Grade 6 in communication arts ($p < .05$) and mathematics ($p < .001$). In addition, scores of students having two years with eMINTS teachers were significantly greater than those of students having an eMINTS teacher for only one year in Grade 6 communication arts ($p < .01$) and Grade 6 mathematics ($p < .001$). Moreover, the variance explained by having two eMINTS teachers was sizable, especially for mathematics (23.8 percent).

Summary of Prior Research

A decade of evaluation on eMINTS consistently has shown promise in changing elementary teachers' practice and raising student achievement. In particular, these results were found to exist in communication arts, mathematics, and social studies among intermediate elementary students representing a range of demographics in more than 40 school districts across Missouri.

Collectively, the studies of eMINTS on teacher and student outcomes suggest that the program makes a difference. Teachers appear to change practices, and students seem to achieve at higher levels. But none of the studies uses a rigorous quasi-experimental design or randomized controlled trial. In addition, there is an overall lack a focus beyond elementary schools; that is, the studies include elementary students, ranging from Grades 3 to 6, and the participating schools represent multiple states and populations (e.g., rural and urban settings). The current evaluation elevates the level of rigor through a randomized controlled trial and expands the reach of research results to middle school students (Grades 7 and 8) in high-poverty rural settings across Missouri.

Research Questions

Unlike previous evaluations of eMINTS professional development, this study focuses on the evaluation of rural middle schools only and uses an experimental design to assess the effectiveness of the eMINTS Comprehensive Program on teacher practices, student engagement, and student achievement. In this report, we assess the impact of the eMINTS Comprehensive Program one year after the full implementation of the program against control schools that are conducting business as usual. We also assess the added impact of teacher professional development supported by the Intel professional development curriculum and instructional resources on teacher practices and student achievement against two groups of schools: those implementing the traditional two-year eMINTS Comprehensive Program and control schools. We investigate the following confirmatory research questions:[7]

1. What is the impact of the eMINTS Comprehensive Program and the eMINTS Comprehensive Program plus Intel Teach on school mean 7th and 8th grade performance in mathematics and communication arts?
2. What is the impact of the eMINTS Comprehensive Program and the eMINTS Comprehensive Program plus Intel Teach on school mean 7th and 8th grade measure of 21st century skills (including communication, technology literacy, and critical thinking)?
3. What is the impact of the eMINTS Comprehensive Program and the eMINTS Comprehensive Program plus Intel Teach on school mean 7th and 8th grade measure of student engagement?

This Year 3 report as also examined the following exploratory questions on impact variation and impacts on teacher practices:

The study also examined the following exploratory questions on impact variation and impacts on teacher practices:

4. Do the impacts of the eMINTS Comprehensive Program and the eMINTS Comprehensive Program plus Intel Teach on school mean performance in mathematics and communication arts, 21st century skills, and engagement vary across subgroups of students?

5. What are the impacts of eMINTS Comprehensive Program and the eMINTS Comprehensive Program plus Intel Teach on seventh- and eighth-grade teachers' instructional practices?

At baseline (before the 2011–12 school year), 60 schools, 191 teachers, and 3,610 seventh- and eighth-grade students were randomly assigned to one of three groups: the traditional eMINTS Comprehensive Program, the eMINTS Comprehensive Program plus a third year of professional development using the Intel Teach Program courses, or business as usual. By design, the eMINTS Comprehensive Program is a two-year professional development program, and eMINTS+Intel adds a third year to the original program length. The MAP state tests are administered annually in the spring. Similarly, the research team administers the 21st Century Skills Assessment (developed by Learning.com) in the spring of each study year.

Roadmap to This Report

Chapter 2 details the study's research design, sample recruitment and characteristics, data collection and outcome measures, data analytic methods, and the limitations of the study design. Chapter 3 discusses the eMINTS intervention as implemented for this study and provides fidelity of implementation findings, including the extent of program delivery by eMINTS staff, the

[7] Confirmatory research questions are the main questions this study is designed to address. This study was designed using an experimental design to provide rigorous estimates of the causal effects of the eMINTS and eMINTS+Intel programs.

extent to which principals and other school staff supported the eMINTS program, and the extent to which teachers participated in eMINTS professional development and consultation services. Similar fidelity of implementation findings are reported for the year of Intel Teach. Chapter 4 addresses the study's main research on impacts on communication arts and mathematics achievement for the study sites. In addition, Chapter 4 presents the impacts of eMINTS and eMINTS+Intel professional development on students' 21st century skills and classroom engagement. Finally, Chapter 5 addresses exploratory impacts of the eMINTS Comprehensive Program and eMINTS Comprehensive Program plus Intel Teach on teacher practices.

Chapter 2: Study Design and Methodology

Study Design

This study employs a three-year cluster-randomized design, assigning 20 schools each to one of three groups (eMINTS only, eMINTS+Intel, or business as usual; 60 schools total) to obtain unbiased estimates of the impact of the eMINTS and eMINTS+Intel programs on middle school students' communication arts and mathematics achievement. A cluster-randomized design randomly assigns clusters of units (i.e., schools) to either a treatment or a control condition. Randomization ensures that the treatment and control groups are, in expectation, equivalent on baseline characteristics and therefore yields unbiased estimates of the causal effects of the intervention on schools. This chapter describes the research design; district and school recruitment; randomization of schools to the treatment or control conditions; the analysis sample; and the baseline characteristics of participating schools, teachers, and students. It also discusses attrition, data collection and measures, methods used for impact estimation, and study limitations.

Participant Eligibility and Recruitment

The participating study schools include high-poverty rural Missouri middle schools with students in Grade 7 and 8 students. To qualify for the study, schools had to meet requirements under Title I (schoolwide or targeted) or Missouri's historical requirements for Title II.D (50 percent of students in poverty) and be part of the Small Rural School Achievement program or the Rural and Low-Income School program authorized under Title VI.B of the Elementary and Secondary Education Act. Grades 7 and 8 were targeted for this study for two reasons: First, as indicated previously, more empirical research is needed to substantiate the effects of eMINTS on teachers and students at the middle school level. Second, high dropout rates are persistent in poor and rural districts (Mac Iver & Mac Iver, 2009). Grades 7 and 8 represent a critical transition period for students who may be at risk of failing or dropping out of high school in later years (Herlihy, 2007). This study represents the first step toward examining the extent to which the eMINTS program improves middle school students' engagement and achievement levels and puts them on track toward high school graduation.

The eMINTS National Center has offered professional development for more than a decade. During that period, eMINTS steadily increased the number of participating schools and teachers across Missouri. Because the demand for eMINTS among high-poverty rural Missouri districts far exceeds the number that have traditionally been funded by federal or state funds, the incentives to participate in this study for many rural districts were self-evident, even if that meant being assigned to the control condition. For example, during the last 10 years, only 36 percent of the eligible Title II.D districts that applied for eMINTS were awarded a grant. Moreover, no additional grants are expected after 2010 because of federal cuts in Title II.D. In addition, 20 percent of the eligible districts submitted applications more than twice, with some districts applying in seven separate years without receiving an award, thus indicating persistent strong interest in eMINTS despite repeated failures to secure an award. Within only two weeks of recruitment, 68 districts or schools provided letters of interest to participate in the study, indicating the high demand for eMINTS, and, consequently, schools' willingness to wait three years if they are guaranteed an opportunity to implement eMINTS. Schools that participated in the study were guaranteed to receive the program either immediately or after a three-year delay. Control group teachers and data points of contact within the control schools received a small monetary incentive of $250 for their participation in data collection activities (teacher survey and classroom observation).

The study team annually collected signed student assent forms and signed consent[8] forms from teachers and parents of Grades 6, 7, and 8 students. eMINTS conducted information meetings with parents and students in the fall of the 2011–12 and 2012–13 school years. As part of these meetings, eMINTS provided disclosure information about the study as required by the University of Missouri's institutional review board and consent and assent forms for both parents and students to complete. The same process is used in both treatment and control schools. Student assent and parent consent is primarily needed for student academic orientation surveys and the 21st Century Skills Assessment because these assessments are being administered to students as part of the study and are not considered a regular part of students' instruction or schooling.

Random Assignment

After receiving signed commitments from the 60 schools, we randomly assigned schools to one of three groups. As shown in Table 2.1, schools assigned to Group 1 were offered the traditional two-year eMINTS Comprehensive Program beginning in fall 2011[9]; Group 2 schools were offered the traditional two-year eMINTS Comprehensive Program plus a third year of Intel Teach professional development (eMINTS+Intel) beginning in fall 2013; and Group 3 was asked to conduct business as usual with no exposure to eMINTS or eMINTS+Intel (control) until fall 2014.

Table 2.1. School Assignment

Summer 2010	Fall 2010		2011–12	2012–13	2013–14
Recruit 60 schools for participation	Random assignment of schools	eMINTS ($N = 20$)	eMINTS	eMINTS	No additional professional development
		eMINTS+Intel ($N = 20$)	eMINTS	eMINTS	Intel Teach
		Control ($N = 20$)	Business as usual	Business as usual	Business as usual

To determine school assignments, we used one blocking variable, placing schools into the following three groups according to their grade configuration:

[8] For students, signed parent consent forms with an affirmative notation allow the students to be in the study. Signed student assent forms with an affirmative notation indicate students' willingness to be in the study.
[9] All schools in the study will receive the full three years of professional development if they continue to want it. The eMINTS-only group will receive the third year of professional development after the third year of data collection is complete, beginning in fall 2014. Similarly, the business as usual schools began receiving the three-year eMINTS+Intel program in fall 2014.

- Block 1 (30 schools): PK–8 schools (coded as 1)
- Block 2 (8 schools): 5–8, 6–8, and 7–8 schools (coded as 2, 3, and 4, respectively)
- Block 3 (22 schools): 6–12 and 7–12 schools (coded as 5 and 6, respectively)

Next, a random number was generated for each school within each block. Within each block, we assigned schools with random number values in the lowest third to Group 1 (eMINTS only), the middle third to Group 2 (eMINTS+Intel), and the highest third to control.

Blocks 2 and 3 were not divisible by three; therefore, they could not be equally distributed into the three groups. To ensure equal sample sizes across study conditions, we assigned one less school to the business as usual condition in block 2 and one additional school to the business as usual condition in Block 3 before generating random numbers for sample schools.

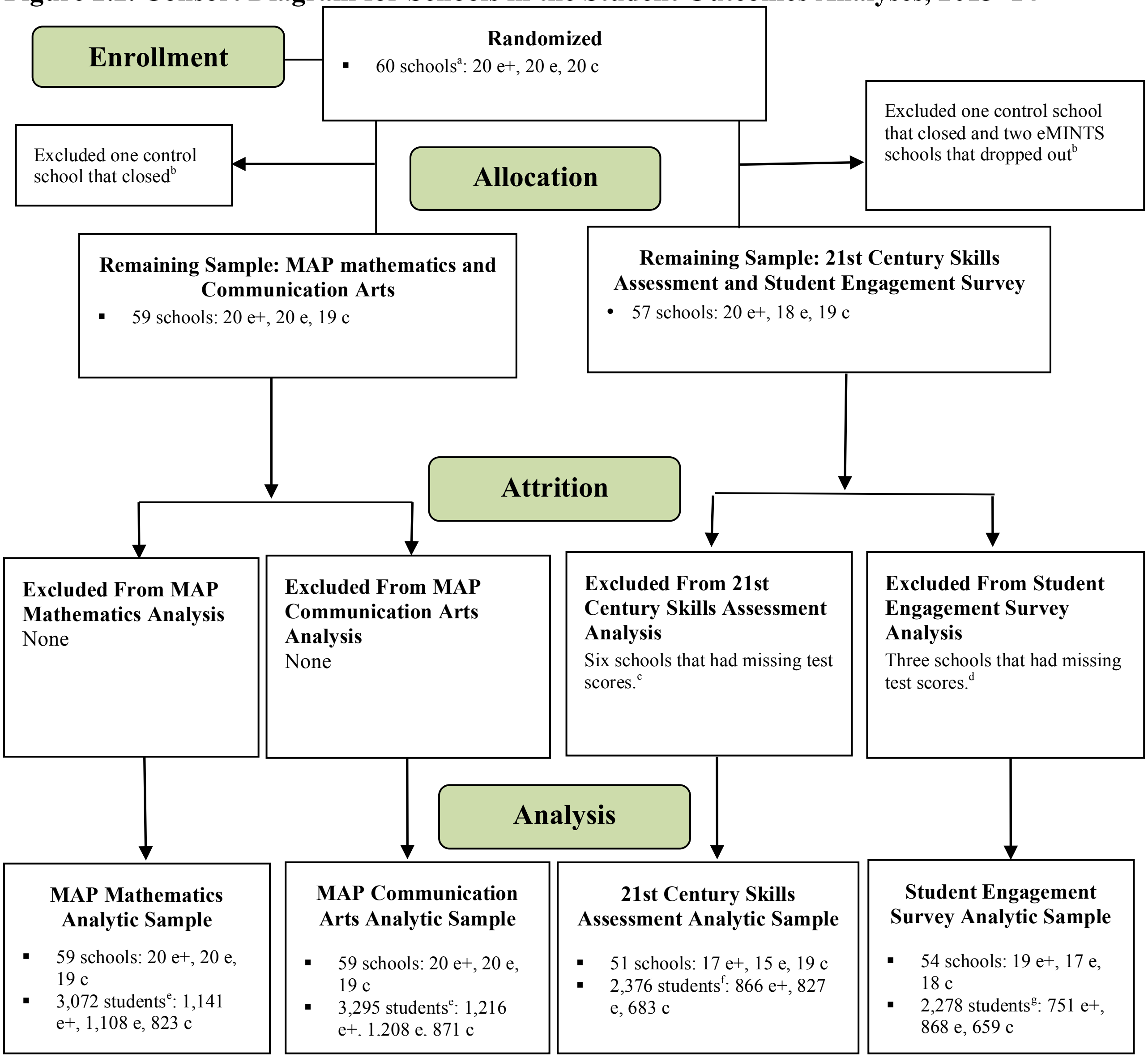
Analytic Sample

Figure 2.2 describes the construction of the Year 3 (2013–14) analytic sample for MAP and the 21st Century Skills Assessment. Of the 60 schools randomly assigned to groups, two withdrew from the study before Year 1 implementation began. Students from these two schools are not included in sample attrition on the MAP mathematics and communication arts outcomes because students in the schools that left the study took MAP in spring 2014. Because these schools were included in random assignment, they are included in the intent-to-treat analysis of the Year 3 sample presented in chapter 4. Students from these two schools did not take the 21st Century Skills Assessment or the student engagement survey and, thus, are included in sample attrition results on this outcome. An additional school closed after Year 1 (the 2011–12 school year). Teachers and students from this school are not included in any Year 3 analyses.

As Figure 2.2 illustrates, 60 schools were randomized to one of three treatment groups. For the purposes of our Year 3 analysis, schools in the eMINTS and eMINTS+Intel groups were analyzed separately because only eMINTS+Intel schools received the additional year of professional development. A total of 59 schools (20 eMINTS+Intel, 20 eMINTS, and 19 control) and 3,072 students (1,141 eMINTS+Intel, 1,108 eMINTS, and 823 control) were included in the final Year 3 MAP mathematics analytic sample; 59 schools (20 eMINTS+Intel, 20 eMINTS, and 19 control) and 3,295 students (1,216 eMINTS+Intel, 1,208 eMINTS, and 871 control) were included in the final Year 3 communication arts analytic sample; 51 schools (17 eMINTS+Intel, 15 eMINTS, and 19 control) and 2,376 students (866 eMINTS+Intel, 827 eMINTS and 683 control) were included in the 21st Century Skills Assessment analytic sample; and, 54 schools (19 eMINTS+Intel, 17 eMINTS, and 18 control) and 2,278 (751 eMINTS+Intel, 868 eMINTS, and 659 control) students were included in the student engagement survey analysis.[10] These analytic samples also were used to examine variations in impact of the two programs across student subgroups.

[10] The 21st Century Skills Assessment and student engagement sample sizes are lower than the MAP student samples because parent consent and student assent with affirmative notation were required for students to complete the 21st Century Skills Assessment and the student engagement survey. The study team needed to collect parent consent and student assent for these measures because neither is part of the school's regular data collection activities.

Figure 2.2. Consort Diagram for Schools in the Student Outcomes Analyses, 2013–14



Note. Prepared from authors' analysis based on data from DESE and the study districts.
[a]Sixty schools were randomized to one of three groups at the beginning of the study: eMINTS, eMINTS + Intel, and control.
[b]One control school closed before the 2012–13 school year, and two schools dropped out - one about four months after random assignment was announced and shortly after baseline data collection began, and the other during the summer before Year 1 implementation. [c]Students who did not provide assent were excluded from the 21st Century Skills Assessment. Nine schools had missing 21st Century Skills Assessment scores: one of these schools closed, two withdrew after randomization, and six schools were unable to administer the test, or their students did not take the test or did not provide student assent or parent consent.
[d]Students who did not provide assent were excluded from the student engagement survey analyses. Six schools had missing student survey data: one of these schools closed, two withdrew after randomization, and three schools did not have any student survey data. [e]Students who took the MAP Alternate Test in mathematics and communication arts were considered ineligible for the study. Those who did not provide assent were included in the MAP mathematics or communication arts analyses provided they had nonmissing MAP mathematics or communication arts scores. More than 3,000 students (3,067 students) are common to the MAP mathematics and communication arts analytic samples. Five students are in the mathematics sample but not in the communication arts sample, and 228 students are in the communication arts sample but not in the mathematics sample. [f]Of the 2,376 students in the 21st Century Skills Assessment analytic sample, 2,180 students are in the mathematics analytic sample, and 2,369 are in the communication arts analytic sample. [g]Of the 2,278 students in the student engagement survey analytic sample, 2,099 students are in the mathematics analytic sample, and 2,273 students are in the communication arts analytic sample.

A similar figure is included in Appendix C to show the construction of the Year 3 analytic sample for teacher outcomes. The analytic sample on teacher survey results included 50 schools (19 eMINTS+Intel, 14 eMINTS, and 17 control) and 97 teachers (35 eMINTS+Intel, 24 eMINTS, and 38 control); the analytic sample on classroom observation results included 54 schools 19 eMINTS+Intel, 17 eMINTS, and 18 control) and 117 teachers (43 eMINTS+Intel, 33 eMINTS, and 41 control).

Baseline Equivalence

Random assignment, particularly when the sample sizes in each group are large,[11] is expected to result in statistically equivalent groups on all observable and unobservable variables. Group equivalence strengthens our capability to attribute any observed differences in outcomes between the groups in the intervention. Although it is not possible to test equivalence for unobservable variables, baseline equivalence of the groups can be assessed for variables on which data are available. In this section, we examine the initial equivalence of eMINTS, eMINTS+Intel, and control group teachers and students on all the observable variables collected.

For the Year 3 student analyses, which are the sole focus of this report, differences in baseline student characteristics are estimated using a two-level model on the pooled sample of seventh- and eighth-grade students, adjusting for a student's grade in Year 3. Differences were estimated separately by randomization block and then pooled into an overall weighted average, using the number of study schools in each block as the weight. (See Appendix D for the model used.)

Comparisons of baseline teacher characteristics for the Year 3 analyses were conducted similarly, except that single-level (OLS) regression models were used in place of two-level models. Similarly, baseline school characteristics were compared using the OLS models, but without the grade indicator ($Grade_{ij}$) as a predictor.

The chi-square test of independence was used to test for group equivalence in the categorical variables, and the two-sided *t* test for equality of means was used for continuous variables.

Baseline School Characteristics

Of the 60 schools, 30 schools serve Grades PK–8 or K–8, eight schools serve Grades 6–8 or 7–8, and the remaining 22 schools serve Grades 6–12 or 7–12. Table 2.2 compares the baseline demographic characteristics and performance of schools in the sample. Across all school demographic characteristics examined, the results showed that eMINTS, eMINTS+Intel, and control schools were relatively similar. School enrollment in all groups was fewer than 200 students, reflecting the small and rural aspects of the sample schools. eMINTS and eMINTS+Intel schools, however, enrolled about 50 more students than control schools did. In our Year 2 report, the difference in enrollment between the treatment groups combined and the control group was

[11] As Bloom (2006) underscores, the randomization process yields, on average, intervention and control groups that are equivalent on all observable and nonobservable characteristics. Randomization applied to a specific sample, however, does not guarantee group equivalence because it is possible to obtain groups that differ simply by chance.

statistically significant (Brandt, Meyers, Molefe, Dhillon, & Zhu, 2013).[12] Across all groups, about 5 percent of students were minorities (although there is a sizeable difference between eMINTS and eMINTS+Intel), about 60 percent qualified for free or reduced-priced lunch, between 1 and 2.5 percent had limited English proficiency, and between 12 percent and 14 percent had individualized education plans. Teachers in eMINTS schools averaged almost one year more of experience than their eMINTS+Intel counterparts who in turn averaged almost one year more than their control counterparts. About 43 percent of teachers in all schools had a master's degree. Average state test results among seventh- and eighth-grade students on the MAP mathematics and communication arts assessments were nonsignificant. In the Year 2 report where eMINTS and eMINTS+Intel were considered one collective treatment group, however, eighth-grade control students scored significantly higher on the communication arts assessment (Brandt, Meyers, Molefe, Dhillon, & Zhu, 2013).

Table 2.2. Characteristics of Study Schools: 2010–11 (Before Year 1 Implementation)

Characteristics[a]	**eMINTS+ Intel**	**eMINTS**	**Control**	***p*-value**
Number of schools	20	20	20	
Average total school enrollment	183.2	180.2	128.8	0.116
Average percentage of students eligible for free or reduced-price lunch	59.7	60.8	59.6	0.938
Average percentage of non-White students	7.3	3.7	5.8	0.227
Average percentage of male students	51.5	49.6	49.9	0.262
Average percentage of LEP students	2.5	1.2	1.2	0.648
Average percentage of students with IEP	13.7	12.8	12.3	0.547
Average years of teaching experience	11.7	12.5	10.9	0.216
Average percentage of teachers with master's degree	43.1	42.9	42.5	0.992
Mean scale score in spring 2011 MAP communication arts[b]				
Grade 7	675.2 (SD[b] = 11.1; *n*[c] = 19)	684.0 (SD = 15.1; *n* = 18)	678.1 (SD = 10.3; *n* = 20)	0.064

[12] In Year 2, both treatment groups were at the same stage – the second year of eMINTS. Thus, baseline comparisons in that report compare the two groups of treatment schools collectively against the control schools. We note differences here without reporting on all characteristics again.

Grade 8	691.7 (SD = 10.6; *n* = 19)	692.9 (SD = 15.4; *n* = 17)	699.4 (SD = 9.4; *n* = 19)	0.083
Mean scale score in spring 2011 MAP mathematics[b]				
Grade 7	685.5 (SD = 13.8; *n* = 19)	691.2 (SD = 16.5; *n* = 18)	682.3 (SD = 13.8; *n* = 20)	0.132
Grade 8	706.1 (SD = 10.0; *n* = 19)	704.4 (SD = 16.0; *n* = 17)	708.3 (SD = 12.6; *n* = 19)	0.633

Note. From authors' analysis based on data from the National Center for Education Statistics Common Core of Data 2010–11 and DESE.
[a]Means were regression adjusted to account for block effects using ordinary least squares regression. For each characteristic, the *p*-value is from an overall chi-square test of whether the means of the three treatment groups (eMINTS+Intel, eMINTS, and control) differed more than what might be expected by chance. [b]SD = standard deviation. [c]When data are missing, *n* is the actual number of teachers used to calculate the average characteristic in each treatment group. Missing data on the spring 2011 school mean MAP scores in communication arts and mathematics result from the fact that DESE does not report grade-level school means when fewer than five students in a grade took the test. As the table shows, fewer than five seventh-grade students took the test in two eMINTS schools and one control school, and fewer than five eighth-grade students took the test in one eMINTS+Intel schools, three eMINTS schools and one control school.

Baseline Teacher Characteristics

The demographic characteristics of participating Grades 7 and 8 teachers are summarized by subject area (communication arts and mathematics, respectively) in Tables 2.3 and 2.4. None of group differences between teacher characteristics was significant.[13] In mathematics, the percentage of male teachers in the eMINTS+Intel group was highest at 29.4 percent versus eMINTS at 11.6 percent and the control group at 24.8 percent. In addition, the percentage of eMINTS+Intel mathematics teachers with a master's degree was highest at 40.9 percent versus 23.8 percent for eMINTS teachers and 27.5 percent for control teachers. eMINTS+Intel teachers had the least experience, however, averaging 7.7 years, while eMINTS teachers averaged 9.3 years and control teachers averaged 9.1 years, respectively. Teachers in the eMINTS group averaged 9.4 years of experience, whereas control teachers averaged 8.3 years of experience.

[13] It should be noted here that teacher characteristics were not included in 21st Century Skills Assessment or student engagement analyses because some students might have multiple eMINTS teachers, so baseline comparisons between teachers for those analyses are not necessary.

Table 2.3. Characteristics of Teachers in the MAP Mathematics Analytic Sample: 2010–11 (Before Year 1 Implementation)

Characteristics	**eMINTS+ Intel**	**eMINTS**	**Control**	***p*-value**
Number of teachers	37	34	29	
Percentage male	29.4 (*n* = 37)	11.6 (*n* = 34)	24.8 (*n* = 29)	0.194
Percentage with graduate degree	40.9 (*n* = 27)	23.8 (*n* = 29)	27.5 (*n* = 22)	0.368
Years of teaching experience	7.7 (*n* = 27)	9.3 (*n* = 29)	9.1 (*n* = 22)	0.701

[a]Means were regression adjusted to account for block effects using ordinary least squares regression. For each characteristic, the *p*-value is from an overall chi-square test of whether the means of the three treatment groups (eMINTS+Intel, eMINTS, and control) differed more than what might be expected by chance. When data are missing, *n* is the actual number of teachers used to calculate the average characteristic in each treatment group.

Among communication arts teachers, the demographic characteristics of the eMINTS+Intel and control groups were similar in terms of the percentage of male and female teachers and percentage of teachers with graduate degrees. The average years of experience between eMINTS and control teachers were also similar. Although there are some notable differences between groups, none of the differences was statistically significant.

Table 2.4. Characteristics of Teachers in the MAP Communication Arts Analytic Sample: 2010–11 (Before Year 1 Implementation)

Characteristics	**eMINTS+ Intel**	**eMINTS**	**Control**	***p*-value**
Number of teachers	34	32	31	
Percentage male	14.8	6.1	13.0	0.524
Percentage with graduate degree	44.1 (*n* = 25)	38.7 (*n* = 23)	48.4 (*n* = 27)	0.794
Years of teaching experience	7.0 (*n* = 25)	10.1 (*n* = 23)	11.4 (*n* = 27)	0.143

[a]Means were regression adjusted to account for block effects using ordinary least squares regression. For each characteristic, the *p*-value is from an overall chi-square test of whether the means of the three treatment groups (eMINTS+Intel, eMINTS, and control) differed more than what might be expected by chance. When data are missing, *n* is the actual number of teachers used to calculate the average characteristic in each treatment group.

Student Characteristics

Tables 2.5 and 2.6 show the characteristics of the seventh- and eighth-grade MAP analytic student samples in the study schools. There are more than 3,000 students in the analytic sample for MAP mathematics testing (Table 2.5) and communication arts testing (Table 2.6). A joint test of overall difference suggests that the student characteristics of eMINTS, eMINTS+Intel, and control group students are equivalent in both samples (*p*-value = .999 in both samples). Similar baseline comparisons of the student analytic group for the 21st Century Skills Assessment and the student engagement survey are in Appendix E. The analytic samples for these outcomes are smaller, primarily because student assent and parent consent were necessary to administer these measures to students. A joint test of overall difference suggests the eMINTS and control group students are equivalent in both samples (student engagement: *p*-value = .999; 21st century skills: *p*-value = 1.000).

Table 2.5. Characteristics of Students in the MAP Mathematics Analytic Sample: 2010–11 (Before Year 1 Implementation)

Characteristics	eMINTS+ Intel	eMINTS	Control	*p*-value
Number of students	1,141	1,108	823	
2011 MAP mathematics *z*-scores	-0.037 (*n* = 942)	0.068 (*n* = 947)	-0.006 (*n* = 669)	0.562
2011 MAP communication arts *z*-scores	-0.017 (*n* = 941)	0.033 (*n* = 947)	-0.023 (*n* = 669)	0.775
Percentage eligible for free or reduced-price lunch	0.563 (*n* = 946)	0.610 (*n* = 949)	0.571 (*n* = 669)	0.505
Percentage White	0.929 (*n* = 946)	0.953 (*n* = 949)	0.941 (*n* = 669)	0.634
Percentage with IEP	0.126 (*n* = 946)	0.141 (*n* = 949)	0.120 (*n* = 669)	0.642
Percentage LEP	0.027 (*n* = 946)	0.000 (*n* = 949)	0.019 (*n* = 669)	0.287
Percentage male	0.482 (*n* = 1,118)	0.539 (*n* = 1,081)	0.511 (*n* = 807)	0.106

Note. From authors' analysis based on student baseline data collected from study districts in spring 2011, when Grade 7 students were in Grade 4 and Grade 8 students were in Grade 5.
[a]Means were regression adjusted to account for block effects, grade, and clustering of students within schools, and weighted by the number of schools in each block. For each characteristic, the *p*-value is from an overall chi-square test of whether the (weighted) means of the three treatment groups (eMINTS+Intel, eMINTS, and control) differed more than what might be expected by chance. Because of zero counts for LEP in some cells, the percentage LEP was estimated on the whole sample (instead of separately by blocks). When data are missing, *n* is the actual number of students used to calculate the average characteristic in each treatment group.

Table 2.6. Characteristics of Students in the MAP Communication Arts Analytic Sample: 2010–11 (Before Year 1 Implementation)

Characteristics	eMINTS+ Intel	eMINTS	Control	*p*-value
Number of students	1,216	1,208	871	
2011 MAP mathematics *z*-scores	-0.002 (n = 1,015)	0.044 (n = 1,037)	-0.020 (n = 711)	0.754
2011 MAP communication arts *z*-scores	0.002 (n = 1,014)	0.024 (n = 1,037)	-0.039 (n = 711)	0.736
Percentage eligible for free or reduced-price lunch	0.543 (n = 1,019)	0.593 (n = 1,039)	0.568 (n = 711)	0.489
Percentage White	0.934 (n = 1,019)	0.951 (n = 1,039)	0.942 (n = 711)	0.634
Percentage with IEP	0.115 (n = 1,019)	0.131 (n = 1,039)	0.115 (n = 711)	0.669
Percentage LEP	0.026 (n = 1,019)	0.000 (n = 1,039)	0.019 (n = 711)	0.293
Percentage male	0.481 (n = 1,192)	0.523 (n = 1,177)	0.494 (n = 855)	0.229

Note. From authors' analysis based on student baseline data collected from study districts in spring 2011, when Grade 7 students were in Grade 4 and Grade 8 students were in Grade 5.
[a]Means were regression adjusted to account for block effects, grade, and clustering of students within schools, and weighted by the number of schools in each block. For each characteristic, the *p*-value is from an overall chi-square test of whether the (weighted) means of the three treatment groups (eMINTS+Intel, eMINTS, and control) differed more than what might be expected by chance. Because of zero counts for LEP in some cells, the percentage LEP was estimated on the whole sample (instead of separately by blocks). When data are missing, *n* is the actual number of students used to calculate the average characteristic in each treatment group.

Differential Attrition

Differential attrition, or differences in the number of participants lost from the treatment and control groups, can introduce violations to the critical assumption of baseline equivalence in experimental designs (McKnight, McKnight, Sidani, & Figueredo, 2007). If severe enough, it can result in seriously biased impact estimates. When data can be assumed missing completely at random or missing at random, then differential attrition is not a problem because the participants who dropped out can be assumed to be representative of the original population sample. In this section, we examine overall and differential attrition after two years of implementation.

During the study's first year, two of the original 60 eMINTS schools dropped out. The first school dropped out about four months after random assignment was announced and shortly after baseline data collection began. A second school dropped out during the summer before Year 1 implementation. A third school was closed after the 2011–12 school year (Year 1), and students were reassigned to one of several districts in the area. Moreover, the following number of schools could not be included in the analyses because they had no data on a particular outcome:

four schools for the 21st Century Skills Assessment analysis, two schools for the student engagement survey analysis, and one school each for the teacher survey and the classroom observation analyses. Because we were able to access MAP scores from the two schools that dropped out, our intent-to-treat analysis of MAP outcomes includes 59 schools; however, all other student and teacher analyses include 56 schools at most.[14] Details on attrition rates are given in Appendix L.

Data Collection and Outcome Measures

Table 2.7 summarizes the study's data collection plan for the entire study, including what we present in this report: the baseline year (2010–11) and subsequent years of eMINTS and eMINTS+Intel implementation (2011–12, 2012–13, and 2013–14). Fidelity measures collected for this study include a school technology coordinator survey, a teacher survey, records of professional development provided (and staff attendance), logs of coaching visits by eMINTS eIS, observations of eMINTS professional development sessions, and teacher portfolios (collected in Years 2 and 3 only). Changes in teachers' instructional practices were measured using annual classroom observations and an annual teacher survey,[15] both of which were administered in the spring semesters of 2011 (baseline), 2012 (Year 1), 2013 (Year 2), and 2014 (Year 3). Student outcome measures included a student engagement survey, the 21st Century Skills Assessment, and annual student assessment results on the MAP communication arts and mathematics. The 21st Century Skills Assessment, the student engagement survey, and MAP were administered in spring 2011 (baseline), spring 2012 (Year 1), spring 2013 (Year 2), and spring 2014 (Year 3). A more detailed description of the fidelity measures and teacher and student outcome measures used in this study is provided later in this report.

[14] Because the cause of the third school's attrition from the study is purely exogenous (i.e., the school's closure was completely unrelated to the school's involvement in the study), the school should not be counted against the sample's overall attrition rate, per the What Works Clearinghouse professional development manual (Institute of Education Sciences, 2011, p. 65).

[15] Only one teacher survey was administered to obtain measures of both implementation fidelity and teachers' instructional practices across the four eMINTS instructional model components. Thus, teachers were asked to complete only one annual teacher survey each spring.

Table 2.7. Data Collection Timeline

	2010–11 (Baseline)		2011–12 (Year 1)		2012–13 (Year 2)		2013–14 (Year 3)	
Teacher and Student Outcome Measures	**Fall**	**Spring**	**Fall**	**Spring**	**Fall**	**Spring**	**Fall**	**Spring**
MAP test (communication arts, mathematics)		X		X		X		X
21st Century Skills Assessment		X		X		X		X
Student survey		X		X		X		X
Teacher survey		X		X		X		X
Teacher observations		X		X		X		X
Implementation Fidelity Measures	**Fall**	**Spring**	**Fall**	**Spring**	**Fall**	**Spring**	**Fall**	**Spring**
Technology coordinator survey				X		X		X
Teacher survey[a]				X		X		X
Professional development records			X	X	X	X	X	X
Staff attendance records			X	X	X	X	X	X
eIS coaching logs			X	X	X	X	X	X
Observations of eMINTS professional development sessions				X		X		X
Teacher portfolios[b]						X		X

[a]Represents the same survey as was administered to measure teacher outcomes. [b]Teacher portfolios will be included in Years 2 and 3 only to assess differences in the quality of teachers' lesson plans between the eMINTS and eMINTS+Intel groups.

Fidelity of Implementation Measures

School Technology Coordinator Survey. The research team developed a survey to assess a school's infrastructure and access to the technology resources associated with the eMINTS program. As noted earlier, eMINTS classrooms must meet minimum school infrastructure requirements (see chapter 1) to support the safe and effective use of the technology resources and equipment provided to eMINTS schools. The key infrastructure elements include Internet connectivity, speed, and capacity; wireless connections; and data storage. This survey asked each school's technology specialist about all the relevant aspects of the school's infrastructure that are deemed necessary to support teachers' technology integration and implementation of the four eMINTS instructional components.

Teacher Survey. The teacher survey included items constructed to measure teachers' implementation of the eMINTS Comprehensive Program. More specifically, these items asked teachers to report on the extent to which they had access to technology, equipment, and professional development associated with the eMINTS Comprehensive Program. It also asked teachers to report on the extent to which they participated in the required professional development and coaching sessions and used the technology available to them.

Records of Professional Development Provided and Staff Attendance. Records of professional development included the number of modules delivered and the hours of professional development and coaching provided to teachers in the eMINTS and eMINTS+Intel groups. These records were used to assess the amount of professional development provided against the full amount required for the eMINTS program. Each eIS submitted attendance records to eMINTS staff, which provided these records to the research team. Attendance records included the total hours of professional development received by each teacher for each session in Years 1 and 2.

Logs of eMINTS Coaching Visits. An important element of eMINTS is the regular coaching visits that an eIS makes to each eMINTS teacher's classroom monthly for two years. Using a predeveloped and validated log instrument (Martin et al., 2008), each eIS records the length of each visit and how much time was spent on each of the following activities:

- **Modeling Instruction.** Activities such as modeling inquiry-based learning techniques or coteaching a lesson
- **Lesson Planning.** Helping review or plan lessons
- **Technology Assistance.** Troubleshooting and modeling
- **Reflective Practice.** Reviewing goals for teaching
- **Problem Solving.** Answering questions or specific issues with program implementation

The logs allowed the research team to examine the types of support offered to teachers during the coaching visits.

Professional Development Observations. Using the eMINTS professional development protocol, each eMINTS area instructional specialist (AIS) observed at least one professional development session for each eMINTS eIS he or she was assigned to supervise in Years 1 and 2.[16] AISs completed two protocols during each observation: the eMINTS Snapshot and the eMINTS Checklist. External researchers developed both instruments (Martin et al., 2008). The checklist was developed as a descriptive tool to confirm content delivery. Researchers developed the eMINTS Snapshot to record information on the ways content was delivered within each professional development. The protocols allow AISs to record the key activities and behaviors conducted by eIS during each professional development session. An expert panel with deep knowledge of the program curriculum and implementation model reviewed the instruments and provided feedback to establish content validity. A prior study on the eMINTS program, and the snapshot tool specifically, established an 88 percent interrater reliability score and internal reliability estimates on the four core constructs—inquiry-based learning, a community of learners, technology integration, and connection to practice—ranging from .60 to .78 (Martin et al., 2008).

[16] AISs are eMINTS National Center employees who supervise eIS, who facilitate the eMINTS professional development sessions and provide coaching to eMINTS teachers in their assigned cohort groups.

Portfolio Scores. Developed by eMINTS, the portfolio scores measure a teacher's knowledge and ability to develop a lesson demonstrating the eMINTS instructional model. Each portfolio submission includes three unique components around a unifying theme: a website, a WebQuest, and a unit plan. A trained reviewer grades each component with a rubric. The rubrics were developed by eMINTS for each portfolio component and designed to capture the aspects of lesson development that eMINTS emphasizes in its training. Each aspect receives a grade of 1–3; an average of those grades provides a mean score for the component. Composite portfolio scores are the average of three component measures. To receive a satisfactory rating on the composite portfolio score, a teacher's average numerical score must be 2.2 or greater, and two or more subcomponents cannot be scored 1.9 or lower. For this study, three raters were trained and demonstrated an interrater agreement of at least .80 for each component before they were assigned individual portfolio components.

Teacher Outcome Measures

Various data sources are used to answer research questions on the impact and the effectiveness of the eMINTS professional development program. Table 2.8 presents the domains, the outcome measures, and the data sources used for assessing teacher practice[17] and student learning. We began with a set of established, psychometrically valid and reliable data collection instruments and augmented them with new items to more closely reflect key eMINTS program components. The research team worked closely in conjunction with the program developers to ensure that the instruments appropriately operationalized eMINTS program features and functions. In addition, we ascertained the psychometric properties of the augmented instruments used in this study. Each instrument was administered annually in the spring of each school year, including once in spring 2011 (baseline) before eMINTS implementation commenced.

Table 2.8. Domains, Outcome Measures, and Data Sources

Domains	Outcome Measures	Data Sources
Teachers		
High-quality lesson planning	Index of a teacher's use of content standards and data to inform instructional planning	Teacher survey
Inquiry-based learning	Index of a teacher's use of inquiry-based learning and assessment strategies	Classroom observations; teacher survey
A community of learners	Index of a teacher's use of strategies encouraging teacher and student collaboration and classroom community	Classroom observations; teacher survey
Technology integration	Index of a teacher's access to and knowledge of skills needed to integrate and use various types of technology for instructional purposes	Teacher survey

[17] Many of the teacher outcome measures double as fidelity-of-implementation measures.

Domains	Outcome Measures	Data Sources
Students		
Academic orientation	Index of a student's academic orientation, measuring a student's sense of engagement, academic efficacy, relevance of school for future success, educational aspirations, self-directed learning, and perception of the quality of emotional and relational support from teachers	Student survey
21st century skills	Measure of 21st century skills—based on the National Educational Technology Standards for Students, developed by the International Society for Technology in Education (ISTE; 2014). Constructs measured in the assessment include creativity and innovation; communication and collaboration; research and information fluency; critical thinking, problem solving, and decision making; digital citizenship; and technology operations and concepts	21st Century Skills Assessment
Academic performance	Mean scaled score for seventh- and eighth-grade students, represented as a continuous measure of students in communication arts and mathematics	MAP communication arts and mathematics assessments

Teacher Survey. Items from the following publicly available surveys were included to measure each eMINTS instructional component:

- SRI Innovative Teaching and Learning Research Survey (Shear, Novais, & Moorthy, 2010)
- eMINTS Teacher Technology Literacy Skill Survey (eMINTS, 2009)
- Intel Teach Program Essentials Course Impact Survey (Intel Corporation, 2006)
- Buck Institute National Survey of High School Reform and Project Based Learning (Buck Institute for Education, 2007)
- Surveys of Enacted Curriculum (Wisconsin Center for Education Research, 2004)
- Special Education Elementary Longitudinal Study (Office of Special Education Programs, 2007)
- MetLife Survey of the American Teacher (MetLife, 2011)

The majority of the items measuring a community of learners and inquiry-based learning practices came from SRI's Teaching and Learning Research Survey and the National Survey of High School Reform and Project Based Learning. Items measuring technology integration primarily came from the eMINTS Technology Literacy Survey and the Intel Teach Program Essentials Course Impact Survey (Martin & Shulman, 2006). Items to measure high-quality lesson planning were developed by the research team or pulled from SRI's survey. We also used a minimum number of items from the other surveys mentioned here. Items are designed to measure teachers' pedagogical beliefs (e.g., beliefs and practices aligned with an inquiry-based, constructivist approach), approaches to lesson planning, and instructional practices.

Content validity was established for this survey in spring 2011. Baseline survey results were used to assess item reliabilities using the Rasch dichotomous model (Rasch, 1980; Wright & Masters, 1982). Internal reliability estimates for each domain ranged between .83 and .94.

Classroom Observations. Classroom observations were conducted in spring 2011, spring 2012, spring 2013, and spring 2014 to measure teaching outcomes across the four eMINTS model components. The observation protocol measures teaching practices, including inquiry-based learning, collaborative teaching, student and teacher use of technology, use of community resources, classroom organization, instructional support, and emotional and relational support. To measure changes in a community of learners and inquiry-based learning constructs, we used the Classroom Assessment Scoring System–Secondary ([CLASS-S], Pianta, La Paro, & Hamre, 2008).

CLASS-S is a theoretically driven and empirically supported conceptualization of classroom interactions organized into three major domains: emotional supports, classroom organization, and instructional supports. CLASS-S scales have been found to be reliable and predictive of student gains in a study of middle and high school professional development (Allen, Pianta, Gregory, Mikami, & Lun, 2011). This protocol also contains a set of scales related to teaching procedures, skills, and content knowledge that can be adapted to infuse a specific content area (e.g., mathematics) and instructional strategies (e.g., inquiry-based learning) into existing CLASS dimensions.

In addition to CLASS-S, numerous items were developed to obtain a comprehensive measure of technology integration. We initially conducted a literature review of technology use and technology integration literature to establish a conceptual framework. After discussion with eMINTS personnel and a review of eMINTS material, we created a brief observation protocol to measure technology use in the classroom by the teacher and students and used preexisting observation tools to measure technology integration in the classroom (Northwest Regional Educational Laboratory, 2004; West Virginia Department of Education, n.d.). In spring 2011, instructional content experts and eMINTS staff members reviewed the complete observation protocol, which included CLASS-S and these additional technology items, to establish content validity.

Student Outcome Measures

Student Engagement Survey. The student engagement survey was used to measure students' engagement and academic orientation. Student surveys included measures associated with students' engagement in school, academic efficacy, the relevance of school for future success, educational aspirations, self-directed learning, and perceptions of emotional and relationship support from teachers. Items were selected primarily from the University of Chicago's Consortium on Chicago School Research student survey (2007a, 2007b). We also reviewed student surveys used in the New Hope Study (Huston et al., 2003), the Research Assessment Package for Schools (Midgley et al., 2000), and other scales of student efficacy and engagement with strong psychometric properties, including Cook et al. (1996), Middleton and Midgley (2002), and Zimmerman, Bandura, and Martinez-Pons (1992). The reported reliability estimates for the scale are .80 or higher.

21st Century Skills Assessment. To measure students' skills in areas identified as 21st century skills, we used the 21st Century Skills Assessment (Condon, Dawson, Molefe, & Swanlund, 2009), which is a 72-item criterion-referenced assessment used to measure the six ISTE National Educational Technology Standards for Students, which include creativity and innovation, communication and collaboration, research and information fluency, critical thinking, problem solving, decision making, digital citizenship, and technology operations and concepts (ISTE, 2014). A Rasch analysis conducted by Condon et al. (2009) demonstrated high levels of reliability and validity on the 21st Century Skills Assessment. The person reliabilities of the pre- and posttests for each grade were consistently greater than .90. In addition, content validity was established by having content experts create the items, followed by a review by a standards-setting panel. The Rasch analysis found that the assessment measured only one construct (i.e., technology literacy), and the items fit the Rasch model well.

Missouri Assessment Program. Primary student outcome data include annual MAP results for Grade 7 and 8 students in communication arts and mathematics. MAP assessments are norm referenced and administered annually in the spring of each school year to students in Grades 3–8. Designed to measure student acquisition of skills and knowledge as described in Missouri's Grade-Level Expectations, the assessment provides information on academic achievement at the student, class, school, and district levels. Student data are provided both as scale scores and performance level (e.g., Below Basic, Basic, Proficient, and Advanced; CTB/McGraw-Hill, 2012).

Data Analysis

This section overviews the analytic strategy used to examine the primary impacts on student outcomes and the exploratory impacts on teacher outcomes. The randomization of schools to an eMINTS, eMINTS+Intel, or control condition provides a strong framework with which to accurately estimate program impact because the randomization of a sufficient number of participants to one of three conditions should, on average, equalize any measured and unmeasured baseline differences among the three groups that may confound impact estimates. Nevertheless, to obtain more precise estimates, the research team directly accounted for student, classroom, and school characteristics in the analytic models.

Analysis of Student Outcomes

This study was designed to obtain statistically unbiased estimates of the effect of eMINTS professional development on school average Grade 7 and 8 student performance within four domains: mathematics achievement, communication arts achievement, 21st century skills, and student engagement and academic orientation. To measure performance in mathematics and communication arts, we used student results from MAP, which was administered in April of every study year. To measure 21st century skills, we used the 21st Century Skills Assessment. We used a student survey to measure students' engagement and academic orientation. The 21st Century Skills Assessment and the student engagement survey were both administered between April and May in each study year.

Unless noted otherwise, our analyses employed two-level hierarchical linear models with students nested within schools. Means and differences were regression adjusted to account for blocked effects, grade, and clustering of students within schools and weighted by the number of schools in each block. Effect sizes were calculated separately by block and then pooled into an overall effect size weighted by the number of schools in each block. Impact results for each domain are calculated separately as pooled averages of scores collected from Grades 7 and 8 students.

For both MAP assessments (mathematics and communication arts), the following covariates were included:

- Student pretest: Spring 2011 MAP scores in mathematics or communication arts
- School pretest: School mean spring 2011 scores in mathematics or communication arts
- Student characteristics:
  - Gender (male/female)
  - White (yes/no)
  - Free- or reduced-price lunch (yes/no)
  - LEP (yes/no)
  - IEP (yes/no)
- Teacher characteristics[18]:
  - Gender (male/female)
  - Graduate degree (yes/no)
  - Years of teaching experience

For 21st century skills and student engagement, the following covariates were included[19]:

- Student pretest: Spring 2011 MAP scores (*z*-scores) in mathematics and communication arts
- School pretest: Spring 2011 school mean scores (*z*-scores) in mathematics and communication arts
- Student characteristics:
  - Gender (male/female)
  - White (yes/no)
  - Free- or reduced-price lunch (yes/no)
  - LEP (yes/no)

[18] In our analysis plan, we stated that we would keep teacher covariates in the model only if they were statistically significant. As it turns out, for the mathematics analysis, the three variables under discussion were statistically significant at either the .05 or .10 levels (*p*-values of .072, .020, and .053, respectively), but none was statistically significant for the communication arts analysis. For consistency, we decided to keep the teacher covariates in both analyses.

[19] We did not include teacher characteristics as covariates in the 21st Century Skills Assessment and student engagement survey analyses because students in both samples often were associated with two different teachers (their mathematics and communication arts teachers).

- IEP (yes/no)

Data were not imputed for missing outcomes on any of the four student outcomes we examined. There were no missing values for student characteristics and school means for any of the analytic samples. For missing teacher characteristics and missing pretest data, the dummy variable approach advanced by Puma, Olsen, Bell, and Price (2009) was used to impute these missing covariate values.

To measure the impact of eMINTS and eMINTS+Intel on student outcomes, we employed the following two-level model. This model estimated impacts separately by randomization block, which were then pooled into an overall impact, weighted by the number of schools in each block.

Level 1 (students)

$$Y_{ij} = \pi_{0j} + \pi_{1j}\text{Grade}_{ij} + \mathbf{X_{ij}}\boldsymbol{\pi}_{\mathbf{2j}} + \mathbf{C_{ij}}\boldsymbol{\pi}_{\mathbf{3j}} + \varepsilon_{ij}$$

Level 2 (schools)

$$\pi_{0j} = \sum_k \beta_{00k} D_k + \sum_k \beta_{01k}\mathrm{T1}_j * D_k + \sum_k \beta_{02k}\mathrm{T2}_j * D_k + r_{0j}$$

$\pi_{1j} = \beta_{10}$, $\pi_{2j} = \beta_{20}$, and $\pi_{3j} = \beta_{30}$

where

- $\mathrm{T1}_j$ = 1 if school *j* belongs to the eMINTS group; 0 otherwise.
- $\mathrm{T2}_j$ = 1 if school *j* belongs to the eMINTS+Intel group; 0 otherwise.
- Y_{ij} is the outcome (i.e., communication arts or mathematics test score or measures of academic orientation or 21st century skills) of student *i* in school *j.*
- D_k = 1 if school *j* is in randomization block *k;* 0 otherwise, $k = 1, \cdots, 3$.
- Grade_{ij} is a grade-level indicator that is equal to 1 for eighth-grade students and 0 for seventh-grade students (centered around the grand mean).
- $\mathbf{X_{ij}}$ is a row vector of background characteristics (e.g., prior academic achievement, race/ethnicity) of student *i* in school *j.*
- $\mathbf{C_{ij}}$ is a row vector of teacher characteristics (e.g., teaching experience, degrees held, gender) for student *i* in school *j.*
- $\mathbf{Z_j}$ is a row vector of school characteristics (e.g., average school achievement, percentage of racial/ethnic minority students, school size) for school *j.*
- ε_{ij} and r_{0j} are random residuals at the student and school levels, respectively.

Grade_{ij} was centered about the grand mean (i.e., the overall proportion of eighth graders in the sample) to control for differences in the proportions of seventh and eighth graders across schools.

With the above formulation, β_{00} is the overall average achievement level of seventh and eighth graders, and β_{10} is the average level of achievement gap between seventh and eighth graders. The parameters of interest are β_{01k} and β_{02k}. The mean of the block-specific impact estimates of β_{01k} ($k = 1, \cdots, K$), weighted by the number of schools in each block, is an estimate of the overall average impact of eMINTS (β_{01}) on the outcomes of seventh and eighth graders relative to the control group. Similarly, the mean of the block-specific impact estimates of β_{02k} ($k = 1, \cdots, K$), weighted by the number of schools in each block, is an estimate of the overall impact of eMINTS+Intel (β_{02}) on student outcomes relative to the control group. A test of the null hypothesis that $\beta_{01} = 0$ is a test of whether eMINTS has a statistically significant impact. Similarly, a test of the null hypothesis that $\beta_{02} = 0$ is a test of whether eMINTS+Intel has a statistically significant impact. We also compared the impact of an extra year of training on student outcomes (eMINTS+Intel) to the impact of eMINTS by testing the hypothesis that $\beta_{01} = \beta_{02}$.

The main goal of this study was to estimate the overall impacts of the eMINTS and eMINTS+Intel programs on student outcomes. Because overall impacts may mask important variations in impacts across different types of students, the study team also estimated the impact of the programs on subsets of students defined by the following baseline characteristics[20]:

- IEP (yes/no)
- Free- or reduced-price lunch (yes/no)
- LEP (yes/no)

The characteristics were chosen in light of prior research the eMINTS program, which suggests that possible interactions could exist for these subgroups.

To examine the differential impacts of eMINTS and eMINTS+Intel across student subgroups, we modified the model presented here to include an interaction of the treatment indicator(s) with the appropriate student subgroup indicator (e.g., racial/minority indicator).

Analysis of Teacher Outcomes

Single-level rather than multilevel models were used to measure teacher outcomes because several schools in the analytic samples had only one teacher, precluding estimation of either the between-school variance or the block-specific impact estimates. Means and impacts reported on domains from the teacher survey and teacher observations were regression adjusted using ordinary least squares to account for block effects and baseline teacher and classroom characteristics and weighted by the number of schools in each block. Effect sizes were calculated separately by block and then pooled into an overall effect size weighted by the number of schools in each block. Block-specific effect sizes were computed using standardized mean differences (Hedges's *g*). *p*-values are from a two-tailed test of the null hypothesis of equality of eMINTS and control means.

To examine the impact of eMINTS Comprehensive Program implementation on teachers' use of inquiry-based learning strategies, a community of learners, and technology integration, results

[20] Differences in impacts between LEP and non-LEP students could not be examined because of the small percentage of LEP students (only about one percent) in the analytic samples.

from teacher surveys and classroom observations were analyzed separately. High-quality lesson design (e.g., teachers' use of data and standards to guide lesson planning) was examined using results from the teacher survey only.

Teacher Survey Analysis

The teacher survey was administered in spring 2011, spring 2012, spring 2013, and spring 2014. Teacher outcomes on each construct—a community of learners, inquiry-based learning, high-quality lesson design, and technology integration—are the logit ability measures from a Rasch analysis of the survey items belonging to each construct.[21] The Rasch analysis was conducted separately for mathematics and communication arts teachers. Teachers who taught both subjects were included in both the mathematics and communication arts analyses, and their scores from the two subjects were averaged. Analysis of impact of eMINTS on teacher survey logit scores was then conducted on the pooled sample of mathematics and communication arts teachers.

- Pretest: Teachers' logit ability scores from Rasch analysis of baseline (spring 2011) teacher survey
- Teacher characteristics
  - Gender (male/female)
  - Graduate degree (yes/no)
  - Years of teaching experience
  - Subject taught (mathematics, communication arts, or both)
  - Grade taught (Grade 7, Grade 8, or both)
- Classroom characteristics
  - Percentage of male students
  - Percentage of White students
  - Percentage of free- or reduced-price lunch students
  - Percentage of LEP students
  - Percentage of IEP students
  - Means of students' spring 2011 MAP scores in mathematics and communication arts

The subject by treatment interaction was initially included in the regression models but was dropped from the analysis because either it was not statistically significant (as were the case for the community of learners, high-quality lesson design, and technology integration outcomes) or

[21] The Rasch model uses (dichotomous or Likert scale) responses to selected survey items (questions) to produce an interval scale that estimates item difficulties and person abilities. The model posits that a person's success on an item is equal to the difference between that person's ability and the item's difficulty. Fitting the Rasch model places items on the scale according to how likely they are to be endorsed (item difficulty). Similarly, a quantitative measure of a person's attitude or ability (person ability) is placed on the same scale. The scale units are called logits (log odds units). Because the logit scales are linear, they are amenable to statistical procedures, such as the calculation of averages and standard deviations. (For more information on the Rasch scale, see Wright and Masters, 1982.)

the subgroup sample sizes were too small to generate reliable subgroup estimates (which was the case for the inquiry-based learning outcome[22]).

Data were not imputed for missing outcomes on any of the teacher domains we examined. There were no missing values for student characteristics or school means for any of the analytic samples. For missing teacher characteristics and missing pretest data, the dummy variable approach advanced by Puma et al. (2009) was used to impute the missing covariate values.

Classroom Observation Analysis

Classroom observations of teachers were conducted with CLASS-S certified observers in spring 2011, spring 2012, spring 2013, and spring 2014. As in the teacher survey outcomes, classroom observation outcomes on each of three constructs[23]—a community of learners, inquiry-based learning, and technology integration—were placed on a Rasch scale for comparison. Each classroom observation was composed of at least two 10-minute observation segments. The Rasch analysis was conducted separately by segment and then the resulting logit scores were averaged across segments. Analyses of the impact of eMINTS on classroom observation scores were conducted on the pooled sample of mathematics and communication arts teachers. The following covariates were used in the classroom observation analysis:

- Pretest: Logit ability scores from Rasch analysis of baseline (spring 2011) classroom observations
- Teacher characteristics
  - Gender (male/female)
  - Graduate degree (yes/no)
  - Years of teaching experience
  - Subject of the classroom observed (mathematics, communication arts)
  - Grade of the classroom observed (Grade 7, Grade 8, or both)
- Classroom characteristics
  - Percentage of male students
  - Percentage of White students
  - Percentage of free- or reduced-price lunch students
  - Percentage of LEP students
  - Percentage of IEP students
  - Means of students' spring 2011 MAP scores in mathematics and reading

---

[22] When the inquiry-based learning analytic sample was broken down by the subject taught, 16 teachers taught mathematics, 21 teachers taught communication arts, and 6 teachers taught both subjects.

[23] No items in the classroom observation protocol measured the high-quality lesson design construct, so this construct was measured only through the teacher survey.

As in the analysis using teacher surveys, a subject-by-treatment interaction was initially included in the regression models but was dropped from the analysis because, in all three outcomes, the interaction was not statistically significant.

Implementation Fidelity

In collaboration with eMINTS staff, we identified five major components of eMINTS Comprehensive Program implementation:

- Technology infrastructure
- Technology use
- Teacher professional development
- Administrative support
- Ongoing technology support

Although each component is necessary to establish and maintain the integrity of eMINTS professional development, we worked with eMINTS personnel to determine the relative weight of each component, with teacher professional development ranking as most important (weighted at 45 percent) and technology use and ongoing technology support as less important (weighted at 10 percent each).

Each component consists of multiple criteria with associated descriptive measures, including technology audits, teacher surveys, technology coordinator surveys, eMINTS teacher attendance files, eMINTS portfolio score records, eMINTS principal attendance files, principal walk-through data, and technology coordinator attendance files. Minimum requirements (e.g., survey score responses per item, rates of professional development attendance) were established for each measure as an item metric, which were then aggregated and divided, producing component-level metrics.

Using the metrics described in Table F1 in Appendix F, we calculated fidelity scores for each component separately and then calculated the overall fidelity score for each treatment school as a weighted average of the individual components. The overall fidelity score in Year 2 is inclusive of measures collected in both Years 1 and 2.

For treatment teachers in the eMINTS+Intel condition who met study requirements in Year 3, the overall fidelity score included only a teacher professional development component. As a result, no weights were required. The overall fidelity score was limited to eMINTS records (e.g., attendance) of eMINTS+Intel teacher participation in Intel Teach. Much like eMINTS Comprehensive Program implementation fidelity calculations, we established an item metric for each measure of Year 3 Intel implementation fidelity. The metrics were then aggregated and divided, producing a component-level metric. After that, we averaged teacher ratings by school, creating an overall fidelity score for each treatment school. See Table F2 in Appendix F for a description of the metrics used to calculate Intel program fidelity in Year 3.

Study Limitations

The key questions addressed in this report pertain to the impact of eMINTS and eMINTS+Intel professional development on communication arts and mathematics achievement and 21st century skills. The strength of this evaluation lies in the randomization of schools to either treatment or control, which allows us to draw causal inferences about the program's impact on school averages of student achievement. It is important to point out some caveats, however, regarding the design and the analysis of Year 3 outcomes that restrict the conclusions that can be drawn from this report.

The first caveat is sample selection. The schools that were recruited and volunteered to be part of the study are not necessarily representative of the universe of schools that are currently using or are intending to implement the eMINTS Comprehensive Program. The sample is limited to rural schools with high levels of poverty. In addition, the study is restricted to schools in Missouri. Thus, conclusions drawn from this report should be limited to the purposeful sample of participating schools and districts.

Second, because participation is voluntary, the observed effects in this study could be different from what might be observed in less motivated settings. In general, the schools involved in the study were motivated to adopt and implement the eMINTS Comprehensive Program. The history of eMINTS Comprehensive Program in Missouri and the anticipation of district and school personnel about receiving the professional development is probably unparalleled elsewhere. Its adoption in other settings may not be transferrable.

Third, although random assignment typically establishes the treatment and control groups as equivalent on observable and unobservable traits and characteristics, the process remains random, which means that an improbable but possible difference between the groups could exist. Considering the significant difference between treatment and control groups at baseline for mathematics achievement, this difference could exist.

The limitations just listed serve to underscore that although we can make causal inferences from this study, the findings reported here are restricted in generalizability.

Chapter 3: Program Implementation

This chapter presents evidence on the eMINTS professional development program implemented in the first two years of the study and the eMINTS+Intel professional development supplement in the third year of the study. Part one of this chapter presents information on the extent to which eMINTS staff implemented the core program components in the treatment schools and offered the required professional development to teachers, administrators, and technology coordinators in Years 1 and 2. Part two examines how faithfully schools implemented the professional development and support that were provided in Years 1 and 2. Part three presents information on the extent to which eMINTS delivered Intel Teach to the 20 eMINTS+Intel schools. Part four describes school receipt of eMINTS+Intel professional development, addressing the extent to which eMINTS+Intel teachers participated in the professional development and completed its core components. This chapter does not attempt to explain why there is variation in the extent to which teachers implemented various program components.

Part 1: Did eMINTS Staff Deliver the eMINTS Comprehensive Program as Designed in Years 1 and 2?

The eMINTS instructional model combines four overarching concepts—high-quality lesson design, inquiry-based learning, a community of learners, and technology integration—to improve teacher instruction and student learning. This involves (1) equipping teachers and school leaders with the knowledge and skills needed to implement the model effectively and that eMINTS personnel engage teachers, principals, and technology coordinators in professional development sessions; (2) establishing the necessary infrastructure within the schools; (3) providing technological resources; and (4) offering instructional support.

eMINTS Implementation: Resources, Professional Development, and Coaching

The developers of the eMINTS Comprehensive Program posit that successful implementation requires that eMINTS staff do the following: (1) monitor the purchase and the installation of essential resources (e.g., student and teacher laptops in each school and interactive whiteboards) through subaward processes approved by the University of Missouri, (2) schedule and deliver teacher professional development modules, (3) schedule and conduct coaching visits, (4) offer each teacher online access to curriculum and professional development materials, (5) review teacher portfolios, (6) provide principals with professional development, (7) conduct walk-throughs of schools with principals, and (8) offer WebEx sessions to help district technology coordinators support eMINTS implementation in their schools. This section focuses on the extent to which eMINTS personnel implemented the program's components as expected in Years 1 and 2 in the schools participating in this study.

The number of teacher professional development sessions, coaching visits, principal professional development sessions, school walk-throughs, and WebEx sessions varied by year of implementation. According to the program developers, full implementation is designed to take two years. As a result, this chapter holistically and collectively examines the first two years of implementation. To support this inclusive review of implementation fidelity, we created a composite metric to assess the delivery of the eMINTS Comprehensive Program in Years 1 and 2 of the study. Table 3.1 lists the program components measured, the delivery criteria, and the total amount of planned and delivered professional development during two years of eMINTS implementation.

eMINTS Technology Delivery. Overall, the eMINTS National Center was successful in monitoring the purchase and the installation of program resources, professional development, and consultation in eMINTS schools. Subawards to purchase and install the equipment necessary for program implementation were made available, as planned, in all 38 treatment schools.[24] Prior to the execution of the i3 subaward contracts that provided funding for the purchase and the installation of equipment, eMINTS National Center conducted technology audits at all the study schools. Decisions were made in consultation with personnel from each school and district about equipment needs. Some schools already possessed some of the necessary technology (e.g., interactive whiteboards in their classrooms). The eMINTS National Center facilitated the execution of subawards to cover the costs of all equipment and installation necessary for each treatment school to comply with the program requirements. eMINTS staff also worked with corporate partners to customize order forms to facilitate the purchase of technology required by the project. Spring 2011 orders included Lenovo ThinkPad T520 teacher laptops, Aruba wireless access points, Xerox printers, Canon digital cameras (from CDW-G), SMART Boards and SMART Ideas site licensing (from KCAV), and BrainPOP site licenses. Fall 2011 orders included Lenovo ThinkPad X120e student laptops (from CDW-G) and laptop charging carts (from CDW-G or Earthwalk). In addition, participation in eMINTS PD provided teachers with access to Tech4Learning's Recipes for Success and School Improvement Network PD360 videos. Across the 38 treatment schools, equipment purchase and implementation included 2,613 student laptops, 179 teacher laptops, 178 wireless access points, 146 interactive whiteboards, 43 printers and scanners, and 173 digital cameras. All eMINTS teachers were also provided accounts to the Intel Online Assessment Library and membership in the Intel Engage Online community. A list of the equipment and software delivered to treatment schools is given in Table in Appendix G.

Who Participated in eMINTS Professional Development? This section focuses on teachers in eMINTS or eMINTS+Intel schools who taught seventh- or eighth-grade communication arts or mathematics (or both grades and subjects) in first two years of the program. A total of 99 teachers met the study requirements and were included in the fidelity analysis.[25] Of these 99 teachers, 84 were returning teachers from first year of the program and 15 were new teachers who began eMINTS training in the second year.[26] Thus, only 84 of the 99 teachers could have

[24] Two treatment schools dropped out before Year 1 implementation began.

[25] The eMINTS staff maintained attendance records of teacher professional development sessions and coaching visits. In Year 1, eMINTS provided professional development and coaching to 128 teachers from the 38 treatment schools participating in this study. Of these 128 teachers, 84 returned in Year 2 and 44 either elected to leave or no longer met the study requirements before the start of Year 2. Fifteen teachers received eMINTS professional development for the first time in Year 2. Of these 15 teachers, five were in eMINTS schools and 10 were in eMINTS+Intel schools. In total, 99 teachers met the study requirements and were included in the Year 2 fidelity analysis (84 teachers completing both years of professional development plus 15 new teachers in Year 2). Of the 99 teachers, 44 were in eMINTS schools, and 55 were in eMINTS+Intel schools.

[26] Teachers who participated in the study during only Year 1 or Year 2 could not achieve 100 percent fidelity because they did not receive two years of training. Eligible teachers who entered the school beginning in Year 2 (2013–14) were invited to participate in Year 1 training activities with other teachers who were new to the eMINTS program.

achieved 100 percent fidelity because full training participation required participation across two years.

Of the 99 teachers who participated in the second year of professional development, 43 teachers (43.4 percent) taught communication arts only, 42 teachers (42.4 percent) taught mathematics only, and 14 teachers (14.1 percent) taught both subjects. At least one administrator from every school participated in principal professional development and eMINTS walk-throughs.

Table 3.1. eMINTS Delivery of Professional Development Sessions and Other Program Components[a]

			All Treatment Schools (N = 38)		eMINTS (N = 18)		eMINTS+Intel (N = 20)	
Program Component	**Year**	**Criteria**	**Planned**	**Delivered**	**Planned**	**Delivered**	**Planned**	**Delivered**
Teachers								
Professional development sessions	Years 1 and 2	47 PLC sessions	47	47	47	47	47	47
Coaching visits	Years 1 and 2	20 coaching visits (10 per year)	1,980	1,657	880	762	1,100	895
Online access to curriculum and professional development materials	Years 1 and 2	All teachers receive access	99	99	44	44	55	55
Portfolio reviews	Spring Year 2	All teacher portfolios are reviewed and scored	99	85	44	39	55	46
Principals								
Professional development sessions	Fall Years 1 and 2	Two 2-day sessions (one session per year)	2	2	2	2	2	2
Walk-throughs	Fall and spring Year 1; fall Year 2	Three school walk-throughs with a certified eMINTS specialist (two in Year 1 and one in Year 2)	114	112	54	53	60	59
Technology Coordinators								
WebEx sessions	Fall and spring Year 1; fall Year 2	Three WebEx sessions (two in Year 1 and one in Year 2)	3	3	3	3	3	3

[a]The delivery of program components is based on the number of participants who qualified for the study in Year 2. Teachers in Year 1 of the study who left the study or those who do meet the requirements for the study are excluded.

What Professional Development Was Delivered? eIS offered all 47 planned professional development modules (27 in Year 1 and 20 in Year 2) to eligible teachers in the 38 treatment schools. (See Appendix A for a list of the modules.) The maximum number of hours of professional development delivered in each year was about 117 hours. In both years, the total number of professional development hours delivered to teachers ranged from 40.5 to 194.1 hours. Some variation results from rounding the length of the individual modules by site; in cases where teachers began professional development for the first time in the second year, the maximum number of hours they could potentially receive was 117.5 hours.

In addition to the professional development modules, eMINTS attempted to schedule 10 coaching visits per teacher per year. Of the 1,980 possible coaching visits in Years 1 and 2, there were 1,657 visits (83.7 percent). The number of visits was approximately five percentage points higher in eMINTS schools than in eMINTS+Intel schools (86.6 percent for eMINTS and 81.4 percent for eMINTS+Intel). The total number of coaching sessions delivered was less than the number planned, in part, because teachers elected not to participate in all the coaching sessions or did not respond to coaches' requests to schedule a coaching session. For teachers who entered the study in the second year, the maximum number of classroom visits delivered was limited to what those teachers could complete during Year 2.

All treatment teachers were offered online access to curriculum and professional development materials.

In Year 2 of the study, eMINTS introduced and planned to conduct one portfolio review per teacher. In year 3, externally trained reviewers reviewed 85 of the 99 portfolios (85.9 percent). Portfolio reviews were five percentage points higher in eMINTS schools than in eMINTS+Intel schools (88.6 percent for eMINTS and 83.6 percent for eMINTS+Intel). The total number of portfolio reviews delivered was less than the number planned because many teachers did not provide eMINTS with reviewable portfolios.

During Years 1 and 2, eMINTS provided 112 of the intended 114 (98.0 percent) school walk-throughs with principals.[27] In addition, eMINTS offered a two-day professional development session to principals in Year 1 and a one-day session in Year 2. Principal professional development sessions were completed in 100 percent of the schools each year. Two WebEx sessions for technology coordinators of districts with a treatment school were offered in Year 1 and one WebEx session was offered in Year 2. Similar to principal professional development, eMINTS offered these sessions to all 38 treatment schools (100 percent) in Years 1 and 2 (see Table 3.1).

Summary of eMINTS Implementation. Across program years, eMINTS staff provided essential technology resources, professional development, and the guidance needed to support the eMINTS Comprehensive Program at the district, school, and classroom levels. In Year 1,

[27] Three walk-throughs per school were conducted in Years 1 and 2. An additional, optional walk-through was offered in the spring of Year 2. However, no school chose to participate in the optional walk-through. This optional walk-through is not considered a required component of implementation and, consequently, is not included in the analysis.

eMINTS staff conducted technology audits in all the treatment schools to identify areas of equipment need. They also followed up with the necessary resources and support to ensure that all schools met the minimum infrastructure and equipment requirements within the expected time frame. Professional development for teachers, administrators, and technology coordinators, which included formal professional development sessions, school visits, portfolio reviews, and coaching sessions, were offered to all eligible participants in all treatment schools. eMINTS professional development was scheduled and conducted in a timely fashion.

Part 2: To What Extent Did Schools Implement the eMINTS Comprehensive Program as Planned in Years 1 and 2?

Teacher professional development builds knowledge and understanding about how to implement inquiry-based learning strategies and classroom communities. Classroom technology integration promotes the use of eMINTS learning strategies (as defined in the eMINTS model) through the availability of a specific set of technology resources, along with minimum school technology infrastructure levels and ongoing support to promote technology integration across time.

The eMINTS instructional model cannot be fully implemented without appropriate classroom technology resources. Minimum levels of school infrastructure and ongoing technology support are needed to promote and sustain long-term technology integration. Technology alone, however, is insufficient to promote students' 21st century skill development. Teacher professional development is essential for understanding how technology tools can be used to optimize learning through inquiry-based instruction and constructive classroom communities. Administrative support also is critical for ensuring that both tools and conditions are optimal for eMINTS program implementation across time and supporting teachers' implementation of the instructional strategies embedded in the eMINTS Comprehensive Program.

To determine the implementation fidelity levels in the treatment schools, we worked with the program developers to identify the core program components and weight them according to their relative importance toward achieving optimal school performance. These components and their weights are presented in Table 3.2. Teacher professional development was determined to be the most central piece to the eMINTS program (45 percent), followed by administrative support (20 percent), technology infrastructure (15 percent), and technology use and ongoing technology support (10 percent each).

Table 3.2. Years 1 and 2 Fidelity Components for Measuring eMINTS Program Receipt[a]

School	Technology Infrastructure[b] (15%)	Technology Use (10%)	Teacher Professional Development (45%)	Administrative Support (20%)	Ongoing Technology Support (10%)	Overall Fidelity Score (100%)
1	0 to 1	0 to 1	0 to 1	0 to 1	0 to 1	0 to 1
2	0 to 1	0 to 1	0 to 1	0 to 1	0 to 1	0 to 1
3	0 to 1	0 to 1	0 to 1	0 to 1	0 to 1	0 to 1

[a]Total score will equal the weighted average of all six components. Total school implementation score = 0 (no implementation) to 1 (full implementation). Individual component values are presented as percentages of total implementation (see Table 3.10). [b]The score weights for each category are given in parentheses.

Whereas the first part of this chapter confirms that eMINTS delivered the program as planned, the second part reports on the extent that schools implemented the program as designed. We examine the extent to which schools met the minimum technology infrastructure requirements, teachers used technology equipment, school staff attended the required training sessions and teachers completed coaching visits, and technology coordinators provided support for technology use. Data from teacher surveys, technology coordinator surveys, teacher attendance files, principal attendance files, and principal walk-through data inform the implementation fidelity determinations at the school level. (Appendix F details the calculations of component and overall fidelity scores.) We discuss the implementation of each fidelity component separately before presenting the findings on overall school and program-level fidelity. As stated earlier, because there is interest in understanding implementation of eMINTS in both years of the study, a composite metric was created to reflect school-level adherence for the duration of program implementation (i.e., Years 1 and 2 collectively).

Technology Infrastructure

The following infrastructure requirements are necessary in all eMINTS schools and require long-term maintenance:

- **Data Storage.** Each teacher and each student must have at least of 1 gigabyte of network-based storage.
- **Backup and Disaster Recovery.** A system or a procedure must be in place for backing up student, teacher, and administrative data stored on the network.
- **Internet Connectivity.** Building connectivity must be a reliable 1–2 Mbps (megabits per second) connection, and a switched network is needed from the Internet connection at the district level to the eMINTS school building.
- **Internet Security.** Schools must have the ability to unblock and approve specific sites, computers must run filtering software that is centrally managed by the district, license agreements must be current and regularly updated, and antivirus software must run on all computers.
- **Wireless Network.** Schools must have wireless network capability, and students must be able to reliably access the network.
- **Shared Folder System or Server.** A shared folder system or server must be available for students to display and share their work.

eMINTS established fidelity cut points signifying high, moderate, and low fidelity. Schools that demonstrated high fidelity on this component implemented 75 percent or more of the required elements by February of each academic year. Schools with moderate fidelity implemented between 50 percent and 74 percent of all the elements, and low-fidelity schools implemented less than 50 percent of all the elements.

To measure the extent to which technology infrastructure elements were implemented in the school, we administered a survey once a year to each school district's technology coordinator[28]

[28] The technology coordinator was asked whether schools met eMINTS basic levels of the following: data storage space, backup and disaster recovery ability, Internet connectivity, wireless connection, and shared folder system.

(one item was taken from the teacher survey). The composite score was calculated by averaging the scores from Years 1 and 2.[29] As Table 3.3 reports, in the two years of implementation, 35 of 38 treatment schools maintained high levels of technology infrastructure, two schools reported moderate levels, and one school reported low levels.

Table 3.3. School Level of Technology Infrastructure by Year

Technology Infrastructure	Years 1 and 2	
	Frequency	Percentage
Low[a]	1	2.6
Moderate	2	5.3
High	35	92.1

[a]In Year 1, no data were available for one treatment school. This school was given a score of zero in Year 1.

Technology Use

The following technology use requirements are necessary in all eMINTS schools and require long-term maintenance:

- **Student Laptops.** Students must use computers in at least one classroom session per week.
- **Teacher Laptops.** Teachers must use computers in at least one classroom session per week.
- **Interactive Whiteboards.** Classrooms (whether teachers or students) must use interactive whiteboards in at least one classroom session per week.

eMINTS schools that demonstrated high fidelity on this component reported implementing at least two of the three required elements. Schools that demonstrated moderate fidelity implemented one of the three elements, and low-fidelity schools implemented none of the elements. To measure the extent to which technology use elements were implemented in a school, we administered a survey once per year to each seventh- and eighth-grade teacher of communication arts or mathematics.[30] The composite technology use metric was created by averaging the metrics from Years 1 and 2.

According to the teacher survey responses, across two years of implementation, 22 of the 38 treatment schools (see Table 3.4) met the criteria for high levels of technology use, including the use of student laptops, teacher laptops, and interactive whiteboards. Five schools reported moderate levels of technology use, and 11 schools reported low levels of technology use.

[29] To create a comparable school level measure, the average responses of all teachers per school were used.
[30] The teacher survey asked how often teachers or their students used computers and interactive whiteboards in the classroom: never or almost never, 3–6 sessions per year, 1–3 sessions per month, 1–3 sessions per week, or every or almost every session.

Table 3.4. School-Level Technology Use by Year

Technology Use	Years 1 and 2	
	Frequency	Percentage
Low	11	28.9
Moderate	5	13.2
High	22	57.9

Teacher-Level Implementation: Teacher Professional Development

The following teacher professional development requirements are necessary in all eMINTS schools:

- Professional development participation
  - **Year 1.** Teachers must attend 27 professional development sessions or approximately 124 hours.
  - **Year 2.** Teachers must attend 20 professional development sessions or approximately 88 hours.
- Participation in coaching sessions
  - **Year 1.** Teachers must participate in 10 coaching visits conducted by eMINTS staff.
  - **Year 2.** Teachers must participate in 10 coaching visits conducted by eMINTS staff.
- Participation in portfolio review
  - **Year 1.** Not applicable.
  - **Year 2.** Teachers must submit one portfolio to eMINTS for review and rating.[31]

To measure the extent to which teacher development activities were implemented in a school, we analyzed eMINTS teacher attendance files for all professional development and coaching sessions as well as eMINTS portfolio records. eMINTS schools that demonstrated high professional development participation had teachers who collectively attended at least 85 percent of the professional development and coaching sessions and averaged 85 percent or higher on their final portfolio scores (website, WebQuest, and unit plan). Schools with total average scores ranging from 75 percent to 84 percent on these three requirements warranted a moderate fidelity rating. Professional development participation was considered low if the total average score was less than 75 percent.

Changing teacher instructional practices is the heart of the eMINTS program. For the program to be effective, eMINTS personnel must properly implement it, and classroom teachers must use the program components and resources. The professional development sessions and coaching visits are intended to prepare teachers to use eMINTS resources in an inquiry-based lesson. The

[31] Teachers must meet criteria on three portfolio components (website/portal, WebQuest, and unit plan). If rated satisfactory and the teacher completes the professional development attendance requirements, teachers can be added to the list of eMINTS certified educators posted on the DESE website.

participation of teachers in eMINTS professional development, coaching visits, and portfolio reviews are critical steps in the change process. In the following subsection, we examine the degree to which teachers participated in eMINTS professional development, coaching visits, and portfolio reviews overall before considering implementation fidelity levels at the school level.

Overall Teacher Participation in Professional Development, Coaching Visits, and Portfolio Review. Having established that the professional development, resources, and support components for eMINTS professional development were offered as planned by eMINTS personnel, this subsection looks at the extent to which teachers adhered to a major requirement of the program model—attendance at 27 professional development sessions in Year 1 and 20 sessions in Year 2. Table 3.5 provides attendance rates at the professional development sessions for all treatment teachers and by treatment group. The overall percentage of professional development sessions completed for all treatment teachers was 76.6 percent for Year 1 and 76.4 percent for Year 2. When Years 1 and 2 are combined, the overall composite is 76.5 percent.[32] Completion rates were similar in each treatment group for each year.

Table 3.5. Teacher-Level Teacher Professional Development

Metric	Years 1 and 2 Criteria	All Treatment Teachers (N = 99)	eMINTS Teachers (N = 44)	eMINTS+Intel Teachers (N = 55)
Professional development	211.5 hours	76.5%	76.6%	76.4%
Classroom visits	20 visits	83.7%	86.6%	81.4%
Portfolio review	1 review	65.9%	68.0%	64.3%

Teachers also were required to participate in 10 coaching visits in Year 1 and 10 coaching visits in Year 2. When Years 1 and 2 are combined, the overall composite is 83.7 percent. In Years 1 and 2, there was a notable drop-off in completion of the 10th classroom visit. In Year 1, completion of the first nine classroom visits ranged from 80.8 percent to 84.9 percent; however, completion of the 10th classroom visit dropped to 42.4 percent. Similarly, in Year 2, completion rates of the first eight classroom visits ranged from 88.9 percent to 97.0 percent, but the ninth classroom visit completion rate dropped to 76.8 percent, and the 10th classroom visit fell to 38.4 percent. Table 3.5 provides classroom visit attendance rates for all treatment teachers and by treatment group. Classroom visit completion rates were approximately five percentage points higher in eMINTS schools than eMINTS+Intel schools.

Starting in Year 2, teachers also were required to participate in portfolio reviews. The portfolio review includes an assessment of three subcomponents: website/portal, WebQuest, and a unit plan. Each subcomponent is given a numerical score (1, 2, or 3) and a status rating (satisfactory, unsatisfactory, or incomplete). To receive a satisfactory rating on the composite portfolio score, the average numerical score must be 2.2 or greater and two or more subcomponents cannot be

[32] In the Year 1 report, the percentage of professional development completed was reportedly higher (all treatment, 95.5 percent; eMINTS, 93.2 percent; and 1 eMINTS+Intel, 97.3 percent). In this report, Year 1, Year 2, and composite scores include data from only those teachers who participated in Year 2 of the study. Teachers in Year 1 of the study who left the study in Year 2 were removed from the analysis. New teachers in Year 2 were given zeroes for Year 1 because they did not participate in the Year 1 professional development courses.

scored less than 2.0. A weighted average of all three subcomponent scores was used to create a single composite portfolio score. Overall, in Year 2, 65.9 percent of the teachers scored at a satisfactory level on the portfolio reviews requirements.[33] Completion rates were approximately four percentage points higher in eMINTS schools than eMINTS+Intel schools (see Table 3.5).

Total School Composite Scores on Teacher Professional Development. At the school level, the composite records from Years 1 and 2 showed that teachers in 14 of 38 treatment schools achieved high levels of fidelity on the teacher professional development component across Years 1 and 2. Ten schools achieved moderate fidelity, and 14 schools achieved low fidelity (see Table 3.6).

Table 3.6. School-Level Teacher Professional Development

Teacher Professional Development	Years 1 and 2	
	Frequency	Percentage
Low	14	36.8
Moderate	10	26.3
High	14	36.8

Administrative Support

The following administrative support requirements are necessary in all eMINTS schools:

- Formal professional development participation
  - **Year 1.** Principals must attend one 2-day professional development session led by eMINTS personnel.
  - **Year 2.** Principals must attend one 1-day professional development session led by eMINTS personnel.
- Principal participation in school walk-throughs
  - **Year 1.** Principals must complete two school walk-throughs with eMINTS personnel to evaluate the school's current adoption of eMINTS technology and professional development.
  - **Year 2.**[34] Principals must complete one school walk-through with eMINTS personnel to evaluate the school's current adoption of eMINTS technology and professional development.

eMINTS schools that demonstrated high fidelity on this component were those in which an administrator (i.e., principal or assistant principal) participated in 80 percent or more of the required professional development sessions, including the two-day professional development

[33] A score of zero was given for each subcomponent that a teacher did not complete. Thirteen of the 87 teachers did not complete any of the portfolio subcomponents.

[34] One additional walk-through in the spring of Year 2 was offered but considered optional. No school chose to participate in the additional walk-through.

session and two walk-throughs (for a total of three sessions) in Year 1 and one 1-day professional development session and one walk-through (for a total of two sessions) in Year 2. Schools that demonstrated moderate fidelity were those in which an administrator participated in 60 percent of all sessions (i.e., three of the five required in both years). Administrators in low-fidelity schools participated in less than 60 percent of the required professional development sessions. To measure the extent to which administrative support elements were implemented in a school, we analyzed attendance files for the formal professional development session and the walk-through visits.[35] Attendance rates for formal professional development sessions and walk-through visits were averaged together to create the composite metric. As Table 3.7 indicates, 38 principals (100 percent) met the criteria for high implementation.

Table 3.7. Principal Professional Development

Administrative Support	Years 1 and 2	
	Frequency	Percentage
Low	0	0.0
Moderate	0	0.0
High	38	100.0

Ongoing Technology Support

eMINTS also attempts to ensure that participating schools have sufficient ongoing technical support from internal district and school-level technology staff. One aspect of this support is identifying a technology coordinator to answer questions and help teachers troubleshoot issues as they arise. eMINTS offers technology coordinators and other district or school personnel the opportunity to participate in two optional online professional development sessions in Year 1 and one optional online professional development session in Year 2. The following ongoing technology support requirements are necessary in all eMINTS schools:

- Formal professional development participation
  - **Year 1.** Technology coordinators were encouraged but not required to attend two WebEx conferences that provide information on the technological components that eMINTS requires for initial and continued use.
  - **Year 2.** Technology coordinators were encouraged but not required to attend one WebEx conference that provides information on the technological components that eMINTS requires for initial and continued use.
- Access to on-site technical assistance
  - **Years 1 and 2.** Teachers must have on-site access to technology coordinators who provide adequate troubleshooting and technical support. In addition, teachers must have support for integrating technology into their instruction.

[35] The principal walk-through survey requires ratings on the observed levels of teacher-facilitated learning, student-centered learning, teacher use of unique teaching pedagogy and learning strategies, student communities of learners, technology richness, and an assessment of student performance in the context of inquiry-based learning.

To determine the extent of implementation on the ongoing technology support component in the two years of implementation, we implemented a three-step process. First, the school received a 2 (high implementing) if a technology coordinator or support staff member from the school attended two of the three online professional development sessions, a 1 (moderately implementing) if the school technology coordinator or staff member attended one of two sessions, or a 0 (low implementing) if no one from the school attended any of the online professional development sessions.

Second, scores from two teacher survey items were used to index the extent to which teachers experienced adequate technology support and instructional support when attempting to integrate technology into their instruction. These survey items are presented by the following example:

- The following statements are about challenges you face integrating technology into classroom teaching. Please indicate the extent to which you agree or disagree with each statement.
  - I do not have adequate technical support. (The item was reverse coded.)
  - I have adequate instructional support.

Response options for these items were *completely disagree, mostly disagree, mostly agree,* and *completely agree.* The average responses for Years 1 and 2 across teachers on both items were used as the composite response for both questions.

Third, to obtain the ongoing technology support component score, we added together the averages of each item metric: formal professional development participation (0–2), the first teacher survey item (1–4), and the second teacher survey item (1–4). The totals per school could range from 2 to 10. Because participation in professional development was optional, we divided each total by 8 instead of the maximum score of 10. High-implementing schools were those with a total component score greater than or equal to .8; moderately implementing schools were those with a total component score between .6 and .79. Low-implementing schools had a total component score less than .6. Eighteen of the 38 treatment schools reported high levels of fidelity, 18 schools reported moderate levels, and two schools were low implementers (see Table 3.8).

Table 3.8. Fidelity of Ongoing Technology Support

Ongoing Technology Support	Years 1 and 2	
	Frequency	Percentage
Low	2	5.3
Moderate	18	47.4
High	18	47.4

Overall Program Level of Implementation Fidelity

To determine the overall program-level fidelity across the first two years of the program, we implemented three steps. First, for each school, we calculated a fidelity score on each core program component by aggregating individual-level data (e.g., teacher surveys) to the school level within each component. Second, we calculated an overall school-level fidelity score by taking the weighted average of the five component scores for each school (see Table 3.2 for the component weights). This gave us the total fidelity score for each school (see column labeled Overall Fidelity Score in Table 3.9). Third, we took the average of all school fidelity scores to determine a program-level fidelity score. Table 3.9 provides the fidelity scores for each school and across the 38 treatment schools. As the table indicates, 25 of 38 treatment schools (65.8 percent) implemented the eMINTS Comprehensive Program at high levels, and 13 schools (34.2 percent) implemented at moderate levels during the two years of implementation.

Overall levels of fidelity were high in three of the five core program components: technology infrastructure, administrative support, and ongoing technology support. Technology use and teacher professional development fell into the moderate range (see the last row in Table 3.9). The primary driver of implementation was teacher professional development (45 percent of overall fidelity) followed by administrative support (20 percent of overall implementation). The high and moderate levels of implementation on these two components explain why many treatment schools continue to demonstrate high levels of overall implementation fidelity.

Table 3.9. Fidelity Implementation by School

	Technology Infrastructure[a] (10%)		Technology Use (10%)		Teacher Professional Development (45%)		Administrative Support (20%)		Ongoing Technology Support (10%)		Overall School Fidelity Score (100%)		Overall School Fidelity Level
Cut Points	High	> .74	High	> .79	High	> .84	High	> .79	High	> .79	High	> .79	High
	Mod.	.50–.74	Mod.	.67–.79	Mod.	.75–.84	Mod.	.60–.79	Mod.	.60–.79	Mod.	.60–.79	Moderate
	Low	< .50	Low	< .67	Low	< .75	Low	< .60	Low	< .60	Low	< .60	Low
Treatment School ID													
1	1.00		0.50		0.88		1.00		0.71		0.87		High
2	1.00		0.83		0.87		1.00		0.75		0.90		High
3	1.00		1.00		0.20		1.00		1.02		0.64		Moderate
4	0.50		0.67		0.84		1.00		0.69		0.79		Moderate
5	1.00		0.50		0.83		1.00		0.61		0.83		High
6	1.00		0.50		0.60		1.00		0.84		0.76		Moderate
7	1.00		0.83		0.88		0.80		0.69		0.86		High
8	1.00		0.33		0.69		1.00		0.61		0.75		Moderate
9	1.00		1.00		0.85		1.00		0.69		0.90		High
10	1.00		0.83		0.87		1.00		1.13		0.94		High
11	1.00		0.33		0.83		1.00		0.88		0.85		High
12	1.00		0.17		0.87		1.00		0.75		0.83		High
13	1.00		0.83		0.82		1.00		1.14		0.92		High
14	1.00		0.17		0.83		1.00		0.70		0.81		High
15	1.00		0.83		0.60		1.00		0.91		0.79		Moderate
16	0.00		0.67		0.91		1.00		0.91		0.77		Moderate
17	1.00		0.00		0.61		1.00		0.59		0.68		Moderate
18	1.00		0.67		0.60		1.00		0.90		0.78		Moderate
19	1.00		0.83		0.83		1.00		0.65		0.87		High
20	1.00		0.83		0.60		1.00		1.04		0.81		High

	Technology Infrastructure[a] (10%)	Technology Use (10%)	Teacher Professional Development (45%)	Administrative Support (20%)	Ongoing Technology Support (10%)	Overall School Fidelity Score (100%)	Overall School Fidelity Level
Cut Points	High > .74	High > .79	High > .84	High > .79	High > .79	High > .79	High
	Mod. .50–.74	Mod. .67–.79	Mod. .75–.84	Mod. .60–.79	Mod. .60–.79	Mod. .60–.79	Moderate
	Low < .50	Low < .67	Low < .75	Low < .60	Low < .60	Low < .60	Low
21	1.00	1.00	0.87	1.00	0.84	0.92	High
22	1.00	0.83	0.75	1.00	0.78	0.85	High
23	1.00	0.67	0.59	1.00	1.05	0.79	Moderate
24	1.00	0.50	0.76	0.80	0.79	0.78	Moderate
25	1.00	1.00	0.85	1.00	1.01	0.93	High
26	1.00	1.00	0.86	0.80	0.69	0.87	High
27	1.00	0.83	0.64	1.00	0.74	0.80	Moderate
28	1.00	0.83	0.56	1.00	1.06	0.79	Moderate
29	1.00	1.00	0.73	1.00	0.53	0.83	High
30	1.00	0.33	0.85	1.00	1.13	0.88	High
31	1.00	0.33	0.93	1.00	0.69	0.87	High
32	1.00	1.00	0.75	1.00	0.84	0.87	High
33	1.00	1.00	0.89	1.00	0.81	0.93	High
34	1.00	1.00	0.58	1.00	0.69	0.78	Moderate
35	0.50	0.67	0.90	1.00	0.90	0.84	High
36	1.00	0.83	0.88	1.00	0.65	0.89	High
37	1.00	1.00	0.87	1.00	0.72	0.91	High
38	1.00	1.00	0.74	1.00	0.81	0.87	High
Overall Program Fidelity Score	**0.95**	**0.71**	**0.76**	**0.98**	**0.81**	**0.84**	**High**

[a]The score weights for each category are given in parentheses.

Part 3: Did eMINTS Staff Deliver the eMINTS+Intel Program as Designed in Year 3?

In addition to the eMINTS Comprehensive Program provided to treatment schools in Years 1 and 2, 20 eMINTS+Intel schools received the Intel Teach Program. The Intel Teach Program extends what teachers learned during the first two years of the eMINTS Comprehensive Program by providing a third year of professional development using Intel Teach courses and online tools to enhance teachers' technology integration skills and offering online resources to support teacher instructional practices aligned with the four eMINTS components. The year of Intel Teach professional development was expected to accelerate outcomes for teachers and students by providing teachers with additional professional development and Web-based tools to build on what they learned during the first two years of the program.

As part of the Intel training, eMINTS National Center staff posited that successful implementation of Intel requires the provision of (1) approximately 28 hours of online professional development to complete the Intel Teach Elements course "Project Based Approaches," (2) 12 hours of face-to-face professional development, (3) four 1-hour in-classroom coaching and mentoring visits, and (4) a unit plan review. This section focuses on the extent to which eMINTS personnel implemented the Intel program's components as expected in Year 3 of this study.

Who Participated in eMINTS+Intel Professional Development? This section focuses on teachers in 20 eMINTS+Intel schools who taught seventh- or eighth-grade communication arts or mathematics (or both grades or both subjects). Teachers had to complete the two years of eMINTS professional development to be eligible to participate in Intel.

A total of 43 teachers met the study requirements and were included in the Year 3 fidelity analysis. Of the 43 teachers who participated in Year 3 Intel program, 20 teachers (46.5 percent) taught communication arts only, 19 teachers (44.2 percent) taught mathematics only, and four teachers (9.3 percent) taught both subjects.

These 43 teachers represent 19 of the 20 eMINTS+Intel schools. One school did not have any teachers who participated[36] and was given zeros across all Year 3 school-level fidelity components as none of its teachers began the Intel model.[37]

What Intel Program Components Were Delivered? Overall, the eMINTS National Center was successful in providing professional development, coaching, and unit plan review to eMINTS+Intel schools. eMINTS provided all 20 eMINTS+Intel schools with the opportunity to participate in Year 3 professional development. Table 3.10 lists the program components measured, the delivery criteria, and the total amount of planned and delivered professional development during Year 3 of Intel implementation.

[36] Teachers could only participate in the year of Intel if they had completed the two years of eMINTS.

[37] Teacher-level analyses are based on 43 teachers in the 19 eMINTS+Intel schools that met study requirements. For the school-level analyses, all 20 eMINTS+Intel schools were included because all eMINTS+Intel schools were eligible and expected to participate.

The eMINTS National Center offered all planned professional development modules to the 43 eligible teachers in the treatment schools. The maximum number of hours of professional development delivered to eMINTS+Intel teachers in the third year ranged from 4 to 40 hours.

In addition to the professional development modules, eMINTS attempted to schedule a minimum of four coaching visits per teacher. Of the 172 one-hour coaching visits planned, 132 visits (76.7 percent) were successfully delivered. The total number of coaching sessions delivered was less than the number planned, in part, because teachers elected not to participate in all of the coaching sessions or did not respond to coaches' requests to schedule a coaching session.

The culminating project for teachers in the Intel program was to complete a unit plan. The unit plan is a measure designed by eMINTS to assess a teacher's knowledge and ability to develop a lesson demonstrating the eMINTS instructional model. Externally trained reviewers reviewed 34 of the 43 unit plans (79.1 percent). The total number of unit plans reviewed was fewer than the number planned (43) because many teachers did not provide eMINTS with reviewable unit plans.

Table 3.10. eMINTS Delivery of Professional Development Sessions and Other Program Components[a]

		eMINTS+Intel (*N* = 20)		
Teacher Component	**Criteria**	**Planned**	**Opportunity Offered**	**Successfully Delivered**
Online professional development	40 hours total per teacher (28 hours online, 12 hours in person)	1,720	1,720	1,400 (81.4%)
Coaching visits	4 one-hour visits	172	172	132 (76.7%)
Unit plan reviews	One review per teacher	43	43	34 (79.1 %)

[a]The delivery of program components is based on the number of participants who qualified for the study in Year 3. eMINTS+Intel teachers who left the study or those who did not meet the requirements for the study are excluded.

Summary of eMINTS Implementation. Across program years, eMINTS staff provided formal professional development sessions, coaching visits, and unit plan reviews to all eligible participants in all eMINTS+Intel treatment schools. eMINTS professional development was scheduled and conducted in a timely fashion. However, not all teachers participated.

Part 4: To What Extent Did Schools Implement the eMINTS+Intel Program as Planned in Year 3?

To determine the implementation fidelity levels in the eMINTS+Intel treatment schools, we worked with the eMINTS National Center staff to identify the core program components. Unlike in previous years where we weighted multiple components (technology infrastructure, technology use, teacher professional development, administrative support, and ongoing technology support), Year 3 focused completely on Intel teacher development activities. Consequently, no weights were used in assessing optimal Year 3 school performance.

In this section, we examine the extent to which schools met the minimum requirements for teacher professional development, coaching visits, and unit plan development. Data from teacher attendance files informed the implementation fidelity determinations at the school level. (Table F2 in Appendix F details the calculations of component and overall fidelity scores for Year 3.) We discuss the implementation of each fidelity component separately before presenting the findings on overall school- and program-level fidelity[38] for Intel.

Teacher Professional Development

The following teacher professional development requirements are necessary in all eMINTS+Intel schools:

- Professional Development
    - **Online.** Complete 28 hours of Intel Teach Elements course "Project Based Approaches"
    - **In Person.** Complete 12 hours of professional development sessions on Thinking With Technology (TWT) Visual Ranking Tool, TWT Seeing Reason Tool, and TWT Showing Evidence Tool
- Coaching Sessions: Participate in 4 one-hour sessions
- Unit Plan Review: Submit one unit plan to eMINTS for review and receive a satisfactory rating

To measure the extent to which teacher development activities were implemented in a school, we analyzed eMINTS teacher attendance files for all professional development and coaching sessions as well as eMINTS unit plan records. eMINTS+Intel schools that demonstrated high professional development participation had teachers who attained a total average fidelity score of 85 percent or more. Schools with total average scores ranging from 75 percent to 84 percent on three requirements—professional development, coaching, and unit plan—warranted a moderate fidelity rating. Overall professional development participation was considered low if the total average score was less than 75 percent.

Below, we elaborate on the degree to which teachers participated in eMINTS professional development, coaching visits, and unit plan reviews individually before considering implementation fidelity levels at the school level.

Professional Development. Teachers were expected to attend 40 hours of professional development—28 hours online and 12 hours in person. The overall percentage of professional development sessions completed by all eMINTS+Intel treatment teachers was 76.7 percent for online training and 67.6 percent for in-person training. On average, eMINTS+Intel treatment teachers completed 74.0 percent of the professional development sessions in Year 3.

[38] School- and program-level fidelity are based on an original school sample of 20 eMINT+Intel schools. One school within the treatment condition did not have any participating teachers. As a result, this school was given a zero for school-level fidelity.

Coaching Sessions. Teachers were expected to complete 4 one-hour coaching sessions. If teachers exceeded their participation requirements, their total coaching sessions were capped at four. In Year 3, eMINTS+Intel treatment teachers completed 68.0 percent of coaching sessions.

Unit Plan. Teachers were expected to submit one unit plan for eMINTS review and rating. Each unit plan is given a numerical score between 0 to 3 and a status rating (satisfactory, unsatisfactory, or incomplete). To receive a satisfactory rating, the average numerical score must be 2.0 or greater. Of all eMINTS+Intel treatment teachers, 79.1 percent submitted and met requirements in Year 3.

Table 3.11 provides an overview of teacher professional development, coaching visits, and unit plan participation for the eMINTS+Intel treatment group.[39]

Table 3.11. Teacher-Level Professional Development

Metric	Year 3 Criteria	eMINTS+Intel Teachers (N = 43)
Professional development	211.5 hours	74.0%
Classroom visits	20 visits	68.0%
Portfolio review	1 review	79.1%

Teachers' Composite Scores. At the teacher level, the composite records from Year 3 show that, overall, 21 of 43 teachers (48.8 percent) in eMINTS+Intel treatment schools achieved high levels of fidelity and 22 teachers (51.2 percent) achieved low fidelity (see Table 3.12). No teachers achieved moderate fidelity.

Table 3.12. Teachers' Composite Scores

School-Level Teacher Professional Development	Year 3	
	Frequency	Percentage
Low	22	51.2%
Moderate	0	0.0%
High	21	48.8%

Overall Program Level of Implementation Fidelity

To determine the overall program-level fidelity in Year 3, we averaged the means of all 20 school fidelity scores to arrive at a program-level fidelity score. Table 3.13 provides the fidelity scores for each school and across the 20 treatment schools. As the table indicates, nine of 20 treatment schools (45 percent) implemented the Intel program at high levels, two schools (10 percent) implemented at moderate levels, and nine schools (45 percent) implemented at low levels.

[39] Based on 43 teachers from 19 eMINTS+Intel schools who met study requirements.

Table 3.13. Fidelity Implementation by School

Treatment School ID[40]	Observation	School Mean	Overall School Fidelity Level
1	1	1.00	High
2	2	0.65	Low
3	1	1.00	High
4	1	1.00	High
5	5	0.56	Low
6	2	0.83	Moderate
7	1	0.52	Low
8	1	0.65	Low
9	4	0.49	Low
10	1	0.38	Low
11	2	1.00	High
12	4	0.98	High
13	7	0.60	Low
14	2	0.38	Low
15	2	0.99	High
16	1	1.00	High
17	2	0.97	High
18	2	1.00	High
19	0	0	Low
20	2	0.83	Moderate
Overall Program Fidelity Score	**43**	**0.74**	**Low**

Summary

In Years 1 and 2 of the study, across all eMINTS and eMINTS+Intel schools, eMINTS provided resources, professional development, and guidance essential to support implementation at the district, school, and classroom levels. As to whether schools implemented the eMINTS Comprehensive Program as designed in Years 1 and 2, 25 of 38 treatment schools (65.8 percent) were considered high implementers, 13 schools (34.2 percent) were moderate implementers, and 0 schools were low implementers. Many schools experienced challenges in reaching high levels of implementation on the teacher professional development component. In fact, 22 of the 38 treatment schools (57.9 percent) fell to moderate or low implementation levels on teacher professional development. In addition, many schools struggled with technology use. In reviewing technology use, 16 of the 38 treatment schools (42.1 percent) implemented at moderate or low levels. Despite the challenges in teacher professional development and technology use, overall program- level implementation at the end of Year 2 remained high (84 percent). Technology

[40] The numbering of schools in this table does not align with the numbering of schools in other tables.

infrastructure and administrative support appear to be areas of strength, with 35 and 38 treatment schools, respectively, meeting the requirements for high implementation in these areas.

In Year 3, across all eMINTS+Intel schools, the eMINTS National Center provided the planned teacher professional development, coaching sessions, and unit plan reviews. However, many schools experienced challenges in reaching high levels of implementation on teacher professional development. Only nine of 20 treatment schools (45 percent) were considered high implementers, whereas two schools (10 percent) were moderate implementers, and nine schools (45 percent) were low implementers. These figures subsequently led to a low overall program fidelity rating of 74 percent for those schools implementing the Intel program.

Please use caution when interpreting the fidelity results in the number of cases for each school. In Year 3 we examined only teachers in the eMINTS+Intel condition. This limitation, coupled with the fact that several teachers did not meet the study requirements or had attrited, reduced the overall pool of teachers. In many cases, school-level fidelity is based on one or two teacher records.

Chapter 4: Impacts on Achievement, Skills, and Engagement

In this chapter, we present findings that address questions about the overall impact of the eMINTS Comprehensive Program (eMINTS) and the eMINTS Comprehensive Program plus Intel Teach (eMINTS+Intel) on school average student achievement in mathematics and communications arts performance, 21st century skills development, and student engagement.

1. What is the impact of eMINTS and eMINTS+Intel on school mean 7th and 8th grade performance in mathematics and communication arts?
2. What is the impact of the eMINTS and eMINTS+Intel on school mean 7th and 8th grade measure of 21st century skills (including communication, technology literacy, and critical thinking)?
3. What is the impact of the eMINTS and eMINTS+Intel on school mean 7th and 8th grade measure of student engagement?

We also present exploratory findings on impact variations across student subgroups:

4. Do the impacts of eMINTS and eMINTS+Intel on school mean performance in mathematics and communication arts, 21st century skills, and engagement vary across subgroups of students (defined by IEP status, free or reduced-price lunch (FRPL) status, and racial/ethnic minority status)?

Table 4.1 summarizes the overall impact estimates on the four student outcomes. Statistically significant impacts were found in mathematics but not in the other three outcomes (communications arts, 21st century skills, and engagement). Specifically, estimated impacts of eMINTS and eMINTS+Intel on school mean mathematics achievement are positive and statistically significant, with effect sizes of 0.128 (p-value = 0.040) and 0.178 (p-value = 0.009), respectively. Both programs had positive impacts on school mean 21st century skills and negative impacts on school mean communication arts performance when compared with the control group, but none of those impacts was statistically significant. Similarly, impacts of the two programs on school mean student engagement were negative but not statistically significant. There also were no statistically significant differences in school mean student outcomes between eMINTS and eMINTS+Intel. A detailed description of estimated regression coefficients from the analysis of overall impacts on each student outcome is in Appendix H. A summary of the findings on differential impacts across student subgroups follows. Detailed findings on subgroup impacts are given in Tables I1 through I4 in Appendix I.

The estimated impacts on school mean student outcomes varied across student subgroups in only two cases, as follows[41]:

- *eMINTS+Intel vs. eMINTS*: The estimated difference in school mean *mathematics achievement* between eMINTS and eMINTS+Intel differed significantly between the IEP group and non-IEP groups (p-value = 0.005, Table X1). Specifically, among students

[41] As with all other findings in this report, statistical significance of subgroup findings were based on a 0.05 significance level.

with IEP, eMINTS+Intel schools had a statistically significantly higher mean achievement than eMINTS schools (effect size = 0.276, *p*-value = 0.001). Among non-IEP students, the difference in mean math performance between eMINTS+Intel and eMINTS schools was not statistically significant (effect size = 0.008, *p*-value = 0.877). These findings indicate that although the additional year of Intel Teach did not improve mathematics achievement of the full sample of students over and above that of eMINTS, there is evidence that it did so for IEP students.

- *eMINTS vs. Control*: The impact of eMINTS on school mean *engagement* was (marginally) significantly different between the IEP group and the non-IEP group (*p*-value = 0.050, Table X4). Specifically, eMINTS had a positive impact among students with IEP (effect size = 0.121, *p*-value = 0.145) and a negative impact among students without IEP (effect size = -0.209, *p*-value = 0.115). However, both subgroup impacts were not statistically significantly different from zero, and therefore are likely due to chance. These subgroup findings coupled with the statistically non-significant impact on the full sample indicate lack of evidence of an eMINTS effect on engagement.

There were no other statistically significant differences in impacts across subgroups (all other *p*-values in the last column of Tables X1 to X4 are greater than 0.05). However, in mathematics, there were statistically significant subgroup-specific impacts, as follows:

- *eMINTS+Intel vs. Control*: The subgroup-specific impacts of eMINTS+Intel on *mathematics* achievement were consistently positive and statistically significant across the following subgroups: IEP (effect size = 0.281) and non-IEP (effect size = 0.152), FRPL (effect size = 0.170) and non-FRPL (effect size = 0.161), White (effect size = 0.155) and Non-White (effect size = 0.200). (See Table X1 for the corresponding *p*-values). These subgroup-specific findings coupled with the statistically significant overall impact of eMINTS+Intel indicate that the program was effective in raising mathematics achievement for the full sample and was equally effective for all the subgroups examined.

- *eMINTS vs. Control*: The subgroup-specific impacts of eMINTS on *mathematics* achievement was positive and statistically significant for non-IEP, FRPL, and non-White students, and was negative but not statistically significant for IEP, non-FRPL, and White students. However, the test for differential impacts across subgroups defined by IEP, FRPL and minority status were all not statistically significant, indicating that the observed differences in the effect of eMINTS might across subgroups simply be due to chance.

Table 4.1. Year 3 Impact of eMINTS on Grades 7 and 8 School Means of Student Outcomes (2013–14)

Outcome	Mean[a]			Estimated Impact[a]			
	eMINTS+ Intel	eMINTS	Control	Impact	Standard Error	*p*-Value	Effect Size[b]
MAP 2013 mathematics *z*-scores[c]							
eMINTS vs. control	---	0.083 (*n* = 1,108)[d]	-0.047 (*n* = 823)	0.129	0.063	0.040*	0.128
eMINTS+Intel vs. control	0.122 (*n* = 1,141)	---	-0.047 (*n* = 823)	0.168	0.065	0.009*	0.178
eMINTS+Intel vs. eMINTS	0.122 (*n* = 1,141)	0.083 (*n* = 1,108)	---	0.039	0.063	0.532	0.037
MAP 2013 communication arts *z*-scores[c]							
eMINTS vs. control	---	0.052 (*n* = 1,208)	0.087 (*n* = 871)	-0.035	0.050	0.484	-0.035
eMINTS+Intel vs. control	0.011 (*n* = 1,216)	---	0.087 (*n* = 871)	-0.076	0.051	0.137	-0.077
eMINTS+Intel vs. eMINTS	0.011 (*n* = 1,216)	0.052 (*n* = 1,208)	---	-0.041	0.047	0.384	-0.041
21st century skills scale scores[e]							
eMINTS vs. control	---	272.1 (*n* = 827)	266.0 (*n* = 683)	6.123	6.700	0.361	0.103
eMINTS+Intel vs. control	270.4 (*n* = 866)	---	266.0 (*n* = 683)	4.374	6.527	0.503	0.059
eMINTS+Intel vs. eMINTS	270.4 (*n* = 866)	272.1 (*n* = 827)	---	-1.749	6.84	0.798	-0.038

Student engagement scores[e]							
eMINTS vs. control	---	0.629 (n = 868)[d]	0.788 (n = 659)	-0.160	0.115	0.166	-0.176
eMINTS+Intel vs. control	0.667 (n = 751)	---	0.788 (n = 659)	-0.121	0.112	0.280	-0.149
eMINTS+Intel vs. eMINTS	0.667 (n = 751)	0.629 (n = 868)	---	0.038	0.037	0.737	0.038

Note. From authors' analysis based on Year 3 implementation (2013–14) data from the study districts and DESE.
[a]Means and impacts were regression-adjusted to account for clustering of students within schools, block effects, and baseline student, teacher, and school characteristics. They were weighted by the number of schools in each block. [b]Effect sizes were calculated separately by block and then pooled into an overall effect size weighted by the number of schools in each block. Block-specific effect sizes were computed using standardized mean differences (Hedges's g).
p-values are from a two-tailed test of the null hypothesis of equality means. [c]Because the student analytic samples are pooled samples of seventh and eighth graders, MAP scale scores were converted into a common metric by standardizing the scores separately by year (i.e., 2011 for the pretest and 2014 for the posttest), grade, subject (mathematics and communication arts), and randomization block. Specifically, each test score was converted into a z-score by subtracting from the test score the average score for a particular year, grade, subject, and block and then dividing by the standard deviation for a particular year, grade, subject, and block. [d]n gives the sample sizes used in the analysis. For the MAP mathematics and communication arts analyses, n includes all eligible students with nonmissing outcomes. For the 21st century skills and student engagement analyses, n includes all eligible students who provided consent to participate and have nonmissing outcomes. [e]Student engagement scores are Rasch logit ability measures from a Rasch analysis of students' responses to the spring 2014 student engagement survey.

Chapter 5: Impacts on Teacher Practices

In this chapter, we analyze teacher survey responses and observations of teachers to assess the impact of eMINTS professional development on instructional practices. Specifically, this chapter answers the following exploratory research question: What is the impact of eMINTS professional development on seventh- and eighth-grade teachers' instructional practices?

Teacher Survey Results

According to our analyses of teacher self-report data (from the spring 2014 teacher survey) in Table 5.1, no statistically significant differences arose between treatment and control teachers for a community of learners after Year 3. The eMINTS Comprehensive Program and the eMINTS Comprehensive Program plus Intel Teach teachers reported significantly higher levels of inquiry-based learning (effect sizes of 0.73 and 0.96, respectively) and technology integration (effect sizes of 1.43 and 1.56, respectively) relative to the control group. The eMINTS Comprehensive Program plus Intel Teach group also reported significantly higher levels of high-quality lesson design than their control group counterparts. However, no significant differences for any domain were found between the eMINTS Comprehensive Program and the eMINTS Comprehensive Program plus Intel Teach. Appendix J gives the estimated regression coefficients for teacher survey outcomes.

Table 5.1. Overall Impact of eMINTS on Four Teacher Survey Outcomes in Implementation Year 3 (2013–14)

Outcome[a]	Mean[b]			Estimated Impact[b]			
	eMINTS+ Intel	eMINTS	Control	Impact	Standard Error	*p*-Value	Effect Size[b]
Community of learners							
eMINTS vs. control	---	2.115 (n = 24)[d]	1.410 (n = 38)	0.705	0.860	0.412	0.237
eMINTS+Intel vs. control	2.561 (n = 35)	---	1.410 (n = 38)	1.151	0.727	0.113	0.346
eMINTS+Intel vs. eMINTS	2.561 (n = 35)	2.115 (n = 24)	---	0.445	0.837	0.595	0.126
Inquiry-based learning							
eMINTS vs. control	---	-0.042 (n = 24)	-0.632 (n = 37)	0.591	0.135	0.000*	0.730
eMINTS+Intel vs. control	0.046 (n = 35)	---	-0.632 (n = 37)	0.679	0.114	0.000*	0.959
eMINTS+Intel vs. eMINTS	0.046 (n = 35)	-0.042 (n = 24)	---	0.088	0.131	0.501	0.122
High-quality lesson design							
eMINTS vs. control	---	0.565 (n = 24)	0.211 (n = 37)	0.355	0.237	0.135	0.371
eMINTS+Intel vs. control	0.702 (n = 35)	---	0.211 (n = 37)	0.491	0.201	0.015*	0.503
eMINTS+Intel vs. eMINTS	0.702 (n = 35)	0.565 (n = 24)	---	0.136	0.230	0.554	0.119

Technology integration							
eMINTS vs. control	---	0.568 (n = 24)	-0.097 (n = 37)	0.665	0.100	0.000*	1.431
eMINTS+Intel vs. control	0.573 (n = 35)	---	-0.097 (n = 37)	0.671	0.084	0.000*	1.551
eMINTS+Intel vs. eMINTS	0.573 (n = 35)	0.568 (n = 24)	---	0.006	0.097	0.952	0.012

Note. From authors' analysis based on Year 3 implementation (2013–14) data from the study districts and DESE.
[a]Teacher outcomes are Rasch logit ability measures from a Rasch analysis of teachers' responses to the spring 2014 teacher survey items that fall under the four constructs: community of learners (12 items), inquiry-based learning (43 items for mathematics and 49 items for communication arts), high-quality lesson design (9 items), and technology integration (96 items). [b]Means and impacts were regression-adjusted using ordinary least squares to account for block effects and baseline teacher and classroom characteristics. Because of the relatively small number of teachers within each block, a constant treatment effect was assumed across blocks. [c]Effect sizes were calculated using standardized mean differences (Hedges's *g*). *p*-values are from a two-tailed test of the null hypothesis of equality of means. [d]*n* gives the sample sizes used in the analysis. It includes all eligible mathematics and communication arts teachers who provided consent to participate and have nonmissing outcomes.
*Statistically significantly different from zero at the .05 level.

Classroom Observation Results

As highlighted in Table 5.2, we found significant positive results for the eMINTS Comprehensive Program and the eMINTS Comprehensive Program plus Intel Teach teachers versus control teachers for each domain observed. Each domain had medium effect sizes—community of learners (0.50 and 0.52, respectively), inquiry-based learning (0.68 and 0.56, respectively), and technology integration (0.58 and 0.78, respectively). There were no significant differences between the eMINTS Comprehensive Program and the eMINTS Comprehensive Program plus Intel Teach, however. The observation instrument did not include a high-quality lesson design domain. Appendix K gives the estimated regression coefficients for classroom observation outcomes.

Table 5.2. Overall Impact of eMINTS on Three Classroom Observation Outcomes in Implementation Year 3 (2013–14)

Outcome[a]	Mean[b]			Estimated Impact[b]			
	eMINTS+ Intel	eMINTS	Control	Impact	Standard Error	*p*-Value	Effect Size[b]
Community of learners							
eMINTS vs. control	---	2.404 (n = 33)[d]	1.692 (n = 41)	0.713	0.340	0.036*	0.498
eMINTS+Intel vs. control	2.506 (n = 43)	---	1.692 (n = 41)	0.815	0.301	0.007*	0.521
eMINTS+Intel vs. eMINTS	2.506 (n = 43)	2.404 (n = 33)	---	0.102	0.329	0.757	0.069
Inquiry-based learning							
eMINTS vs. control	---	-0.425 (n = 33)	-1.776 (n = 41)	1.351	0.526	0.010*	0.684
eMINTS+Intel vs. control	-0.534 (n = 43)	---	-1.776 (n = 41)	1.242	0.465	0.008*	0.555
eMINTS+Intel vs. eMINTS	-0.534 (n = 43)	-0.425 (n = 33)	---	-0.109	0.506	0.083	-0.045
Technology integration							
eMINTS vs. control	---	0.387 (n = 33)	-0.186 (n = 41)	0.573	0.210	0.006*	0.579
eMINTS+Intel vs. control	0.592 (n = 41)	---	-0.186 (n = 41)	0.779	0.182	0.000*	0.777
eMINTS+Intel vs. eMINTS	0.592 (n = 41)	0.387 (n = 33)	---	0.206	0.203	0.311	0.234

Note. From authors' analysis based on Year 3 implementation (2013–14) data from the study districts and DESE.

[a]Teacher outcomes are Rasch logit ability measures from a Rasch analysis of items in the spring 2014 classroom observation protocol that belong to three constructs: a community of learners (6 items), inquiry-based learning (5 items), and technology integration (15 items). There were no items in the classroom observation protocol measuring the high-quality lesson design construct, so this construct was measured through only the teacher survey. [b]Means and impacts were regression adjusted using ordinary least squares to account for block effects and baseline teacher and classroom characteristics. Because of the relatively small number of teachers within each block, a constant treatment effect was assumed across blocks. [c]Effect sizes were calculated using standardized mean differences (Hedges's *g*). *p*-values are from a two-tailed test of the null hypothesis of equality of eMINTS and control means. [d]*n* gives the sample sizes used in the analysis. It includes all eligible mathematics and communication arts teachers who provided consent to participate and have nonmissing outcomes.
*Statistically significantly different from zero at the .05 level.

References

Allen, J. P., Pianta, R. C., Gregory, A., Mikami, A. Y., & Lun, J. (2011). An interaction-based approach to enhancing secondary school instruction and student achievement. *Science, 333*(6045), 1034–1037.

Barron, B., & Darling-Hammond, L. (2008). *Teaching for meaningful learning: A review of research on inquiry-based and cooperative learning.* San Rafael, CA: Edutopia. Retrieved from http://www.edutopia.org/pdfs/edutopia-teaching-for-meaningful-learning.pdf (Excerpted from Darling-Hammond, L., Barron, B., Pearson, P. D., Schoenfeld, A. H., Stage, E. K., Zimmerman, T. D., et al. (Eds.), *Powerful learning: What we know about teaching for understanding*. San Francisco, CA: John Wiley & Sons.)

Bellanca, J., & Brandt, R. (2010). *21st century skills: Rethinking how students learn.* Bloomington, IN: Solution Tree.

Belland, B., Glazewski, K., & Richardson, J. (2011). Problem-based learning and argumentation: Testing a scaffolding framework to support middle school students' creation of evidence-based arguments. *Instructional Science, 39*(5)*,* 667–694.

Bloom, H. S. (2006). *The core analytics of randomized experiments for social research* (MDRC Working Papers on Research Methodology). New York, NY: MDRC. Retrieved from http://www.mdrc.org/sites/default/files/full_533.pdf

Brandt, W.C., Meyers, C., Molefe, A., Dhillon, S., & Zhu, B. (2013). *The impact of eMINTS professional development on teacher instruction and student achievement: Year 2 report*. Washington, DC: American Institutes for Research.

Buck Institute for Education. (2007). *National survey of high school reform and project based learning.* Novato, CA: Author.

Chinn, C., & Hung, C. (2007, April). *Learning to reason about the methodology of scientific studies: A classroom experiment in the middle school.* Paper presented at the annual meeting of the American Educational Research Association, Chicago, IL.

Condon, C., Dawson, M., Molefe, A., & Swanlund, A. (2009). *Results of the Rasch analysis of items based on the ISTE-NETS-S technology literacy standards for 5th and 8th grades* (pretest). Naperville, IL: Learning Point Associates.

Consortium on Chicago School Research at the University of Chicago. (2007a). *Elementary student survey form.* Chicago, IL: University of Chicago. Retrieved from http://ccsr.uchicago.edu/downloads/42592007_elem_student_codebook.pdf

Consortium on Chicago School Research at the University of Chicago. (2007b). *High school teacher survey form.* Chicago, IL: University of Chicago. Retrieved from http://ccsr.uchicago.edu/downloads/44132007_hs_teacher_codebook.pdf

Cook, T. D., Church, M. B., Ajanaku, S., Shadish, W. R. Jr., Jeong-Ran, K., & Cohen, R. (1996). The development of occupational aspirations and expectations among inner-city boys. *Child Development, 67*(6), 3368–3385.

CTB/McGraw-Hill. (2012). *Missouri Assessment Program: Grade-level assessments* (Technical report 2012: Final). Monterey, CA: Author. Retrieved from http://dese.mo.gov/divimprove/assess/tech/documents/asmt-gl-2012-tech-report.pdf

Drake, S. M. (2007). *Creating standards-based integrated curriculum: Aligning curriculum, content, assessment, and instruction.* Thousand Oaks, CA: Corwin.

eMINTS National Center. (2009). *eMINTS teacher technology literacy skill survey.* Columbia, MO: University of Missouri, eMINTS National Center.

Finkelstein, N., Hanson, T., Huang, C.-W., Hirschman, B., & Huang, M. (2010). *Effects of problem based economics on high school economics instruction* (NCEE 2010-4002). Washington, DC: U.S. Department of Education, Institute of Education Sciences, National Center for Education Evaluation and Regional Assistance. Retrieved from http://ies.ed.gov/ncee/edlabs/regions/west/pdf/REL_20104012.pdf

Geier, R., Blumenfeld, P., Marx, R., Krajcik, J., Fishman, B., Soloway, E., et al. (2008). Standardized test outcomes for students engaged in inquiry-based science curricula in the context of urban reform. *Journal of Research in Science Teaching, 45*(8), 922–939.

Guthrie, J. T., Wigfield, A., & VonSecker, C. (2000). Effects of integrated instruction on motivation and strategy use in reading. *Journal of Educational Psychology*, *92*(2), 331–341.

Hager, D. (2004). *An exploratory study of the black-white achievement gap in eMINTS-Student MAP results.* Columbia, MO: Office of Social and Economic Data Analysis.

Hamilton, L., Halverson, R., Jackson, S. S., Mandinach, E., Supovitz, J. A., & Wayman, J. C. (2009). *Using student achievement data to support instructional decision making* (NCEE 2009-4067). Washington, DC: U.S. Department of Education, Institute of Education Sciences, National Center for Education Evaluation and Regional Assistance. Retrieved from http://ies.ed.gov/ncee/wwc/pdf/practice_guides/dddm_pg_092909.pdf

Heritage, M. (2007). Formative assessment: What do teachers need to know and do? *Phi Delta Kappan, 89*(2), 140–145.

Herlihy, C. (2007). *State and district-level supports for successful transition into high school graduation rates.* Washington, DC: National High School Center. Retrieved from http://www.betterhighschools.org/docs/NHSC_PolicyBrief_TransitionsIntoHighSchool.pdf

Hmelo-Silver, C., Duncan, R., & Chinn, C. (2007). Scaffolding and achievement in problem-based and inquiry learning: A response to Kirschner, Sweller, and Clark (2006). *Educational Psychologist, 42*(2), 99–107.

Houston, J. S., & Ferstl, K. (2007, May–June). *Future skill demands, from a corporate consultant perspective.* Presentation at the National Academies Workshop on Research Evidence Related to Future Skill Demands, Washington, DC.

Hunter, L., & Greever-Rice, T. (2007). *Analysis of 2005 MAP results for eMINTS students.* Columbia, MO: Office of Social and Economic Data Analysis.

Huston, A., Miller, C., Richburg-Hayes, L., Duncan, G., Eldred, C., Weisner, T. S., et al. (2003). *New hope for families and children: Five-year results of a program to reduce poverty and reform welfare.* New York, NY: MDRC.

Institute of Education Sciences. (2010). *The nation's report card: Mathematics 2009: National Assessment of Educational Progress* (NCES 2010–451). Washington, DC: U.S. Department of Education, Institute of Education Sciences. Retrieved from http://nces.ed.gov/nationsreportcard/pdf/main2009/2010451.pdf

Institute of Education Sciences. (2011). *What Works Clearinghouse reviewer training manual.* Washington, DC: U.S. Department of Education, Institute of Education Sciences, National Center for Education Evaluation and Regional Assistance.

Intel Corporation. (2006). *Intel Teach Program Essentials Course: Impact survey.* Santa Clara, CA: Author. Retrieved from http://download.intel.com/education/EvidenceOfImpact/IntelTeach_ImpactSurvey.pdf

International Society for Technology in Education. (2014). *NETS for students*. Washington, DC: Author. Retrieved from http://www.iste.org/docs/pdfs/20-14_ISTE_Standards-S_PDF.pdf

Johnson, D. W., & Johnson, R. (2005). New developments in social interdependence theory. *Psychology Monographs, 131*(4), 285–358.

Johnson, D. W., & Johnson, R. (2009). Educational psychology success story: Social interdependence theory and cooperative learning. *Educational Researcher, 38*(5), 365–379.

Joyce, B., & Showers, B. (1995). *Student achievement through staff development: Fundamentals of school renewal* (2nd ed.). White Plains, NY: Longman.

Kohn, A. (2006). *The homework myth: Why our kids get too much of a bad thing*. Cambridge, MA: Da Capo Press.

Lawless, K. A., & Pelligrino, J. W. (2007). Professional development in integrating technology into teaching and learning: Knowns, unknowns, and ways to pursue better questions and answers. *Review of Educational Research, 77*(4), 575–614.

Lembke, M., & Gonzalez, P. (2006). *U.S. student and adult performance on international assessments of educational achievement* (NCES 2006-073). Washington, DC: U.S.

Department of Education, Institute of Education Sciences, National Center for Education Statistics. Retrieved from http://nces.ed.gov/pubs2006/2006073.pdf

Lewis, C., Perry, R., & Hurd, J. (2004). A deeper look at lesson study. *Educational Leadership, 61*(5), 18–23.

Li, Q., & Ma, X. (2010). A meta-analysis of the effects of computer technology on school students' mathematics learning. *Educational Psychology Review, 22*(3), 215–243.

Mac Iver, M. A., & Mac Iver, D. J. (2009). *Beyond the indicators: An integrated school-level approach to dropout prevention.* Arlington, VA: Mid-Atlantic Equity Center, George Washington University Center for Equity and Excellence in Education. Retrieved from http://diplomasnow.org/wp-content/uploads/2013/06/dropout-report-8-11-09.pdf

Martin, W., & Shulman, S. (2006). *Intel Teach Essentials instructional practices and classroom use of technology survey report.* New York, NY: Education Development Center, Center for Children and Technology. Retrieved from http://download.intel.com/education/EvidenceOfImpact/IntelTeach-Essentials-Sep2006-Report.pdf

Martin, W., Strother, S., Beglau, M., Bates, L., & Reitzes, T. (2010). Connecting instructional technology professional development to teacher and student outcomes. *Journal of Research on Technology in Education, 43*(1), 55–76.

Martin, W., Strother, S., & Reitzes, T. (2009). *eMINTS 2009 program evaluation report: An analysis of the persistence of program impact on student achievement.* New York, NY: Education Development Center, Center for Children and Technology. Retrieved from http://www.emints.org/wp-content/uploads/2013/09/programeval2009.pdf

Martin, W., Strother, S., Weatherholt, T., & Dechaume, M. (2008). *eMINTS program evaluation report: An investigation of program fidelity and its impact on teacher mastery and student achievement.* New York, NY: Education Development Center, Center for Children and Technology. Retrieved from http://www.emints.org/wp-content/uploads/2013/09/programeval.pdf

Maxwell, N., Mergendoller, J., & Bellisimo, Y. (2005). Problem-based learning and high school macroeconomics: A comparative study of instructional methods. *Journal of Economic Education, 36*(4), 315–331.

McClure, C., & Reeves, C. (2004). *Rural teacher recruitment and retention: Review of the research and practice literature.* Charleston, WV: Appalachian Educational Laboratory. Retrieved from http://eric.ed.gov/?id=ED484967

McKnight, P. E., McKnight, K. M., Sidani, S., & Figueredo, A. J. (2007). *Missing data: A gentle introduction.* New York, NY: Guilford Press.

MetLife. (2011). *The MetLife survey of the American teacher: Preparing students for college and careers.* New York, NY: Author. Retrieved from http://eric.ed.gov/PDFS/ED519278.pdf

Middleton, M. J., & Midgley, C. (2002). Beyond motivation: Middle school students' perceptions of press for understanding in math. *Contemporary Educational Psychology, 27*(3), 373–391.

Midgley, C., Maehr, M. L., Hruda, L. Z., Anderman, E., Anderman, L., Freeman, K. E., et al. (2000). *Manual for the Patterns of Adaptive Learning Scales (PALS).* Ann Arbor, MI: University of Michigan.

Minner, D., Levy, A., & Century, J. (2010). Inquiry-based science instruction—what is it and does it matter? Results from a research synthesis years 1984 to 2002. *Journal of Research in Science Teaching, 47*(4), 474–496.

Monk, D. (2007). Recruiting and retaining high-quality teachers in rural areas. *Future of Children, 17*(1), 155–174.

National Research Council. (2000). *Inquiry and the National Science Education Standards.* Washington DC: National Academies Press.

National Research Council. (2003). *How people learn: Brain, mind, experience, and school*. Washington, DC: National Academy Press.

Northwest Regional Educational Laboratory. (2004). *Observation protocol for technology integration in the classroom.* Portland, OR: Author. Retrieved from http://www.netc.org/images/pdf/observation.rubric.pdf

Office of Educational Technology. (2012). *Understanding the implications of online learning for educational productivity.* Washington, DC: U.S. Department of Education, Office of Educational Technology. Retrieved from http://www2.ed.gov/about/offices/list/os/technology/implications-online-learning.pdf

Office of Social and Economic Data Analysis. (2001a). *Final results from the eMINTS teacher survey.* Columbia, MO: Author.

Office of Social and Economic Data Analysis. (2001b). *A general typology of eMINTS lessons.* Columbia, MO: eMINTS National Center.

Office of Social and Economic Data Analysis. (2002). *Analysis of 2001 MAP results for eMINTS students.* Columbia, MO: Author.

Office of Social and Economic Data Analysis. (2003a). *Analysis of 2002 MAP results for eMINTS students.* Columbia, MO: Author.

Office of Social and Economic Data Analysis. (2003b). *Assessing instructional practices in eMINTS classrooms.* Columbia, MO: Author.

Office of Social and Economic Data Analysis. (2003c). *Classroom climate, instructional practices and effective behavior management in eMINTS expansion classrooms.* Columbia, MO: Author.

Office of Social and Economic Data Analysis. (2004). *Analysis of 2003 MAP results for eMINTS students.* Columbia, MO: Author.

Office of Social and Economic Data Analysis. (2005). *Analysis of 2004 MAP results for eMINTS students.* Columbia, MO: Author.

Office of Special Education Programs. (2007). *Special education elementary longitudinal study.* Menlo Park, CA: SRI International.

Olson, L. (2006). Skills for work, college readiness are found comparable. *Education Week, 25*(36), 1–2.

Partnership for 21st Century Skills. (2006). *Professional development for the 21st century.* Washington, DC: Author. Retrieved from http://www.p21.org/storage/documents/ProfDev.pdf

Pianta, R. C., La Paro, K. M., & Hamre, B. K. (2008). *Classroom Assessment Scoring System* (*CLASS).* Baltimore, MD: Brookes.

Pulakos, E. D., Arad, S., Donovan, M. A., & Plamondon, K. E. (2000). Adaptability in the workplace: Development of a taxonomy of adaptive performance. *Journal of Applied Psychology, 85,* 612–624.

Puma, M. J., Olsen, R. B., Bell, S. H., & Price, C. (2009). *What to do when data are missing in group randomized controlled trials* (NCEE 2009-0049). Washington, DC: U.S. Department of Education, Institute of Education Sciences, National Center for Education Evaluation and Regional Assistance. Retrieved from http://ies.ed.gov/ncee/pdf/20090049.pdf

Rasch, G. (1980). *Probabilistic models for some intelligence and attainment tests* (Exp. ed.). Chicago, IL: University of Chicago Press.

Rohrbeck, C., Ginsburg-Block, M., Fantuzzo, J., & Miller, T. (2003). Peer-assisted learning interventions with elementary school students: A meta-analytic review. *Journal of Educational Psychology, 95*(2), 240–257.

Sadler, D. R. (1989). Formative assessment and the design of instructional systems. *Instructional Science, 18*(2), 119–144.

Schroeder, C., Scott, T., Tolson, H., Huang, T., & Lee, Y. (2007). *Journal of Research in Science Teaching, 44*(10), 1436–1460.

Shavelson, R. J. (2006). *On the integration of formative assessment in teaching and learning with implications for teacher education.* Stanford, CA: Stanford Education Assessment

Laboratory. Retrieved from http://www.stanford.edu/dept/SUSE/SEAL/Reports_Papers/On%20the%20Integration%20of%20Formative%20Assessment_Teacher%20Ed_Final.doc

Shear, L., Novais, G., & Moorthy, S. (2010). *Innovation teaching and learning research: Executive summary.* Microsoft. Retrieved from http://www.itlresearch.com/images/stories/reports/ITL%20Research%20Pilot%20Year%20Executive%20Summary%20Oct%202010%20F.pdf

Simons, K. D., & Klein, J. D. (2007). The impact of scaffolding and student achievement levels in a problem based learning environment. *Instructional Science, 35*(1), 41–72.

Slavin, R. (2010). Co-operative learning: What makes groupwork work? In Centre for Educational Research and Innovation (Ed.), *The nature of learning: Using research to inspire practice* (pp. 161–178). Paris, France: Organisation for Economic Co-operation and Development.

Strange, M., Johnson, J., Showalter, D., & Klein, R. (2012). *Why rural matters 2011–2012: The condition of rural education in the 50 states.* Washington, DC: Rural School and Community Trust. Retrieved from http://files.ruraledu.org/wrm2011-12/WRM2011-12.pdf

Strother, S., Martin, W., & Dechaume, T. (2006). *Analysis of 2006 MAP results for eMINTS and non-eMINTS students.* New York, NY: Education Development Center, Center for Children and Technology. Retrieved from http://www.emints.org/wp-content/uploads/2011/07/2006MAP.pdf

Tamim, R. M., Bernard, R. M., Borokhovski, E., Abrami, P. C., & Schmid, R. F. (2011). What forty years of research says about the impact of technology on learning: A second-order meta-analysis and validation study. *Review of Educational Research, 81*(1), 4–28.

Taraban, R., Box, C., Myers, R., Pollard, R., & Bowen, C. W. (2007). Effects of active-learning experiences on achievement, attitudes, and behaviors in high school biology. *Journal of Research in Science Teaching, 44*(7), 960–979.

Tharp, S. (2004). *Classroom climate, instructional practice, and mentoring experience in the eMINTS expansion classrooms: A two-year study.* Columbia, MO: Office of Social and Economic Data Analysis. Retrieved from http://www.emints.org/wp-content/uploads/2013/09/expansionclassroom-climate.pdf

Tharp, S. (2006). *Principals exhibiting student-centered leadership correlates with positive school climate indicators in Title I eMINTS schools and higher MAP math scores in Title I eMINTS classrooms.* Columbia, MO: Office of Social and Economic Data Analysis.

Tharp, S., Bickford, A., & Hager, D. (2005). *The effects of eMINTS enrollment on Title I students.* Columbia, MO: Office of Social and Economic Data Analysis.

Thomas, J. (2000). *A review of research on project-based learning.* San Rafael, CA: Autodesk Foundation.

Tomlinson, C., Brighton, C., Hertbery, H., Callahan, C., Moon, T., Brimijjoin, K., et al. (2003). Differentiating instruction in response to student readiness, interest, and learning profiles in academically diverse classrooms: A review of literature. *Journal for the Education of the Gifted, 27*(2–3), 119–145.

West Virginia Department of Education. (n.d.). *Classroom observation instrument scoring rubric.* Charleston, WV: Author. Retrieved from http://wvde.state.wv.us/teach21/documents/ClassroomObservationInstrumentScoringRubric.doc

Wiggins, G., & McTighe, J. (1998). *Understanding by design.* Upper Saddle River, NJ: Prentice-Hall.

Wiliam, D., & Thompson, M. (2007). Integrating assessment with instruction: What will it take to make it work? In C. A. Dwyer (Ed.), *The future of assessment: Shaping teaching and learning* (pp. 53–82). Mahwah, NJ: Erlbaum.

Wisconsin Center for Education Research. (2004). *Surveys of enacted curriculum: State collaborative on assessments and student standards* (K–8 mathematics and English language arts surveys; website). Madison, WI: Author. Retrieved from https://secure.wceruw.org/seconline/secWebReference.htm

Wright, B. D., & Masters, G. N. (1982). *Rating scale analysis: Rasch measurement.* Chicago, IL: MESA Press.

Zimmerman, B. J., Bandura, A., & Martinez-Pons, M. (1992). Self-motivation for academic attainment: The role of self-efficacy beliefs and personal goal setting. *American Educational Research Journal, 29*(3), 663–676.

Appendix A. eMINTS Professional Development Timeline

Tables A1 through A3 present the delivery timeline for eMINTS professional development and eMINTS+Intel professional development. The timeline is based on a history of 10 years of professional development delivery and follows the sequence that groups have used in the program.

Table A1. Year 1 eMINTS+Intel Professional Development Timeline

Year 1 Professional Development Activity Titles	For Groups 1 and 2 From Fall 2011 to Spring 2012									
	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb.	Mar.	Apr.	May
Getting Started	X									
Transforming Learning With Technology	X									
Constructivism	X									
Questioning Strategies	X									
Cooperative Learning		X								
Effective Uses of Productivity Tools		X								
Peer Visit		X								
Interactive Whiteboards			X							
Finding and Organizing Internet Resources			X							
Evaluating and Using Internet Resources				X						
Using Presentations in Inquiry-Based Learning				X						
Learning Communities and Technology				X						
Planning a Classroom Website					X					
Inquiry-Based Lessons					X					
Intro to WebQuests						X				
Visual Literacy						X				
Creating and Editing Digital Images						X				
Creating a Classroom Website							X			
Tools for Thinking							X			
Website Work Session							X			

Year 1 Professional Development Activity Titles	For Groups 1 and 2 From Fall 2011 to Spring 2012									
	Aug.	**Sept.**	**Oct.**	**Nov.**	**Dec.**	**Jan.**	**Feb.**	**Mar.**	**Apr.**	**May**
Modifying a WebQuest								X		
Collaboration Session 1—Troubleshooting								X		
Classroom Communication									X	
Collaboration Session 2									X	
Connections Between Inquiry-Based Teaching and State Assessment										X
File Management										X
Writing a WebQuest										X
eMINTS Coaching Visits	X	X	X	X	X	X	X	X	X	X

Table A2. Year 2 eMINTS+Intel Professional Development Timeline

Year 2 Professional Development Activity Titles	For Groups 1 and 2 From Fall 2011 to Spring 2012									
	Aug.	**Sept.**	**Oct.**	**Nov.**	**Dec.**	**Jan.**	**Feb.**	**Mar.**	**Apr.**	**May**
Classroom Management	X									
Website Enhancement		X								
Working With Authentic Data		X								
Peer Visit		X								
Assessment			X							
Interdisciplinary Teaching and Learning			X							
Collaboration Session 1			X							
Revisiting WebQuests				X						
Collaboration Session 2				X						
Mapping a Multimedia Project					X					
Creating Multimedia Products					X					
Assessing Student Technology Products						X				
Lesson Design 1						X				
Lesson Design 2							X			

Year 2 Professional Development Activity Titles	For Groups 1 and 2 From Fall 2011 to Spring 2012									
	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb.	Mar.	Apr.	May
Collaboration Session 3							X			
eMINTS Winter Conference								X		
Online Projects								X		
Lesson Design 3									X	
Collaboration Session 4									X	
Planning for Year 2 Professional Development										X
eMINTS Coaching Visits	X	X	X	X	X	X	X	X	X	X

Table A3. Year 3 eMINTS Plus Intel Professional Development Timeline

Year 3 Professional Development Activity Titles	For Group 2 Only Beginning Fall 2013–Spring 2014									
	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb.	Mar.	Apr.	May
Targeting Thinking	X									
Designing Standards-Based Projects		X								
Creating Curriculum-Framing Questions to Support Thinking Skills			X							
Planning Student-Centered Assessment				X						
Visual Ranking Tool						X	X			
Seeing Reason Tool							X			
Showing Evidence Tool								X	X	
Completing Unit Plan, Evaluation, and Feedback										X

Appendix B. eMINTS Instructional Model: Component Definitions

eMINTS professional development is designed to help teachers implement the eMINTS instructional model, which is based on four components: inquiry-based instruction, high-quality lesson design, a community of learners, and technology integration. More detailed descriptions of the four components follow.

Inquiry-Based Instruction

A core goal of the eMINTS professional development model is to increase teachers' capacity to use student-centered, inquiry-based instructional practices in the classroom (Minner, Levy, & Century, 2010; Schroeder, Scott, Tolson, Huang, & Lee, 2007). Inquiry-based learning activities require students to construct knowledge through meaningful investigations that require higher-order thinking skills, such as reasoning analysis, judgment, and decision making (Barron & Darling-Hammond, 2008; National Research Council, 2000). Instruction is typically framed around open-ended questions that are authentic and relevant to life outside the classroom (Barron & Darling-Hammond, 2008; Hmelo-Silver, Duncan, & Chinn, 2007). Investigations are intended to spark curiosity and encourage students to probe for deeper meaning (Barron & Darling-Hammond, 2008; Maxwell, Mergendoller, & Bellisimo, 2005; Schroeder et al., 2007). As part of their classroom pursuits, students typically have to draw on knowledge and skills from more than one academic discipline (Thomas, 2000). In addition, learners may be responsible for designing their own investigations, collecting relevant data, making inferences, drawing logical conclusions, supporting conclusions with valid information, and communicating their findings (Barron & Darling-Hammond, 2008; Minner et al., 2010; Taraban, Box, Myers, Pollard, & Bowen, 2007). Teachers strive to ensure that (1) the learning process is individualized to meet different learning needs, (2) new material is connected to students' past experiences and current understanding of concepts, and (3) students have access to a variety of appropriate learning tools and resources (Schroeder et al., 2007). As a result, the teacher's role evolves from being a giver of knowledge to a facilitator of learning (Barron & Darling-Hammond, 2008; Finkelstein, Hanson, Huang, Hirschman, & Huang, 2010).

eMINTS professional development illustrates for teachers how marrying standards-based instruction with interdisciplinary inquiry-based learning improves student performance. Teachers progress through stages from direct instruction to guided inquiry and finally to open inquiry, using technology to skillfully guide students toward content knowledge needed for success on local and state assessments. Authentic assessments help students develop higher-order thinking skills and are critical to developing 21st century skills; however, assessment in standards- and inquiry-based classrooms can be challenging for teachers new to the practice. In eMINTS professional development, in sessions dealing with assessment and content woven throughout the program, teachers learn about and practice using multiple types of assessment.

High-Quality Lesson Design

High-quality lesson design emphasizes that student learning is best facilitated through a teacher's ability to engage students in lessons with meaningful content and inquiry. In this model, effective facilitation is based on a teacher's ability to select instructional strategies that best meet student needs (Tomlinson et al., 2003) and their ability to help students build personal understanding of lesson content through processes such as reflection and metacognition (National Research Council, 2003). To support student learning, teachers must be attentive to what students currently know and what they are capable of so as to be able to identify areas in which students require additional support (Wiliam & Thompson, 2007). Wiggins and McTighe (1998) suggest that assessing depth of understanding is best approached by using assessments in multiple contexts, which vary in when and how they are administered. The use of multiple data sources enables teachers to improve student achievement by prioritizing instructional time, targeting instruction for struggling students, identifying student strengths, gauging the instructional effectiveness of classroom lessons, and refining instructional methods (Hamilton et al., 2009; Shavelson, 2006). In addition, feedback from assessments can help students focus their own efforts on areas for growth (Heritage, 2007; Sadler, 1989) and reasoning ability (Belland, Glazewski, & Richardson, 2011; Chinn & Hung, 2007) and increase their achievement levels (Simons & Klein, 2007). Professional development focusing on lesson design enables teachers to plan standards-based instruction and inform instruction with formative and summative assessments. eMINTS professional development guides teachers with lesson design processes created by eMINTS staff that are based on a rigorous Japanese lesson study that researchers suggest improves instruction (Lewis, Perry, & Hurd, 2004).

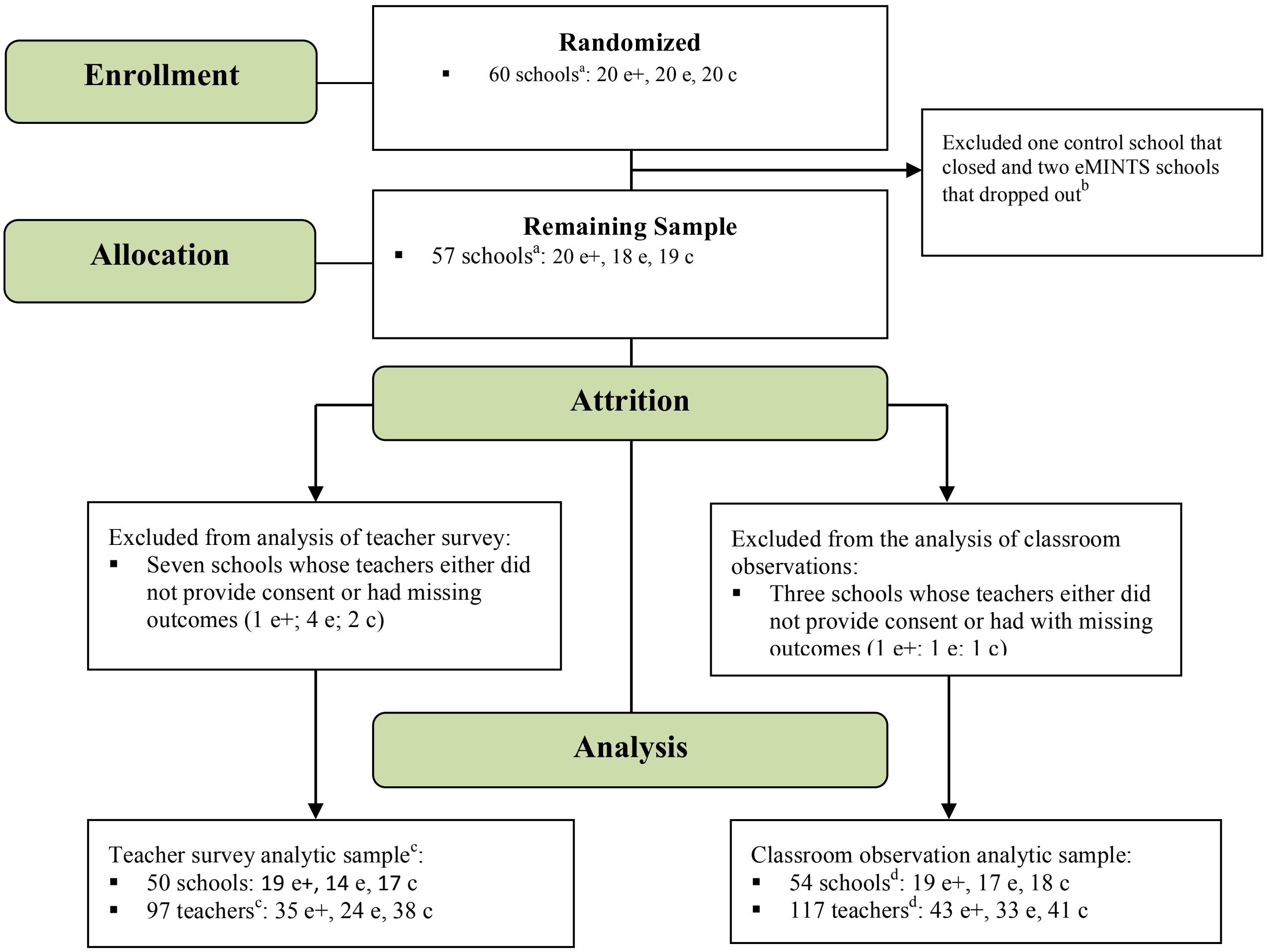
A Community of Learners

eMINTS professional development focuses on helping teachers learn how to establish a community of learners in support of standards- and inquiry-based instruction in their classrooms. In effective communities of learners, teachers expect students to support learning by respectfully communicating and exhibiting a positive regard for diversity (Houston & Ferstl, 2007; Pulakos, Arad, Donovan, & Plamondon, 2000). The open-minded atmosphere, in which students feel comfortable taking risks and sharing ideas and experiences, allows for responsible social interactions to occur that underpin the development of 21st century life and career skills (Johnson & Johnson, 2009; Pulakos et al., 2000, Slavin, 2010). Students use interpersonal skills to guide others and respond to each other in a dependable, reliable, and trustworthy manner. Work is expected to be efficiently managed, such that each student is responsible for his or her contribution and makes sure that projects are completed on time (Johnson & Johnson, 2005; Slavin, 2010). Students are actively involved in setting and managing personal goals (Guthrie, Wigfield, & VonSecker, 2000; Rohrbeck, Ginsburg-Block, Fantuzzo, & Miller, 2003), working independently to become self-directed learners (Houston & Ferstl, 2007). Throughout individual and group projects, students adapt to a variety of roles and responsibilities, learning to exhibit flexibility in different situations (Houston & Ferstl, 2007; Pulakos et al., 2000). Prior research suggests that students who learn in classrooms where decisions are made collaboratively display more creativity and higher-order thinking skills (Kohn, 2006).

Technology Integration

Lawless and Pelligrino (2007) argue that introducing teachers to new technologies for teaching and learning can support changes in teaching practices. eMINTS professional development centers on a suite of technologies and weaves technology integration into the practice of standards- and inquiry-based teaching. Instead of approaching teachers with a need to change their teaching, teacher buy-in for instructional change is facilitated by adding technology to the classroom. The eMINTS model develops the technological skills of its participating teachers so that they can integrate technology into their inquiry-based instructional practices and student activities. The technology itself is intended to enhance—not simply replace—teacher instruction to achieve desired instructional and learning goals; the use of technology should enable the teacher to instruct or students to demonstrate understanding in ways that would not be possible without the technology (Tamim, Bernard, Borokhovski, Abrami, & Schmid, 2011). Technology underpins the other three tenants of the eMINTS model by providing new modalities and tools for students to explore and construct knowledge as part of inquiry-based learning (e.g., manipulating digital artifacts, completing online drills, engaging in simulations or games; Geier et al., 2008; Li & Ma, 2010) and serving as a means of interaction to support collaborative and cooperative learning inside and outside the classroom (i.e., a community of learners; Office of Educational Technology, 2012).

Appendix C. Consort Diagram for Schools in the Teacher Outcomes Analyses, 2013–14



Note. From authors' analysis based on data from DESE and the study districts.
[a]Sixty schools were randomized to one of three groups at the beginning of the study: eMINTS, eMINTS + Intel, and control.
[b]One control school closed before the 2012–13 school year, and two eMINTS schools dropped out - one about four months after random assignment was announced and shortly after baseline data collection began, and the other during the summer before Year 1 implementation. [c]The teacher survey was analyzed separately for four constructs of teacher practice: a community of learners, inquiry-based learning, high-quality lesson design, and technology integration. The analytic sample for the community of learners construct had 97 teachers. The analytic samples for the, inquiry-based learning, high-quality lesson design, and technology integration constructs are the same and had 96 teachers. (because one teacher had a missing outcome on these three constructs). [d]The classroom observation sample was analyzed separately for three constructs of teacher practice: a community of learners, inquiry-based learning, and technology integration. (There were no items in the observation protocol that measured high-quality lesson design.) The analytic samples for community of learners and inquiry-based learning constructs had 117 teachers, and the analytic sample for technology of integration construct had 115 teachers (because two eMINTS+Intel teachers had a missing outcome on this construct).

Appendix D. Model for Baseline Equivalence Testing

For the Year 3 analyses, the following model was employed for comparing baseline student characteristics:

Level 1 (students)

$$Ypre_{ij} = \pi_{0j} + \pi_{1j}\text{Grade}_{ij} + \varepsilon_{ij}$$

Level 2 (schools)

$$\pi_{0j} = \sum_k \beta_{00k} D_k + \sum_k \beta_{01k} \text{T1}_j * D_k + \sum_k \beta_{02k} \text{T2}_k * D_k + r_{0j}$$

$$\pi_{1j} = \beta_{10}$$

where

- $\text{T1}_j = 1$ if school j belongs to the eMINTS group; 0 otherwise.
- $\text{T2}_k = 1$ if school j belongs to the eMINTS+Intel group; 0 otherwise.
- $Ypre_{ij}$ is a baseline characteristic (e.g., pretest MAP mathematics or communication arts score) of student i in school j.
- $D_k = 1$ if school j is in randomization block k; 0 otherwise; $k = 1, \ldots, 3$.
- $\text{T}_j = 1$ if school j belongs to either the eMINTS or eMINTS+Intel groups; 0 otherwise.
- Grade_{ij} is a grade-level indicator that is equal to 1 for eighth-grade students and 0 for seventh-grade students (centered around the grand mean).
- ε_{ij} and r_{0j} are random residuals at the student and school levels, respectively.

Appendix E. Baseline Characteristics of Students in 21st Century Skills and Student Engagement Analytic Samples

Table E1. Characteristics of Students in the 21st Century Skills Analytic Sample: 2010–11 (Before Year 1 Implementation)[a]

Characteristics	eMINTS+ Intel	eMINTS	Control	*p*-value
Number of students	866	827	683	
2011 MAP mathematics *z*-scores	-0.029 (*n* = 762)	0.018 (*n* = 750)	0.006 (*n* = 582)	0.858
2011 MAP communication arts *z*-scores	-0.003 (*n* = 761)	-0.025 (*n* = 750)	0.000 (*n* = 582)	0.960
Percentage eligible for free or reduced-price lunch	0.531 (*n* = 763)	0.568 (*n* = 752)	0.525 (*n* = 582)	0.566
Percentage White	0.946 (*n* = 763)	0.959 (*n* = 752)	0.948 (*n* = 582)	0.852
Percentage with IEP	0.122 (*n* = 763)	0.128 (*n* = 752)	0.090 (*n* = 582)	0.142
Percentage LEP	0.019 (*n* = 763)	0.000 (*n* = 752)	0.019 (*n* = 582)	0.708
Percentage male	0.474 (*n* = 817)	0.497 (*n* = 817)	0.493 (*n* = 675)	0.713

Note. From authors' analysis based on student baseline data collected from study districts in spring 2011, when Grade 7 students were in Grade 4 and Grade 8 students were in Grade 5.
[a]180 of the 2,376 students in the 21[st] century skills analytic sample are in the MAP mathematics analytic sample, and 2,369 are in the MAP communication arts analytic sample. [b]Means were regression adjusted to account for block effects, grade, and clustering of students within schools, and weighted by the number of schools in each block. For each characteristic, the *p*-value is from an overall chi-square test of whether the (weighted) means of the three treatment groups (eMINTS+Intel, eMINTS, and control) differed more than what might be expected by chance. Because of zero counts for LEP in some cells, the percentage LEP was estimated on the whole sample (instead of separately by blocks). When data are missing, *n* is the actual number of students used to calculate the average characteristic in each treatment group.
*Statistically significantly different from zero at the .05 level.

Table E2. Characteristics of Students in the Student Engagement Analytic Sample: 2010–11 (Before Year 1 Implementation)[a]

Characteristics	eMINTS+ Intel	eMINTS	Control	*p*-value
Number of students	751	868	659	
2011 MAP mathematics *z*-scores	-0.013 (*n* = 661)	0.055 (*n* = 790)	-0.028 (*n* = 569)	0.600
2011 MAP communication arts *z*-scores	-0.015 (*n* = 660)	0.027 (*n* = 790)	-0.025 (*n* = 569)	0.800
Percentage eligible for free or reduced-price lunch	0.506 (*n* = 662)	0.546 (*n* = 792)	0.542 (*n* = 569)	0.580
Percentage White	0.930 (*n* = 662)	0.954 (*n* = 792)	0.941 (*n* = 569)	0.649
Percentage with IEP	0.104 (*n* = 662)	0.128 (*n* = 792)	0.105 (*n* = 569)	0.359
Percentage LEP	0.030 (*n* = 662)	0.000 (*n* = 792)	0.022 (*n* = 569)	0.299
Percentage male	0.461 (*n* = 740)	0.508 (*n* = 859)	0.491 (*n* = 652)	0.270

Note. From authors' analysis based on student baseline data collected from study districts in spring 2011, when Grade 7 students were in Grade 4 and Grade 8 students were in Grade 5.
[a]2,099 of the 2,278 students in the student survey analytic sample are in the MAP mathematics analytic sample, and 2,273 are in the MAP communication arts analytic sample. [b]Means were regression adjusted to account for block effects, grade, and clustering of students within schools, and weighted by the number of schools in each block. For each characteristic, the *p*-value is from an overall chi-square test of whether the (weighted) means of the three treatment groups (eMINTS+Intel, eMINTS, and control) differed more than what might be expected by chance. Because of zero counts for LEP in some cells, the percentage LEP was estimated on the whole sample (instead of separately by blocks). When data are missing, *n* is the actual number of students used to calculate the average characteristic in each treatment group.
*Statistically significantly different from zero at the .05 level.

Appendix F. Calculating Implementation Fidelity Received

Table F1. Calculating Implementation Fidelity in the First Two Years of the eMINTS Comprehensive Program

Components	Criteria	Measures	Item Metric	Component Metric
Technology Infrastructure				
Adequate data storage	1 gigabyte	TCoordinator Survey Q1, Q2 For responses, code "yes" = 1; "no" = 0; "don't know" = 0	If tc_stor (Q1) + tc_stor(Q2) = 2, then tc_stor_tot = 1 If tc_stor (Q1) + tc_stor(Q2) = 1, then tc_stor_tot =.5 If tc_stor (Q1) + tc_stor(Q2) = 0, then tc_stor_tot = 0	tech_infstr_ave = (tc_stor_tot + tc_bckp_tot + tc_intcomm_tot + tc_sec_tot + wrlss_tot + tc_shrfldr_tot)/6 Calculate tech_infstr_ave index separately for Years 1 and 2. Final index value will equal the average of the Years 1 and 2 indexes (tech_infstr _ave_y1 + tech_infstr _ave_y2)/2). If tech_infstr _ave ≥ .75, then tech_insfstr_tot = 1 If tech_infstr _ave ≥ .5 and < .75, then tech_insfstr_tot = .5 If tech_infstr _ave < .5, then tech_insfstr_tot = 0 High: tech_infstr_tot ≥ .75 Moderate: tech_infstr_tot ≥ .5 < .75 Low: tech_infstr_tot < .5
Backup and disaster recovery	Backups for students, teachers, and administrators	TCoordinator survey Q3 = tc_bckp_adm TCoordinator survey Q4 = tc_bckp_tch TCoordinator survey Q5 = tc_bckp_std For responses, code "yes" = 1; "no" = 0; "don't know" = 0	If tc_bckp_adm (Q3) + tc_backup_tch (Q4) + tc_backup_std (Q5) = 3, then tc_bckp_tot = 1 If tc_backup (Q3) + tc_backup(Q4) + tc_backup(Q5) = 2, then tc_bckp_tot =.5 If tc_backup (Q3) + tc_backup(Q4) + tc_backup(Q5) < 2 , then tc_bckp_tot = 0	
Internet connectivity	1 Mbps connection speed or higher	TCoordinator Survey Q6 = tc_intcomm For responses, code "1 Mbps or higher" = 1; "less than 1 Mbps" = 0	If tc_intcomm = 1 then tc_intcomm_tot = 1; otherwise tc_intcomm_tot = 0	

Components	Criteria	Measures	Item Metric	Component Metric
Internet security	Fully implemented security criteria (filtering software, antivirus, automatic antivirus updates, Web blocking, and malware protection)	TCoordinator Survey Q9/Q18 = tc_fltr (code "yes" = 1, "no" = 0) TCoordinator Survey Q10/20 = tc_antivrs (code "yes" = 1, "no" = 0) TCoordinator Survey Q11/22 = tc_vrsdef (code "yes" = 1, "no" = 0) TCoordinator Survey Q12/19 = tc_weblck (code "yes" = 1, "no" = 0) TCoordinator Survey Q14/Q21 = tc_mlwr (code "yes" = 1,"no" = 0)	If tc_fltr + tc_antivrs + tc_vrsdef + tcweblck + tc_mlwr ≥ 3, then tc_sec_tot = 1; otherwise tc_sec_tot = 0	
Wireless connection	Wireless must be available; wireless must have adequate speed	TCoordinator Survey Q15/23 = tc_wrlss (code "yes" = 1, "no" = 0) Teacher survey (5.9e) = tch_intspd_ave (code "completely disagree" = 1, "mostly disagree" = 2, "mostly agree" = 3, "completely agree" = 4)	Teacher survey: If scl_intspd_ave (Q5.9e) ≥ 3, then scl_intspd_tot = 1; otherwise, scl_intspd_tot = 0 If scl_intspd_tot + tc_wrlss = 2 then wrlss_tot = 1 If scl_intspd_tot + tc_wrlss = 1 then wrlss_tot = .5 If scl_intspd_tot + tc_wrlss = 0 then wrlss_tot = 0	
Shared folder system	Shared folder system available to all eMINTS teachers and their students	TCoordinator Survey Q24 = tc_shrfldr (code "yes" = 1,"no" = 0)	If tc_shrfldr = 1, then tc_shrfldr_tot = 1; otherwise tc_shrfldr_tot = 0	

Components	Criteria	Measures	Item Metric	Component Metric
Technology Use				
Student laptops	Student's use of laptop	Teacher Survey Q5.6b = stulap_use (code "never or almost never" = 1, "3–6 sessions per year" = 2, "1–3 sessions per month" = 3, "1–3 sessions per week" = 4, "almost every or every session" = 5)	If stulap_use ≥ 4, then X1 = 1; otherwise X1 = 0	tech_use_ave= (X1+X2+X3)/3 Calculate tech_infstr_ave index separately for Years 1 and 2. Final index value will equal the average of the indexes for Years 1 and 2. High: tech_use_ave ≥ .8 Moderate: tech_use_ave ≥.67 < .8 Low: tech_use_ave < .67
Teacher laptop	Teacher's use of laptop	Teacher Survey Q5.6a = tchlap_use (code "never or almost never" = 1, "3–6 sessions per year" = 2, "1–3 sessions per month" = 3, "1–3 sessions per week" = 4, "almost every or every session" = 5)	If tchlap_use ≥ 4, then X2 = 1; otherwise X2 = 0	
Interactive whiteboard	Teacher's use of whiteboard	Teacher Survey Q5.6g = whtbd_use (code "never or almost never" = 1, "3–6 sessions per year" = 2, "1–3 sessions per month" = 3, "1–3 sessions per week" = 4, "almost every or every session" = 5)	If whtbd_use ≥ 4, then X3 = 1; otherwise X3 = 0	
Scanner/printer	Teacher's use of scanner or printer	Teacher Survey Q5.6d, e	Not included in index for Years 1 and 2.	
Digital camera	Teacher's use of digital camera	Teacher Survey Q5.6f	Not included in index for Years 1 and 2.	
Technology Professional Development				
PLC professional development participation	27 sessions in Year 1 (123.5 hours) and 20 sessions in Year 2 (88 hours) = 47 sessions (211.5 hours) total	eMINTS teacher attendance file. Because the total number of hours varies across trainers and teachers (i.e., some sessions, although estimated to last 4 hours, actually take only 3.5 hours to complete), calculate the proportion of total hours that each teacher attended (tch_fmltrng).	tch_fmltrng = (number of hours a teacher attended)/211.5 hours* *Total hours for individual teachers may vary slightly by eMINTS eIS. Calculate the proportion of hours attended (total attended/total required) for each individual teacher.	tch_trng_tot tch_trng_ave = tch_fmltrng + tch_cch + portfolio/3 High: tch_trng_ave ≥ .85 Moderate: tch_trng_ave ≥ .75 < .85

Components	Criteria	Measures	Item Metric	Component Metric
Participation in coaching sessions	10 sessions each year for two years = 20 sessions total	eMINTS teacher attendance file. Calculate the proportion of sessions a teacher attended out of 20 sessions (tch_cch).	tch_cch = (number of sessions a teacher attended)/20 sessions	
Teacher portfolio	To become eMINTS certified, a teacher must complete three assignments that meet minimum criteria: a unit plan, a WebQuest, and a website.	The unit plan, the WebQuest, and the website are graded using a rubric. The total score on each assignment ranges from 0 to 3.	tch_prtfl = (unit_plan)(.5) + (WebQuest)(.25) + website(.25)/3	
Administrative Support				
Formal professional development participation	Two sessions total; one 2-day session in Year 1 and one 1-day session in Year 2	eMINTS principal attendance file (code prcpl_fmltrng_yr1 and prcpl_fmltrng_yr2: 0 = "attended", 1 = "did not attend")	If prcpl_fmltrng_yr1 + prcpl_fmltrng_yr2 = 2, then prcpl_fmltrng = 2 If prcpl_fmltrng_yr1 + prcpl_fmltrng_yr2 = 1, then prcpl_fmltrng = 1 If prcpl_fmltrng_yr1 + prcpl_fmltrng_yr2 = 0, then prcpl_fmltrng = 0	prcpl_trng_tot = prcpl_fmltrng + prcpl_wlkthrgh/5 High: prcpl_trng_tot ≥ .8 Moderate: prcpl_trng_tot = 0.6 Low: prcpl_trng_tot < 0.6
Participation in walk-throughs	Three sessions total; two sessions in Year 1 and one session in Year 2. Note: One additional session is optional during spring Year 2.	Vovici principal walk-through survey data (code prcpl_wlkthrgh = number of walk-throughs in which the principal participated).	If prcpl_wlkthrgh = 4 then prcpl_wlkthrgh_tot = 3 If prcpl_wlkthrgh = 3 then prcpl_wlkthrgh_tot = 3 If prcpl_wlkthrgh = 2, then prcpl_wlkthrgh_tot = 2 If prcpl_wlkthrgh = 1, then prcpl_wlkthrgh_tot = 1 If prcpl_wlkthrgh = 0, then prcpl_wlkthrgh_tot = 0	

Components	Criteria	Measures	Item Metric	Component Metric
Ongoing Technology Support				
Formal professional development participation	Three optional WebEx conference sessions total: two sessions in Year 1 and one session in Year 2.	eMINTS technology coordinator attendance file. For each school, code tech_trng_yr 1 and tech_trng_yr2 as 0 = "did not attend," 1 = "attended 1 session," 2 = "attended 2 sessions," and 3 = "attended 3 sessions." A school is considered to have attended if anyone from the school attended the professional development session.	If tech_trng_y1 +tech_trng_y2 ≥ 2, then tech_trng_tot = 2 If tech_trng_y1 + tech_trng_y2 = 1, then tech_trng_tot = 1 If tech_trng_y1 + tech_trng_y2 = 0, then tech_trng_tot = 0	(tech_sup_yr1 + tech_sup_yr2/2) + (inst_sup_yr1 + inst_sup_yr2/2) + tech_trng_tot /8 = schl_techspt_ave High: schl_techspt_ave ≥ 0.8 Moderate: schl_techspt_ave > 0.6 < 0.8 Low: schl_techspt_ave ≤ 0.6
Access to on-site technical assistance	Access to a technology coordinator for troubleshooting and instructional support	Teacher survey Q5.9j (5.9j REVERSE CODED; code tech_sup: "completely disagree" = 4, "mostly disagree" = 3, "mostly agree" = 2, "completely agree" = 1) Teacher survey 5.9l (code inst_sup: "completely disagree" = 1, "mostly disagree" = 2, "mostly agree" = 3, "completely agree" = 4)	sch_techsup_y1 (item 5.9j, Year 1 survey) Sch_techsup_y2 (item 5.9j, Year 2 survey) Sch_instsup_y1 (item 5.9j, Year 1 survey) Sch_instsup_y2 (item 5.9j, Year 2 survey)	

Table F2. Calculating Implementation Fidelity of the eMINTS+Intel Program (Year 3)

Components	Criteria	Measures	Item Metric	Component Metric
Teacher Professional Development				
Professional learning community professional development participation	Four sessions in Year 3 (28 hours online and 12 hours in person)	eMINTS teacher attendance file. Because the total number of hours varies across trainers and teachers (i.e., some sessions, although estimated to last 4 hours, actually take only 3.5 hours to complete), calculate the proportion of total hours that each teacher attended (tch_fmltrng).	tch_fmltrng = (number of hours a teacher attended)/total hours required)40 hours* *Total hours for individual teachers was designed to be 40 hours, but the totals may vary slightly.	tch_trng_tot tch_trng_ave = tch_fmltrng + tch_cch + portfolio/3 High: tch_trng_ave ≥ .85 Moderate: tch_trng_ave ≥ .75 < .85 Low: tch_trng_ave < .75
Participation in coaching sessions	4 one-hour sessions in Year 3	eMINTS teacher attendance file. Calculate the proportion of sessions a teacher attended out of four sessions (tch_cch).	tch_cch = (number of sessions a teacher attended)/4 sessions** ** Maximum score achievable is 1	
Teacher portfolio	Teacher must meet minimum criteria for a unit plan	The unit plan is graded using a rubric. The total score ranges from 0 to 3. Measure based on whether or not minimum criteria were met (tch_prtfl).	If teacher meet criteria tch_prtfl = 1 if not tch_prtfl = 0	

Appendix G. Equipment and Software Delivered to Treatment Schools

Table G1. Technology Devices and Software Delivered to Participating Schools

Technology Devices and/or Software	Description	Vendor / Supplier
Purchased by participating districts		
Teacher laptop (includes Mini Dock Series 3 docking station, case, asset tag, fully imaged, MS Office 2010 and MS Forefront antivirus software)	Lenovo ThinkPad T520 4242 – Core 15-2540M 2.5 GHz – 15.6" (9 cell)	CDW-G*
SMART Board/Projector System (includes board, projector, installation and SMART Ideas site license)	SB680i4 interactive whiteboard system	KCAV/SMART Technologies*
Printer (toner and 3-year service warranty)	Xerox 6505N MFP	CDW-G*
Digital camera	Canon A300	CDW-G
Aruba Wireless Access Point, one per classroom (includes cable and install kit for carts or classrooms)		CDW-G
Student Laptops (includes asset tag, fully imaged, laser etching, MS Office 2010 and MS Forefront antivirus software)	Lenovo ThinkPad X120e 2876 - Fusion E-240	CDW-G*
Laptop Carts		CDW-G,* Earthwalk*
BrainPOP (Site license for 24 months)	A suite of digitally animated products (e.g., animated movies, learning games, interactive quizzes) that presents curriculum-based topics for students	BrainPOP*
Available through eMINTS		
Tech4Learning*	Recipes for Success	eMINTS
School Improvement Network PD360*	On-demand series of videos that supports teachers in using technology in the classroom to increase student learning	eMINTS

*In-kind product discount donations were provided by the vendors to comply with the i3 requirement for private-sector matching funds.

Appendix H. Estimated Regression Coefficients for the Analyses of the Four Student Outcomes

Table H1. Estimates of Regression Coefficients for the Impact Analyses of eMINTS on the Four Student Outcomes in Implementation Year 3 (2013-14)

Parameter	MAP Mathematics *z*-Scores			MAP Communication Arts *z*-Scores			21st Century Skills Scores			Student Engagement Scores		
	Estimate	Standard Error	*p*-Value	Estimate	Standard Error	*p*-Value	Estimate	Standard Error	*p*-Value	Estimate	Standard Error	*p*-Value
Block 1												
Intercept (control)	-0.081	0.07	0.256	0.141	0.06	0.018*	267.294	6.63	0.000*	0.804	0.12	0.000*
Impact: eMINTS—control	0.165	0.10	0.084	-0.092	0.08	0.243	6.697	9.57	0.484	-0.114	0.17	0.509
Impact: eMINTS+Intel—control	0.298	0.10	0.003*	-0.162	0.08	0.047*	18.507	9.67	0.056	-0.004	0.17	0.983
Block 2												
Intercept (control)	-0.060	0.11	0.585	-0.032	0.07	0.656	269.794	12.36	0.000*	0.763	0.21	0.000*
Impact: eMINTS—control	0.139	0.14	0.321	0.021	0.09	0.821	7.896	15.84	0.618	-0.120	0.27	0.658
Impact: eMINTS+Intel—control	0.078	0.14	0.582	0.165	0.09	0.076	-7.968	16.26	0.624	-0.347	0.30	0.249
Block 3												
Intercept (control)	0.008	0.07	0.911	0.054	0.06	0.331	262.234	7.05	0.000*	0.776	0.13	0.000*
Impact: eMINTS—control	0.074	0.10	0.457	0.026	0.08	0.733	4.411	11.59	0.704	-0.235	0.18	0.199
Impact: eMINTS+Intel—control	0.018	0.10	0.856	-0.044	0.07	0.556	-11.434	10.30	0.267	-0.201	0.17	0.245
Pretests												

Spring 2011 MAP mathematics *z*-scores	0.720	0.02	0.000*	—	—	—	14.050	1.32	0.000*	0.099	0.03	0.000*
Spring 2011 MAP communication arts *z*-scores	—	—	—	0.666	0.01	0.000*	22.855	1.32	0.000*	0.517	0.20	0.012*
Spring 2011 MAP school mean mathematics *z*-scores	0.102	0.08	0.191	—	—	—	-2.577	16.25	0.874	-0.004	0.03	0.874
Parameter	**MAP Mathematics *z*-Scores**			**MAP Communication Arts *z*-Scores**			**21st Century Skills Scores**			**Student Engagement Scores**		
	Estimate	**Standard Error**	***p*-Value**	**Estimate**	**Standard Error**	***p*-Value**	**Estimate**	**Standard Error**	***p*-Value**	**Estimate**	**Standard Error**	***p*-Value**
Spring 2011 MAP school mean communication arts *z*-scores	—	—	—	0.004	0.08	0.962	28.405	15.63	0.069	-0.406	0.24	0.094
Student Demographics												
Grade: Grade 8—Grade 7	0.000	0.03	0.990	0.003	0.03	0.906	18.335	1.74	0.000*	-0.058	0.04	0.102
Free or reduced-price lunch: yes—no	-0.022	0.03	0.449	-0.073	0.03	0.013*	-3.537	1.97	0.073	-0.061	0.04	0.129
Race/ethnicity: White—non-White	-0.015	0.06	0.802	0.063	0.06	0.296	2.492	4.30	0.562	0.021	0.09	0.810
Individualized education plan: yes—no	-0.220	0.04	0.000*	-0.329	0.05	0.000*	-0.654	3.20	0.838	0.160	0.07	0.015*
LEP: yes—no	-0.227	0.14	0.107	0.008	0.14	0.958	-3.246	10.12	0.748	0.180	0.18	0.324

Gender: male—female	0.047	0.03	0.069	0.201	0.03	0.000*	12.139	1.77	0.000*	0.215	0.04	0.000*
Teacher Demographics												
Gender: male—female	-0.085	0.04	0.044*	-0.244	0.06	0.000*	—	—	—	—	—	—
Graduate degree: with—without	0.064	0.04	0.140	0.046	0.04	0.224	—	—	—	—	—	—
Years of teaching experience	0.003	0.00	0.251	0.005	0.00	0.036*	—	—	—	—	—	—

Note. The analytic regression models were adjusted for clustering of students within schools, block effects, and baseline student and school characteristics. For the analysis of MAP mathematics and communication arts scores, baseline teacher characteristics also were included as covariates.

*Statistically significantly different from zero at the .05 significance level.

Appendix I. Subgroup Findings

Table I1. Year 3 Impact of eMINTS on Mathematics Achievement, Overall and By Subgroup

Contrast	Subgroup	Mean[a]			Estimated Impact Within Subgroup[a]				Difference in Impacts Between Subgroups: *p*-Value*
		eMINTS+ Intel	eMINTS	Control	Impact	Standard Error	*p*-Value	Effect Size[b]	
eMINTS vs. control	Full sample	---	0.083 (n = 1,108)[d]	-0.047 (n = 823)	0.129	0.063	0.040*	0.128	N/A
eMINTS+Intel vs. control	Full sample	0.122 (n = 1,141)	---	-0.047 (n = 823)	0.168	0.065	0.009*	0.178	N/A
eMINTS+Intel vs. eMINTS	Full sample	0.122 (n = 1,141)	0.083 (n = 1,108)	---	0.039	0.063	0.532	0.037	N/A
eMINTS vs. control	Non-IEP	---	0.136	-0.010	0.145	0.063	0.021*	0.143	0.204
	IEP	---	-0.742	-0.748	0.006	0.088	0.948	0.005	
eMINTS+Intel vs. control	Non-IEP	0.145	---	-0.010	0.155	0.064	0.016*	0.152	0.230
	IEP	-0.462		-0.748	0.286	0.089	0.001*	0.281	
eMINTS+Intel vs. eMINTS	Non-IEP	0.145	0.136	---	0.010	0.062	0.877	0.008	0.005*
	IEP	-0.462	-0.742		0.280	0.082	0.001*	0.276	
eMINTS vs. control	Non-FRPL	---	0.130	0.011	0.119	0.066	0.071	0.118	0.759
	FRPL	---	-0.060	-0.199	0.139	0.067	0.037*	0.138	
eMINTS+Intel vs. control	Non-FRPL	0.176	---	0.011	0.164	0.067	0.014*	0.161	0.890
	FRPL	-0.026		-0.199	0.173	0.068	0.011*	0.170	
eMINTS+Intel vs. eMINTS	Non-FRPL	0.176	0.130	---	0.045	0.065	0.491	0.043	0.853
	FRPL	-0.026	-0.060		0.034	0.066	0.607	0.032	
eMINTS vs. control	Non-White	---	-0.072	-0.288	0.216	0.075	0.004*	0.213	0.180
	White	---	0.076	-0.032	0.108	0.064	0.091	0.107	
eMINTS+Intel vs. control	Non-White	-0.084	---	-0.288	0.204	0.074	0.006*	0.200	0.543
	White	0.126		-0.032	0.158	0.066	0.016*	0.155	
eMINTS+Intel vs. eMINTS	Non-White	-0.084	-0.072	---	-0.012	0.073	0.872	-0.013	0.405
	White	0.126	0.076		0.050	0.064	0.436	0.048	

*This column gives the p-value of the test of difference in impacts between subgroups (e.g., whether the impact on IEP students differs from the impact on non-IEP students).

Note: Means and impacts were regression-adjusted to account for clustering of students within schools, block effects, and baseline student, teacher, and school characteristics. They were weighted by the number of schools in each block. Effect sizes were calculated separately by block and then pooled into an overall effect size weighted by the number of schools in each block. Block-specific effect sizes were computed using standardized mean differences (Hedges's *g*).

p-values are from a two-tailed test of the null hypothesis of equality means. Because the student analytic samples are pooled samples of seventh and eighth graders, MAP scale scores were converted into a common metric by standardizing the scores separately by year (i.e., 2011 for the pretest and 2014 for the posttest), grade, subject (mathematics and communication arts), and randomization block. Specifically, each test score was converted into a *z*-score by subtracting from the test score the average score for a particular year, grade, subject, and block and then dividing by the standard deviation for a particular year, grade, subject, and block. *n* gives the sample sizes used in the analysis. For the MAP mathematics analyses, *n* includes all eligible students with nonmissing outcomes.

Source: From authors' analysis based on Year 3 implementation (2013–14) data from the study districts and DESE.

Table I2. Year 3 Impact of eMINTS on Communication Arts Achievement, Overall and By Subgroup

Contrast	Subgroup	Mean			Estimated Impact Within Subgroup				Difference in Impacts Between Subgroups: *p*-Value*
		eMINTS+ Intel	eMINTS	Control	Impact	Standard Error	*p*-Value	Effect Size	
eMINTS vs. control	Full sample	---	0.052 (*n* = 1,208)	0.087 (*n* = 871)	-0.035	0.050	0.484	-0.035	--
eMINTS+Intel vs. control	Full sample	0.011 (*n* = 1,216)	---	0.087 (*n* = 871)	-0.076	0.051	0.137	-0.077	--
eMINTS+Intel vs. eMINTS	Full sample	0.011 (*n* = 1,216)	0.052 (*n* = 1,208)	---	-0.041	0.047	0.384	-0.041	--
eMINTS vs. control	Non-IEP	---	0.104	0.138	-0.034	0.051	0.501	-0.034	0.938
	IEP	---	-0.896	-0.871	-0.025	0.081	0.754	-0.025	
eMINTS+Intel vs. control	Non-IEP	0.047	---	0.138	-0.091	0.051	0.076	-0.092	0.135
	IEP	-0.795		-0.871	0.076	0.082	0.355	0.074	
eMINTS+Intel vs. eMINTS	Non-IEP	0.047	0.104	---	-0.057	0.047	0.225	-0.058	0.109
	IEP	-0.795	-0.896		0.101	0.073	0.166	0.102	
eMINTS vs. control	Non-FRPL	---	0.136	0.156	-0.020	0.054	0.706	-0.020	0.658
	FRPL	---	-0.151	-0.102	-0.049	0.055	0.369	-0.049	
eMINTS+Intel vs. control	Non-FRPL	0.072	---	0.156	-0.084	0.054	0.119	-0.085	0.755
	FRPL	-0.166		-0.102	-0.064	0.056	0.255	-0.065	
eMINTS+Intel vs. eMINTS	Non-FRPL	0.072	0.136	---	-0.064	0.050	0.201	-0.065	0.408
	FRPL	-0.166	-0.151		-0.015	0.051	0.773	-0.015	
eMINTS vs. control	Non-White	---	-0.126	-0.195	0.070	0.065	0.282	0.069	.106
	White	---	0.040	0.101	-0.061	0.051	0.233	-0.060	
eMINTS+Intel vs. control	Non-White	-0.241	---	-0.195	-0.046	0.063	0.465	-0.047	0.614
	White	0.016		0.101	-0.084	0.052	0.105	-0.085	
eMINTS+Intel vs. eMINTS	Non-White	-0.241	-0.126	---	-0.116	0.060	0.053	-0.117	0.216
	White	0.016	0.040		-0.024	0.048	0.619	-0.024	

*This column gives the p-value of the test of difference in impacts between subgroups (e.g., whether the impact on IEP students differs from the impact on non-IEP students).

Note: Means and impacts were regression-adjusted to account for clustering of students within schools, block effects, and baseline student, teacher, and school characteristics. They were weighted by the number of schools in each block. Effect sizes were calculated separately by block and then pooled into an overall effect size weighted by the number of schools in each block. Block-specific effect sizes were computed using standardized mean differences (Hedges's *g*).
p-values are from a two-tailed test of the null hypothesis of equality means. Because the student analytic samples are pooled samples of seventh and eighth graders, MAP scale scores were converted into a common metric by standardizing the scores separately by year (i.e., 2011 for the pretest and 2014 for the posttest), grade, subject (mathematics and communication arts), and randomization block. Specifically, each test score was converted into a *z*-score by subtracting from the test score the average score for a particular year,

grade, subject, and block and then dividing by the standard deviation for a particular year, grade, subject, and block. *n* gives the sample sizes used in the analysis. For the MAP communication arts analyses, *n* includes all eligible students with nonmissing outcomes.

Source: From authors' analysis based on Year 3 implementation (2013–14) data from the study districts and DESE.

Table I3. Year 3 Impact of eMINTS on 21st Century Skills, Overall and By Subgroup

Contrast	Subgroup	Mean			Estimated Impact Within Subgroup				Difference in Impacts Between Subgroups: *p*-Value*
		eMINTS+ Intel	eMINTS	Control	Impact	Standard Error	*p*-Value	Effect Size	
eMINTS vs. control	Full sample	---	272.1 (*n* = 827)	266.0 (*n* = 683)	6.123	6.700	0.361	0.103	--
eMINTS+Intel vs. control	Full sample	270.4 (*n* = 866)	---	266.0 (*n* = 683)	4.374	6.527	0.503	0.059	--
eMINTS+Intel vs. eMINTS	Full sample	270.4 (*n* = 866)	272.1 (*n* = 827)	---	-1.749	6.84	0.798	-0.038	--
eMINTS vs. control	Non-IEP	---	275.5	268.0	7.499	6.761	0.267	0.126	0.075
	IEP	---	237.8	244.7	-6.861	8.149	0.400	-0.116	
eMINTS+Intel vs. control	Non-IEP	273.1	---	268.0	5.114	6.582	0.437	0.073	0.253
	IEP	240.8		244.7	-3.890	7.965	0.625	-0.087	
eMINTS+Intel vs. eMINTS	Non-IEP	273.1	275.5	---	-2.385	6.897	0.730	-0.049	0.440
	IEP	240.8	237.8		2.971	7.898	0.707	0.045	
eMINTS vs. control	Non-FRPL	---	280.6	274.1	6.587	6.835	0.335	0.111	0.917
	FRPL	---	261.9	255.8	6.119	6.885	0.374	0.103	
eMINTS+Intel vs. control	Non-FRPL	274.9	---	274.1	0.865	6.648	0.896	-0.003	0.084
	FRPL	264.3		255.8	8.501	6.723	0.206	0.132	
eMINTS+Intel vs. eMINTS	Non-FRPL	274.9	280.6	---	-5.722	6.965	0.411	-0.107	0.054
	FRPL	264.3	261.9		2.381	6.981	0.733	0.034	
eMINTS vs. control	Non-White	---	260.0	253.4	6.573	7.452	0.378	0.111	0.926
	White	---	274.2	268.2	6.009	6.729	0.372	0.101	
eMINTS+Intel vs. control	Non-White	260.0	---	253.4	6.599	7.119	0.354	0.099	0.620
	White	272.1		268.2	3.860	6.561	0.556	0.050	
eMINTS+Intel vs. eMINTS	Non-White	260.0	260.0	---	0.026	7.491	0.997	-0.007	0.710
	White	272.1	274.2		-2.150	6.869	0.754	-0.045	

*This column gives the p-value of the test of difference in impacts between subgroups (e.g., whether the impact on IEP students differs from the impact on non-IEP students).

Note: Means and impacts were regression-adjusted to account for clustering of students within schools, block effects, and baseline student, teacher, and school characteristics. They were weighted by the number of schools in each block. Effect sizes were calculated separately by block and then pooled into an overall effect size weighted by the number of schools in each block. Block-specific effect sizes were computed using standardized mean differences (Hedges's *g*).
p-values are from a two-tailed test of the null hypothesis of equality means. Because the student analytic samples are pooled samples of seventh and eighth graders, MAP scale scores were converted into a common metric by standardizing the scores separately by year (i.e., 2011 for the pretest and 2014 for the posttest), grade, subject (mathematics and communication arts), and randomization block. Specifically, each test score was converted into a *z*-score by subtracting from the test score the average score for a particular year, grade, subject, and block and then dividing by the standard deviation for a particular year, grade, subject, and block. *n* gives the sample sizes used in the analysis. For the 21st century skills and student engagement analyses, *n* includes all eligible students who provided consent to participate and have nonmissing outcomes.

Source: From authors' analysis based on Year 3 implementation (2013–14) data from the study districts and DESE.

Table I4. Year 3 Impact of eMINTS on Student Engagement, Overall and By Subgroup

Contrast	Subgroup	Mean: eMINTS+ Intel	Mean: eMINTS	Mean: Control	Estimated Impact Within Subgroup: Impact	Standard Error	*p*-Value	Effect Size	Difference in Impacts Between Subgroups: *p*-Value*
eMINTS vs. control	Full sample	---	0.629 (n = 868)[d]	0.788 (n = 659)	-0.160	0.115	0.166	-0.176	--
eMINTS+Intel vs. control	Full sample	0.667 (n = 751)	---	0.788 (n = 659)	-0.121	0.112	0.280	-0.149	--
eMINTS+Intel vs. eMINTS	Full sample	0.667 (n = 751)	0.629 (n = 868)	---	0.038	0.037	0.737	0.038	--
eMINTS vs. control	Non-IEP	---	0.602	0.793	-0.191	0.115	0.097	-0.209	0.050*
	IEP	---	0.766	0.650	0.116	0.145	0.424	0.121	
eMINTS+Intel vs. control	Non-IEP	0.650	---	0.793	-0.143	0.112	0.202	-0.173	0.168
	IEP	0.735		0.650	0.085	0.147	0.561	0.084	
eMINTS+Intel vs. eMINTS	Non-IEP	0.650	0.602	---	0.048	0.113	0.670	0.049	0.592
	IEP	0.735	0.766		-0.031	0.141	0.828	-0.039	
eMINTS vs. control	Non-FRPL	---	0.640	0.832	-0.192	0.118	0.104	-0.211	0.447
	FRPL	---	0.597	0.721	-0.124	0.119	0.299	-0.138	
eMINTS+Intel vs. control	Non-FRPL	0.682	---	0.832	-0.150	0.115	0.193	-0.181	0.504
	FRPL	0.633		0.721	-0.088	0.117	0.452	-0.112	
eMINTS+Intel vs. eMINTS	Non-FRPL	0.682	0.640	---	0.042	0.116	0.716	0.042	0.940
	FRPL	0.633	0.597		0.036	0.117	0.760	0.035	
eMINTS vs. control	Non-White	---	0.661	0.691	-0.030	0.133	0.822	-0.037	0.205
	White	---	0.615	0.800	-0.185	0.117	0.112	-0.204	

eMINTS+Intel vs. control	Non-White	0.667	---	0.691	-0.024	0.128	0.849	-0.040	0.303
	White	0.655		0.800	-0.145	0.114	0.204	-0.175	
eMINTS+Intel vs. eMINTS	Non-White	0.667	0.661	---	0.006	0.131	0.965	0.001	0.770
	White	0.655	0.615		0.041	0.115	0.723	0.040	

*This column gives the p-value of the test of difference in impacts between subgroups (e.g., whether the impact on IEP students differs from the impact on non-IEP students).

Note: Means and impacts were regression-adjusted to account for clustering of students within schools, block effects, and baseline student, teacher, and school characteristics. They were weighted by the number of schools in each block. Effect sizes were calculated separately by block and then pooled into an overall effect size weighted by the number of schools in each block. Block-specific effect sizes were computed using standardized mean differences (Hedges's *g*).
p-values are from a two-tailed test of the null hypothesis of equality means. Because the student analytic samples are pooled samples of seventh and eighth graders, MAP scale scores were converted into a common metric by standardizing the scores separately by year (i.e., 2011 for the pretest and 2014 for the posttest), grade, subject (mathematics and communication arts), and randomization block. Specifically, each test score was converted into a *z*-score by subtracting from the test score the average score for a particular year, grade, subject, and block and then dividing by the standard deviation for a particular year, grade, subject, and block. *n* gives the sample sizes used in the analysis. Student engagement scores are Rasch logit ability measures from a Rasch analysis of students' responses to the spring 2014 student engagement survey.

Source: From authors' analysis based on Year 3 implementation (2013–14) data from the study districts and DESE.

Appendix J. Estimated Regression Coefficients for the Teacher Survey Analysis

Table J1. Estimates of Regression Coefficients for the Impact Analyses of eMINTS on Teacher Survey Outcomes in Implementation Year 3 (2013-14)

Parameter	Community of Learners Logit Scores			Inquiry-Based Learning Logit Scores			High Quality Lesson Design Logit Scores			Technology Integration Logit Scores		
	Estimate	Standard Error	*p*-Value	Estimate	Standard Error	*p*-Value	Estimate	Standard Error	*p*-Value	Estimate	Standard Error	*p*-Value
Intercepts (control)												
Block 1	3.089	1.12	0.006*	-0.739	0.18	0.000*	0.215	0.31	0.486	-0.212	0.13	0.103
Block 2	3.687	1.51	0.015*	-0.631	0.24	0.008*	-0.004	0.42	0.993	-0.251	0.17	0.150
Block 3	2.949	1.34	0.028*	-1.056	0.21	0.000*	-0.180	0.37	0.627	-0.411	0.15	0.008*
Impacts												
Impact: eMINTS—control	0.705	0.86	0.412	0.591	0.13	0.000*	0.355	0.24	0.135	0.665	0.10	0.000*
Impact: eMINTS+Intel—control	1.151	0.73	0.113	0.679	0.11	0.000*	0.491	0.20	0.015*	0.671	0.08	0.000*
Pretest												
Spring 2011 teacher survey logit scores	0.687	0.19	0.000*	0.728	0.08	0.000*	0.457	0.16	0.005*	0.728	0.11	0.000*
Teacher Characteristics												
Gender: male—female	-0.297	1.09	0.785	0.359	0.17	0.036*	-0.060	0.30	0.842	-0.036	0.13	0.773
Graduate degree: with—without	-0.098	0.72	0.892	-0.120	0.11	0.291	0.168	0.20	0.400	0.047	0.08	0.571
Years of teaching experience	0.029	0.04	0.502	-0.001	0.01	0.903	0.002	0.01	0.864	-0.002	0.01	0.641
Grade(s) taught: Grade 8—both grades	-0.676	1.02	0.507	-0.143	0.16	0.372	-0.120	0.28	0.672	-0.078	0.12	0.507
Grade(s) taught: Grade 7—both grades	-1.130	1.02	0.267	0.143	0.16	0.375	0.068	0.29	0.811	0.005	0.12	0.968

Parameter	Community of Learners Logit Scores			Inquiry-Based Learning Logit Scores			High Quality Lesson Design Logit Scores			Technology Integration Logit Scores		
	Estimate	Standard Error	*p*-Value	Estimate	Standard Error	*p*-Value	Estimate	Standard Error	*p*-Value	Estimate	Standard Error	*p*-Value
Subject(s) taught: communication arts—both subjects	-2.008	1.22	0.100	0.226	0.19	0.237	0.590	0.34	0.079	0.253	0.14	0.072
Subject(s) taught: mathematics—both subjects	-1.288	1.26	0.306	0.005	0.20	0.981	-0.184	0.35	0.596	0.146	0.14	0.314
Student Demographics												
Percentage of students with free or reduced-price lunch	0.665	2.17	0.759	0.488	0.34	0.153	1.672	0.60	0.005 *	0.101	0.25	0.685
Percentage of White students	2.109	6.03	0.727	-0.156	0.95	0.869	-0.005	1.70	0.997	-0.299	0.70	0.669
Percentage of students with IEP	-3.007	2.52	0.232	0.189	0.40	0.635	-0.989	0.71	0.162	0.181	0.30	0.541
Percentage of LEP students	3.932	8.96	0.661	0.092	1.41	0.948	-0.705	2.50	0.778	-0.483	1.04	0.644
Percentage of male students	2.045	2.62	0.435	-0.927	0.42	0.026 *	-0.086	0.74	0.907	-0.599	0.30	0.048 *
Classroom mean MAP mathematics and communication arts *z*-scores	1.071	0.92	0.246	0.196	0.15	0.180	0.150	0.26	0.561	0.005	0.11	0.962

Note. The analytic regression models were adjusted for block effects and baseline teacher and classroom characteristics. From authors' analysis based on Year 3 (2013–14) data from the study districts and DESE.
*Statistically significantly different from zero at the .05 significance level.

Appendix K. Estimated Regression Coefficients for the Classroom Observation Analysis

Table K1. Estimates of Regression Coefficients for the Impact Analyses of eMINTS on Classroom Observation Outcomes in Implementation Year 3 (2013–14)

	Community of Learners Logit Scores			Inquiry-Based Learning Logit Scores			Technology Integration Logit Scores		
	Estimate	Standard Error	*p*-Value	Estimate	Standard Error	*p*-Value	Estimate	Standard Error	*p*-Value
Intercepts (control)									
Block 1	1.770	0.49	0.000*	-1.233	0.76	0.103	-0.730	0.30	0.014*
Block 2	0.513	0.62	0.405	-2.491	0.95	0.009*	-1.006	0.37	0.007*
Block 3	0.715	0.59	0.229	-1.968	0.92	0.032*	-0.892	0.26	0.013*
Impacts									
Impact: eMINTS—control	0.713	0.34	0.036*	1.351	0.53	0.010*	0.573	0.21	0.006*
Impact: eMINTS+Intel—control	0.815	0.30	0.007*	1.242	0.47	0.008*	0.779	0.18	0.000*
Pretest									
Spring 2011 classroom observation logit scores	0.203	0.13	0.112	0.411	0.14	0.004*	-0.013	0.03	0.621
Teacher Characteristics									
Gender: male—female	0.043	0.39	0.913	0.299	0.60	0.621	-0.085	0.24	0.720
Graduate degree: with—without	0.016	0.30	0.958	-0.200	0.47	0.667	0.051	0.18	0.783
Years of teaching experience	0.016	0.02	0.361	0.024	0.03	0.396	-0.003	0.01	0.775
Classroom Characteristics									
Grade observed: Grade 8—both grades	0.638	0.48	0.186	-0.292	0.74	0.693	0.443	0.29	0.127
Grade observed: Grade 7—both grades	0.358	0.47	0.447	-0.272	0.73	0.709	0.505	0.29	0.077
Subject observed: comm. arts—math	0.217	0.26	0.396	0.296	0.40	0.464	0.401	0.16	0.011*
Percentage of students with free or reduced-price lunch	-1.126	0.92	0.220	0.159	1.40	0.910	-0.272	0.54	0.615
Percentage of White students	-3.238	2.35	0.168	-2.409	3.65	0.509	-1.791	1.54	0.244

	Community of Learners Logit Scores			Inquiry-Based Learning Logit Scores			Technology Integration Logit Scores		
	Estimate	**Standard Error**	***p*-Value**	**Estimate**	**Standard Error**	***p*-Value**	**Estimate**	**Standard Error**	***p*-Value**
Percentage of IEP students	-0.344	1.51	0.820	-0.489	2.34	0.835	-3.657	0.93	0.000*
Percentage of LEP students	-4.617	3.54	0.192	-0.748	5.55	0.893	-1.130	2.23	0.612
Percentage of male students	1.776	1.22	0.147	0.836	1.87	0.656	-0.588	0.73	0.423
Classroom mean MAP mathematics and communication arts *z*-scores	0.391	0.43	0.358	1.120	0.65	0.083	0.178	0.25	0.479

Note. The analytic regression models were adjusted for block effects and baseline teacher and classroom characteristics. From authors' analysis based on Year 3 (2013–14) data from the study districts and DESE.
*Statistically significantly different from zero at the .05 significance level.

Appendix L. Attrition Rates

Table L1. Attrition Rates on Student Outcomes, 2013-14

Outcome	Group	Number of Students			Attrition Rate (Percentage)	Chi-Square Test of Equality of Rates
		Observed	Missing	Total		
MAP 2012 mathematics *z*-scores	Overall	3,072	311	3,383	9.2%	$\chi^2 = 4.5$, df = 2, *p*-value = .107
	eMINTS +Intel	1,141	105	1,246	8.4%	
	eMINTS	1,108	131	1,239	10.6%	
	Control	823	75	898	8.4%	
MAP 2012 communication arts *z*-scores	Overall	3,295	88	3,383	2.6%	$\chi^2 = 0.8$, df = 2, *p*-value = .665
	eMINTS +Intel	1,216	30	1,246	2.4%	
	eMINTS	1,208	31	1,239	2.5%	
	Control	871	27	898	3.0%	
21st century skills scale scores	Overall	2,376	923	3,299	28.0%	$\chi^2 = 11.3$, df = 2, *p*-value = .004*
	eMINTS +Intel	866	380	1,246	30.5%	
	eMINTS	827	328	1,155	28.4%	
	Control	683	215	898	23.9%	
Student engagement scores	Overall	2,278	1,021	3,299	30.9%	$\chi^2 = 72.9$, df = 2, *p*-value = .000*
	eMINTS +Intel	751	495	1,246	39.7%	
	eMINTS	868	287	1,155	24.8%	
	Control	659	239	898	26.6%	

*Statistically significantly different from zero at the .05 level.

Table L2. Attrition Rates on Teacher Outcomes, 2013-14

Outcome	Group	Number of Students			Attrition Rate (Percentage)	Chi-Square Test of Equality of Rates
		Observed	Missing	Total		
Teacher Survey[a]						
Community of learners	Overall	97	72	169	42.6%	$\chi^2 = 6.5$, df = 2, *p*-value = .039*
	eMINTS +Intel	35	28	63	44.4%	
	eMINTS	24	28	52	53.8%	
	Control	38	16	54	29.6%	
Inquiry-based	Overall	96	73	169	43.2%	$\chi^2 = 5.5$, df = 2,

learning/high-quality lesson design/ technology integration	eMINTS +Intel	35	28	63	44.4%	*p*-value = .065
	eMINTS	24	28	52	53.8%	
	Control	37	17	54	31.5%	
Classroom Observation[b]						
Community of learners/inquiry-based learning	Overall	117	52	169	30.8%	$\chi^2 = 2.0$, df = 2, *p*-value = .372
	eMINTS +Intel	43	20	63	31.7%	
	eMINTS	33	19	52	36.5%	
	Control	41	13	54	24.1%	
Technology integration	Overall	115	54	169	32.0%	$\chi^2 = 2.3$, df = 2, *p*-value = .317
	eMINTS +Intel	41	22	63	34.9%	
	eMINTS	33	19	52	36.5%	
	Control	41	13	54	24.1%	

[a]There are four outcomes from the teacher survey. The analytic samples for an inquiry-based learning, high-quality lesson design and technology integration outcomes are the same, and the analytic sample for the community of learners outcome had one more teacher than those of the other samples. [b]There are three outcomes from the classroom observations. (There were no items in the observation protocol that measured high-quality lesson design.) The analytic samples for the community of learners and inquiry-based learning outcomes are the same, and the analytic sample for the technology of integration outcome had two fewer teachers.

AMERICAN INSTITUTES FOR RESEARCH®

1120 East Diehl Road, Suite 200
Naperville, IL 60563-1486
800.356.2735 | 630.649.6500
www.air.org

Making Research Relevant